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UNIVERSITY of WEST FLORIDA

Electric Jet-Driven Surfboard
with Modular Propulsion Box

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We certify that this assignment is the result of our own efforts.

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Abstract

This project presents the design and development of a low-cost, electric jet-driven surfboard capable of high-speed planing on open water. Commercial electric surfboards are cost-prohibitive and rely on proprietary systems, creating a need for an accessible, modular alternative. The proposed design integrates a 25 kW brushless outrunner motor, a jet drive, and a 20s10p Molicel P45B lithium-ion battery pack into a removable propulsion cartridge housed within a CNC-machined XPS foam core board laminated with fiberglass and epoxy. Performance modeling predicts a top speed of 28–32 mph with a ride duration of 15–30 minutes at a total system weight under 100 lbs. Material testing of the fiberglass-epoxy composite shell — via four-point flexural bending and tensile testing — confirmed structural properties within acceptable design margins. The system targets a minimum IP-67 waterproof rating for all electronics and was designed within a $4,000 budget. Testing phases include static thrust, waterproof enclosure validation, tethered water trials, and open-water speed and endurance testing to verify that the prototype meets all design objectives.

1. Introduction

The objective of this capstone project was to design and develop an electric jet-driven surfboard capable of achieving high speed planing on water while maintaining safe, efficient, and environmentally responsible operation. Electric surfboards represent an emerging segment within the personal watercraft industry; however, commercially available models often exceed $10,000 and provide limited opportunities for customization. Given these constraints, this project focused on the development of a modular, high-performance prototype that could be fabricated using accessible resources at UWF and at a significantly lower cost.

The final design was produced for approximately $4,000 using a combination of Sea3D Lab additive manufacturing capabilities, composite fabrication techniques, and commercially available electric propulsion components. The Fall 2025 semester was dedicated to conceptual design and system development, including requirements definition, propulsion system selection, battery architecture design, hydrodynamic analysis, and initial CAD modeling.

During the Spring 2026 semester, the team transitioned to fabrication and assembly. This phase included manufacturing a modular propulsion housing unit through 3D printing, followed by fiberglass reinforcement to enhance structural integrity. The surfboard itself was constructed by CNC machining a foam core, which was then layered with fiberglass to create a durable composite structure. After fabrication and assembly were completed, the prototype underwent waterproof testing and full assembly testing to confirm overall functionality and durability.

2. Problem Definition

2.1 Problem/Need

Commercial electric surfboards are expensive, difficult to repair, and use proprietary propulsion systems. There is a need for a modular, customizable, low-cost electric board that can be replicated by engineering teams and scaled for future small-business production. This project fills that gap by creating an open-architecture design using accessible manufacturing methods.

2.2 Intended user(s) and use(s)

The intended users include recreational riders, surfers, and engineering students seeking a replicable capstone project. The final surfboard is meant to operate in lakes, bays, and near-shore coastal waters, providing a stable planing platform capable of moderate-to-high speeds. The design prioritizes accessibility, repairability, and performance suitable for riders of various experience levels.

2.3 Assumptions and limitations

Several assumptions guided the initial design. The expected rider weight ranges from 150 to 220 pounds, and the board will typically be used in relatively calm waters to reduce drag and improve handling. It was also assumed that standard charging equipment would be available and that the motor and ESC must remain fully sealed during operation. Limitations included the maximum printing size of Sea3D Lab equipment, the weight of the battery pack relative to board buoyancy, and the electrical constraints imposed by the 300 A continuous-rated ESC.

2.4 Design objectives

The core objectives of the design were to achieve a minimum speed of 25 mph using a 100 mm jet drive, provide at least 15 to 20 minutes of continuous ride time, and develop a propulsion cartridge that is both waterproof and modular. Additionally, the board had to be manufacturable using CNC-cut XPS foam laminated with fiberglass and epoxy, suitable for fabrication within university facilities.

3. Design Specifications

The technical specifications established for this semester include a 25 kW, 110 kV brushless outrunner motor paired with a 100 mm jet pump. The propulsion system is powered by a 20s10p Molicel P45B lithium-ion battery pack capable of supplying approximately 45 Ah at 74 V nominal. The board is designed to be between six and seven feet in length and must remain under 100 pounds to maintain efficient planing characteristics. All electronic components must meet a minimum waterproofing requirement of IP67, and the ESC is rated for 300 A continuous operation with water-cooling.

4. End-Product Description

The final product is a fully electric jet-driven surfboard capable of achieving top speeds of approximately 30 mph, with an estimated continuous duration of 45 minutes. A key feature is its removable modular propulsion unit, which integrates the motor, ESC, jet drive, cooling system, and battery pack into a single self-contained assembly.

This modular design enables the propulsion system to be transferred between compatible watercraft platforms designed with a standardized mounting interface, allowing for adaptation into applications such as electric kayaks, personal watercraft, or small boats. In addition to enhancing versatility, the modular unit simplifies maintenance, troubleshooting, and future system upgrades.

The surfboard itself is constructed from a CNC-machined extruded polystyrene (XPS) foam core, reinforced with a fiberglass and epoxy composite layup. This construction provides a balance of strength, durability, and lightweight performance, resulting in a watercraft suitable for both recreational use and experimental applications.

5. Design constraints

The electric jet-driven surfboard design was shaped by several practical and engineering constraints that directly impacted material selection, propulsion decisions, safety considerations, and the final board geometry. These constraints guided the direction of the project.

5.1 Weight and Buoyancy

The overall weight of the surfboard, including the battery, motor, jet pump, and enclosure, is expected to be around 92 lb. When combined with a typical rider, the total operating weight approaches 270 lb. This limits how small or thin the board can be, because it must provide enough buoyancy to support the combined weight without sinking too deeply in the water. This constraint influenced the board thickness, the internal foam core volume, and where the rider stands relative to the board's center of buoyancy.

Based on the board's approximate displacement, it should generate approximately 300 lbf of buoyant force in fresh water. This is more than enough to keep the board itself and the average rider afloat.

5.2 Propulsion System

The jet pump and electric motor must work together efficiently to push the board onto plane and maintain speed. The 100 mm jet pump can only move a certain amount of water, and the electric motor and ESC can only deliver a certain amount of power without overheating. These limits influenced our choice of motor size, battery configuration, and jet pump diameter. The propulsion system had to be powerful enough to overcome water drag while still fitting inside a compact waterproof enclosure.

5.3 Battery and Thermal

The 20s10p Molicel P45B battery pack has limits on how much current it can safely deliver and how much heat it produces. Because the surfboard draws high power, battery temperature, ESC temperature, and cooling all became important constraints. The pack must provide enough runtime without exceeding safe operating temperatures. These considerations affected the enclosure design, placement of cooling lines, and decisions about wire sizing and connectors.
Cooling will be handled by tapping some of the high pressure water at the turbine exhaust to route through the cooling system. Using the turbine itself as the coolant pump will ensure that cooling water is always flowing through the components while they are under load and require the most cooling. An additional air-to-water heat exchanger may also be added to the inside of the drive unit to allow cooling of the air to actively provide cooling to other components without their own dedicated cooling solution. Providing ample cooling will prolong the life of all driveline components and electronics.

5.4 Structural and Material

The board must be strong enough to handle rider weight, impacts from waves, and the forces coming from the jet pump, all while staying as light as possible. Working with CNC-machined XPS foam and fiberglass limited what shapes and thicknesses could be made easily within the Sea3D Lab's capabilities. The propulsion box also needed to be rigid and waterproof, which constrained its shape, wall thickness, and mounting locations inside the board.

5.5 Electrical and Safety

The surfboard uses high-current components that can be dangerous if not handled properly. All wiring, connectors, and fuses had to be selected to prevent overheating or electrical failure. Safety constraints also required the system to remain waterproof, avoid short circuits, and allow the rider to cut power quickly in the event of a fall. These needs directly shaped the interior layout of the propulsion box.

5.6 Manufacturing and Budget

The project budget limited which components could be purchased, requiring cost-efficient choices for the motor, ESC, and battery cells. The physical manufacturing process was constrained by the CNC machine size, the availability of fiberglass and epoxy, and the amount of 3D printing the Sea3D Lab could support. Large parts needed to be split into multiple sections or assembled in stages due to equipment limitations.

5.7 Hydrodynamic and Stability

The board must remain stable when accelerating, turning, and riding at high speeds. Stability requirements influenced the board's width, rocker profile, and stance area. The center of gravity had to be placed carefully so the board would not nose-dive, lose control, or become difficult to steer. These constraints shaped the overall board geometry more than any other factor.

5.8 Waterproofing and Durability

The propulsion components must remain completely sealed from water. This limited the number of openings allowed in the propulsion box and dictated the type of seals, gaskets, and cable pass-throughs used. Because the craft will repeatedly be submerged or splashed, waterproofing drove decisions about enclosure design, materials used, and connector types.

5.9 Economic

The total project budget is limited to approximately $3,000. This constraint requires careful selection of components that balance performance with cost. Commercial electric surfboards sell for over $10,000, so keeping the prototype affordable while maintaining functionality is a primary goal. The budget affected decisions whether to purchase pre-made components or manufacture parts in-house using university resources.

6. Safety

Safety was considered throughout every phase of the design. The most significant risks involve the lithium-ion battery pack, which can experience thermal runaway if punctured or short-circuited. To address this, the pack design includes insulation layers, proper fusing, and robust mechanical protection. High-current wiring between the battery, ESC, and motor poses electrocution and fire hazards; therefore, appropriate conductor sizing and secure high-current connectors are required. Composite manufacturing introduces chemical exposure risks, making the use of gloves, respirators, and ventilation mandatory. Water-testing will be carried out only after controlled on-land functional tests verify the reliability of the propulsion system, ensuring rider safety during early trials.

7. Engineering Standards

Several engineering standards guided our design to ensure the surfboard was safe, reliable, and built using good engineering practices. For the battery pack, we followed safety principles from UL and IEC lithium-ion standards, which emphasize proper insulation, cell spacing, and protection against short circuits. These guidelines influenced how we arranged the cells and designed the battery enclosure. For the electrical system, we referenced ABYC marine wiring standards. These standards helped us choose appropriate wire sizes, connectors, and fuse ratings so the high-current system could safely operate in a wet environment. The design of the fiberglass board structure was informed by common ASTM composite material guidelines, which helped us determine how many fiberglass layers were needed for strength and durability. We also considered ISO recommendations for small watercraft regarding rider safety and stability. These principles helped shape the overall board layout and ensured the craft would behave predictably on the water. Overall, these standards provided a foundation that helped us make safe and informed engineering decisions during the design process.

For the mechanical design of the drive unit, we chose to follow the Ingress Protection (IP) dust and water resistance rating to ensure that the electronics are protected from the elements. We plan to at a minimum meet the IP-67 rating requirements (no dust ingress, no water ingress submerged at a depth of 1m for at least 30 minutes), but are targeting an IP-68 rating (no dust ingress, no water ingress submerged at 3m for a long period of time). These water and dust protection ratings give a clear guideline of how to analyze our design to see if it will perform well in the real world for protecting the electronics.

8. Technical Approach

The technical foundation of the Fall 2025 work relied heavily on modeling the hydrodynamics of the surfboard, the thrust characteristics of the 100 mm jet pump, and the electrical/thermal behavior of the propulsion system. Further testing was then conducted below:

8.1 Technical Details of Implementation

8.1.1 Four-Point Flexural Bending Test Results - Fiberglass/Epoxy Shell

8.1.1.1 Equipment and Specimens

This experiment utilized a PASCO Scientific Materials Testing System Model ME-8230 with the Four-Point Load Anvil Model ME-8249 attachment to conduct the flexural tests. The Materials Testing System measures force up to 1600 lbs with an attached load cell and measures displacement with an optical encoder.

Figure A - The Four-Point Load Anvil Model ME-8249 used in this experiment.

Figure B - Flexural Testing Samples - Before Testing

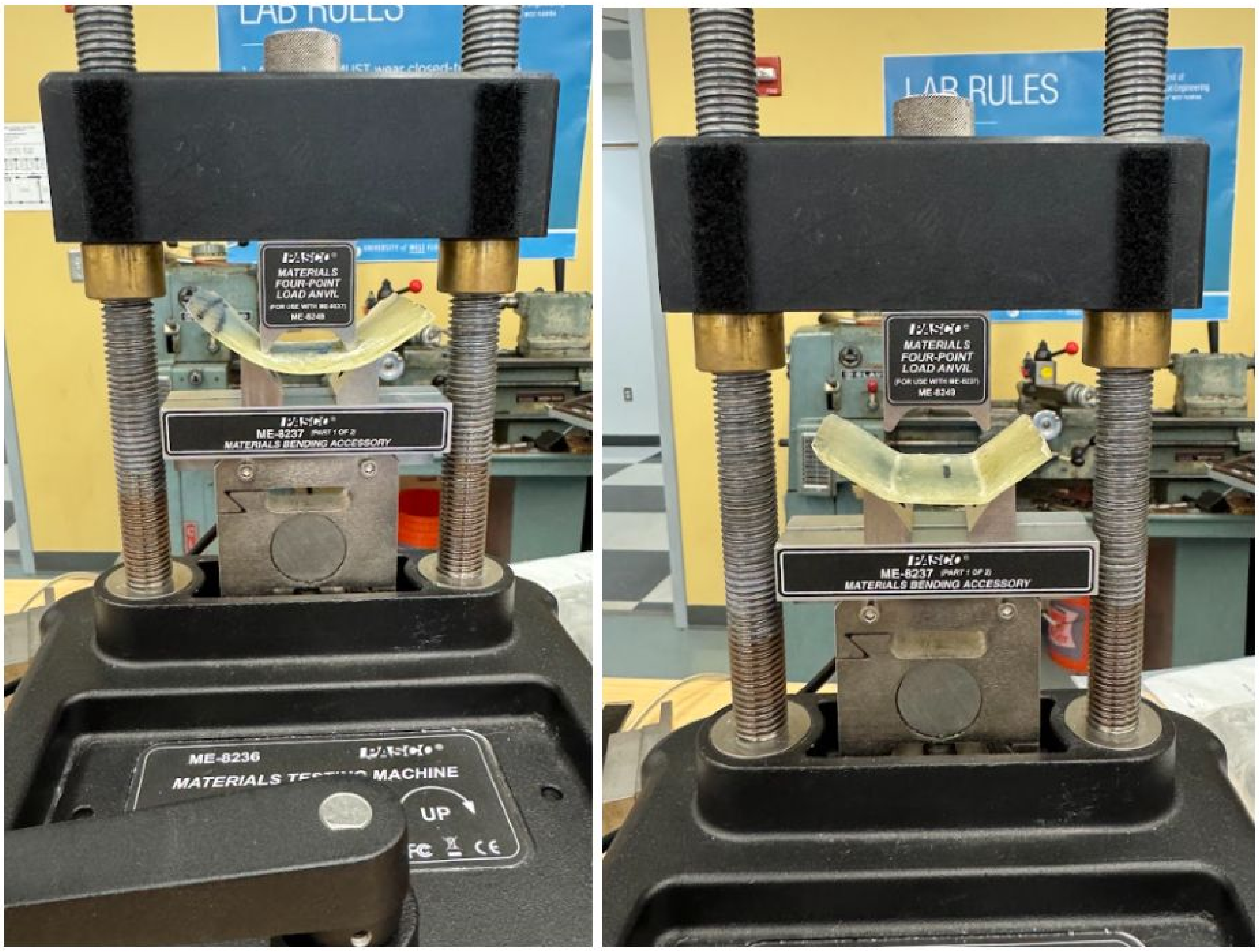


Figure C - Flexural Testing Samples - After Testing

8.1.1.2 Test Setup and Variables

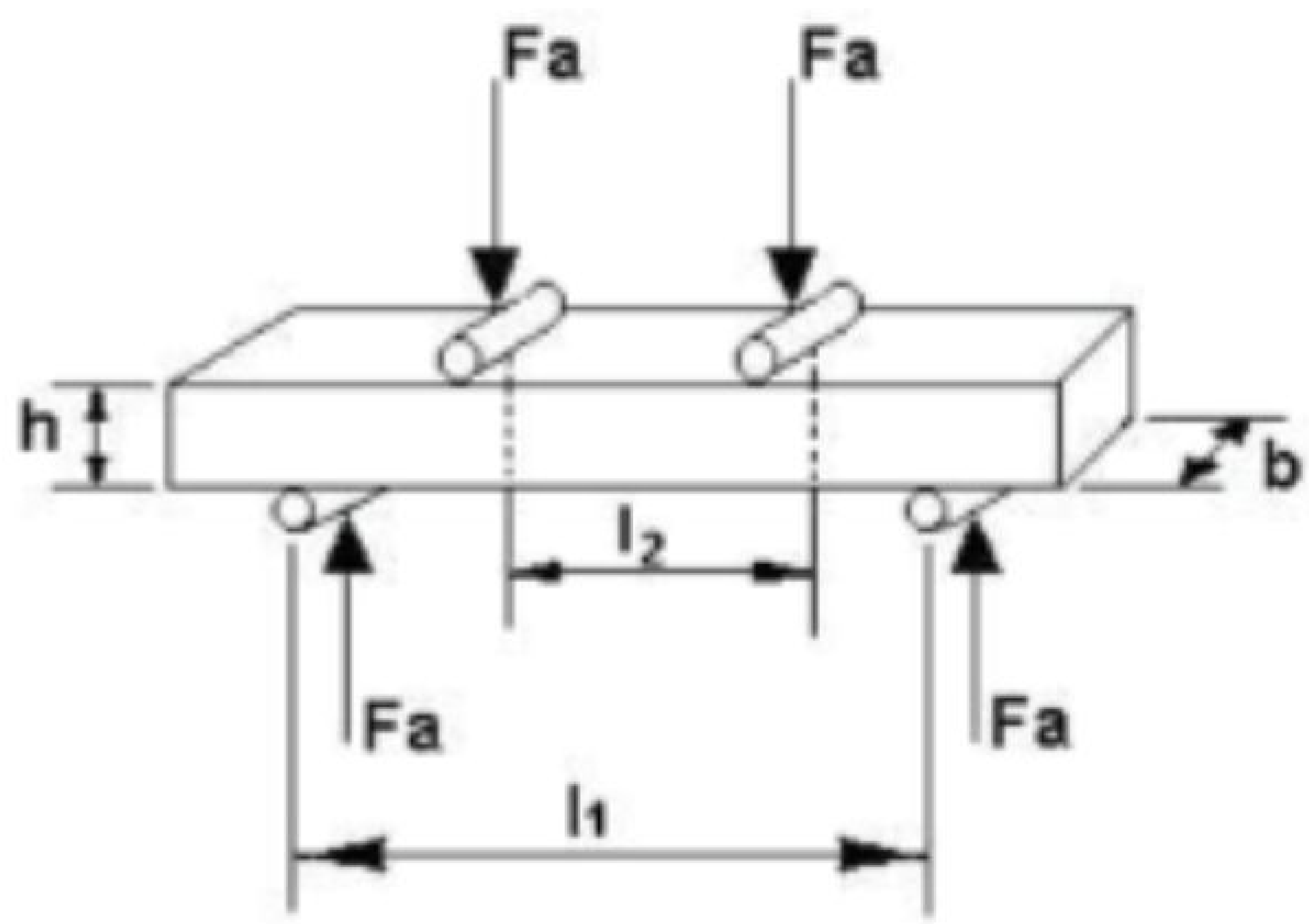


Figure D - Four-Point Bending Configuration

Figure D: Four-point bending test configuration showing outer span l_1, inner loading span l_2, beam width b, and thickness h

- l_1 = outer support span (distance between lower supports)
- l_2 = inner loading span (distance between upper loading points)
- b = specimen width

- h = specimen thickness
- F_a = applied load at each loading point
- F = total applied load (sum of both F_a loads)

8.1.1.3 Determine the Slope K from Load-Deflection Data

The slope K represents the stiffness of the specimen in the elastic (linear) region of the load-deflection curve.

$$K = \frac{\Delta F}{\Delta \delta}$$

where:

- ΔF = change in applied force (N)
- $\Delta \delta$ = change in mid-span deflection (m or mm)

Procedure to find K:

1. Plot the load F versus mid-span deflection δ data from your test
2. Identify the initial linear (elastic) region of the curve—typically the first 20–30% of the maximum load
3. Select two points (F_1, δ_1) and (F_2, δ_2) within this linear region
4. Calculate the slope:

$$K = \frac{F_2 - F_1}{\delta_2 - \delta_1}$$

Figures X-X: Load-deflection curves from the four-point bending test. The initial linear region will be used to determine the slope K.

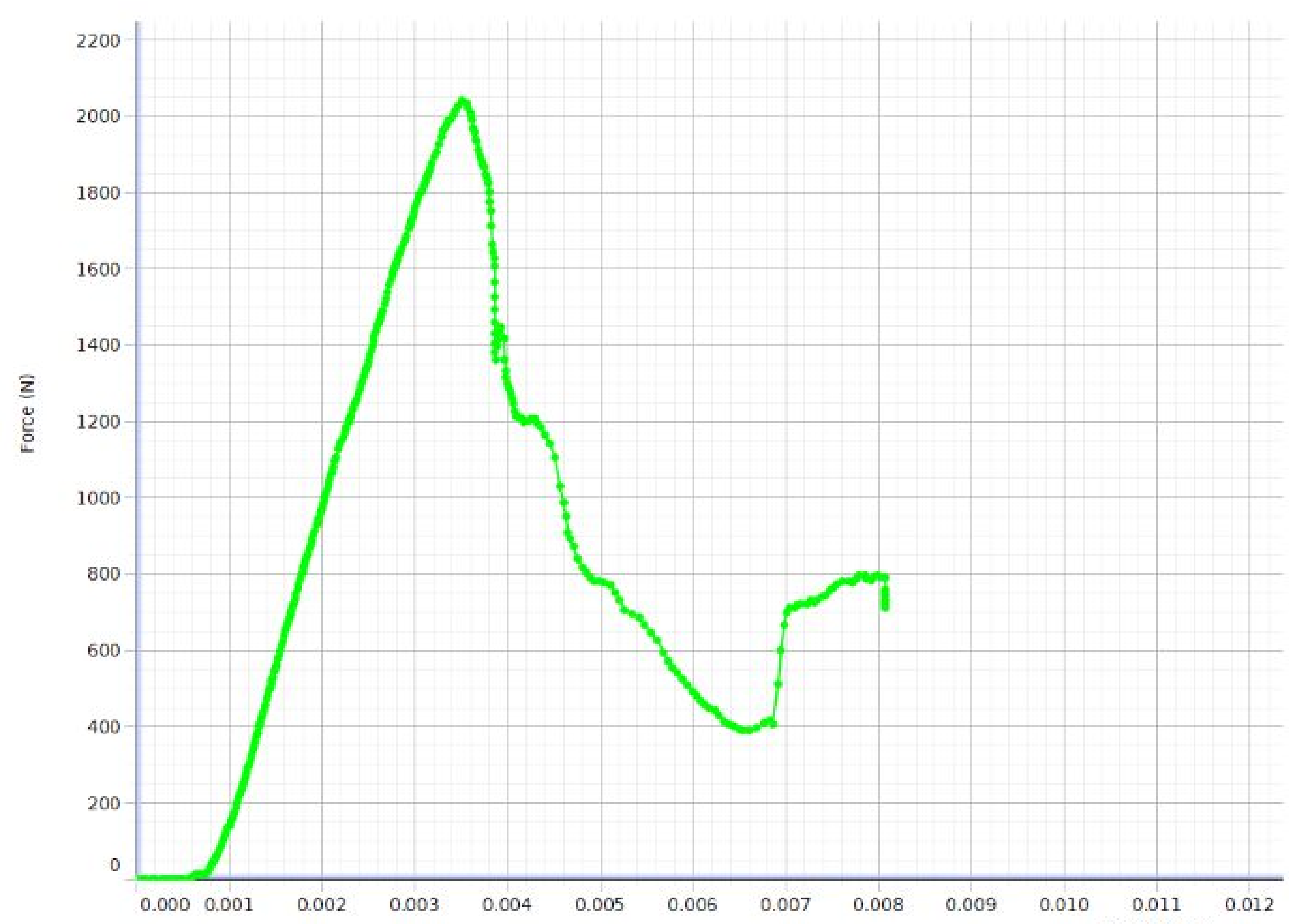


Figure E - Test Sample 1a - Run #1

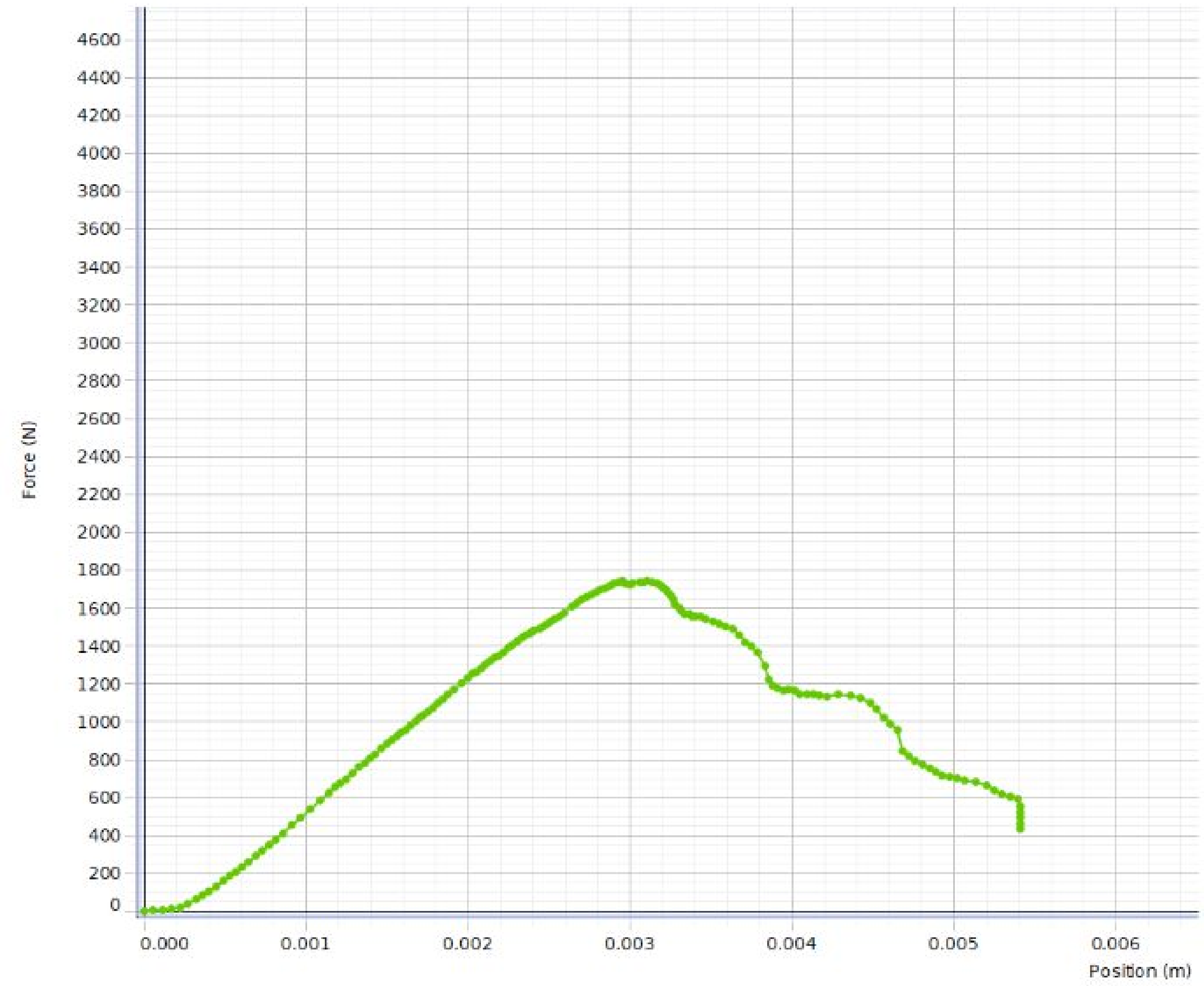


Figure F - Test Sample 1b - Run #6

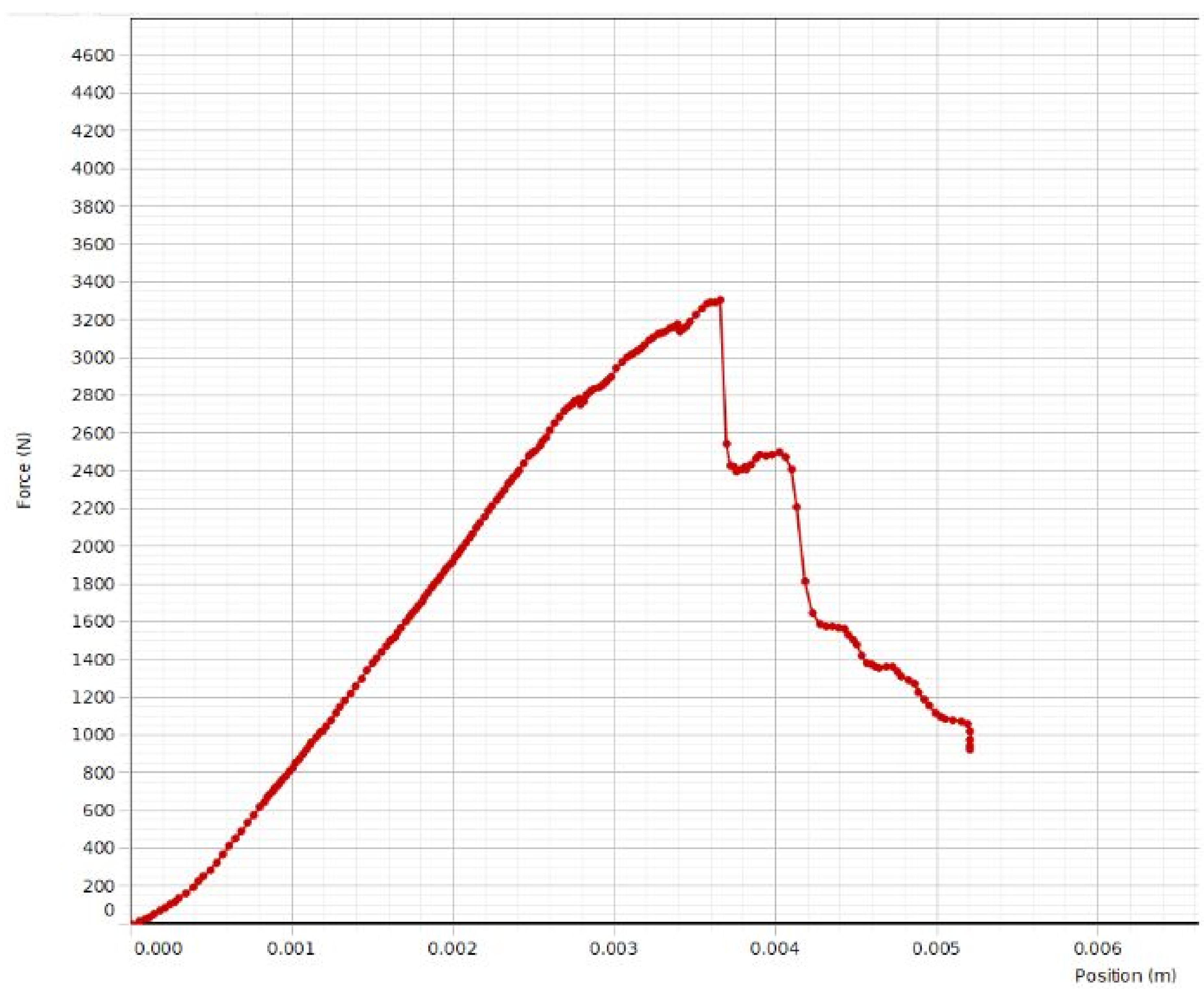


Figure G - Test Sample 2a - Run #1

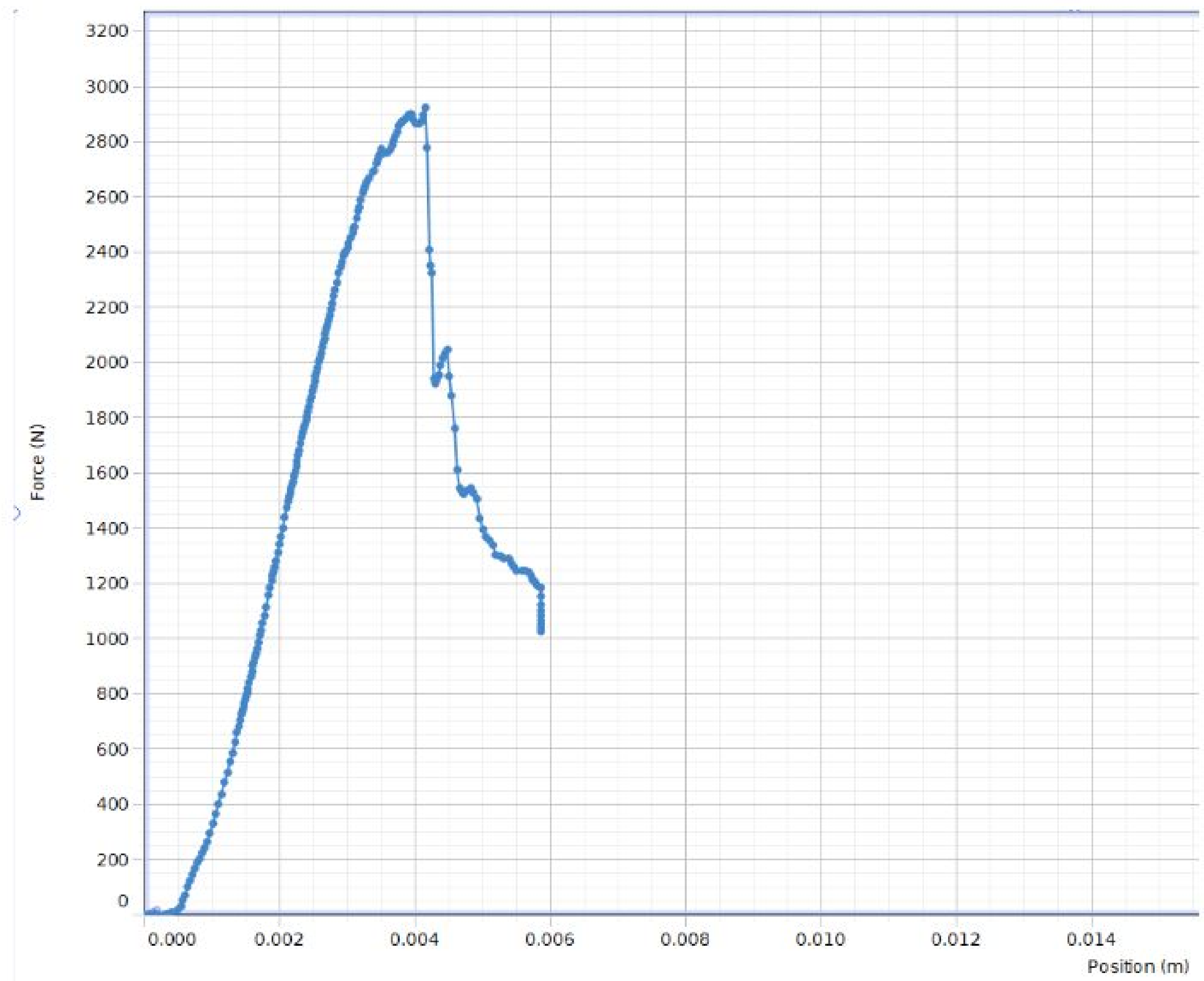


Figure H - Test Sample 2b - Run #3

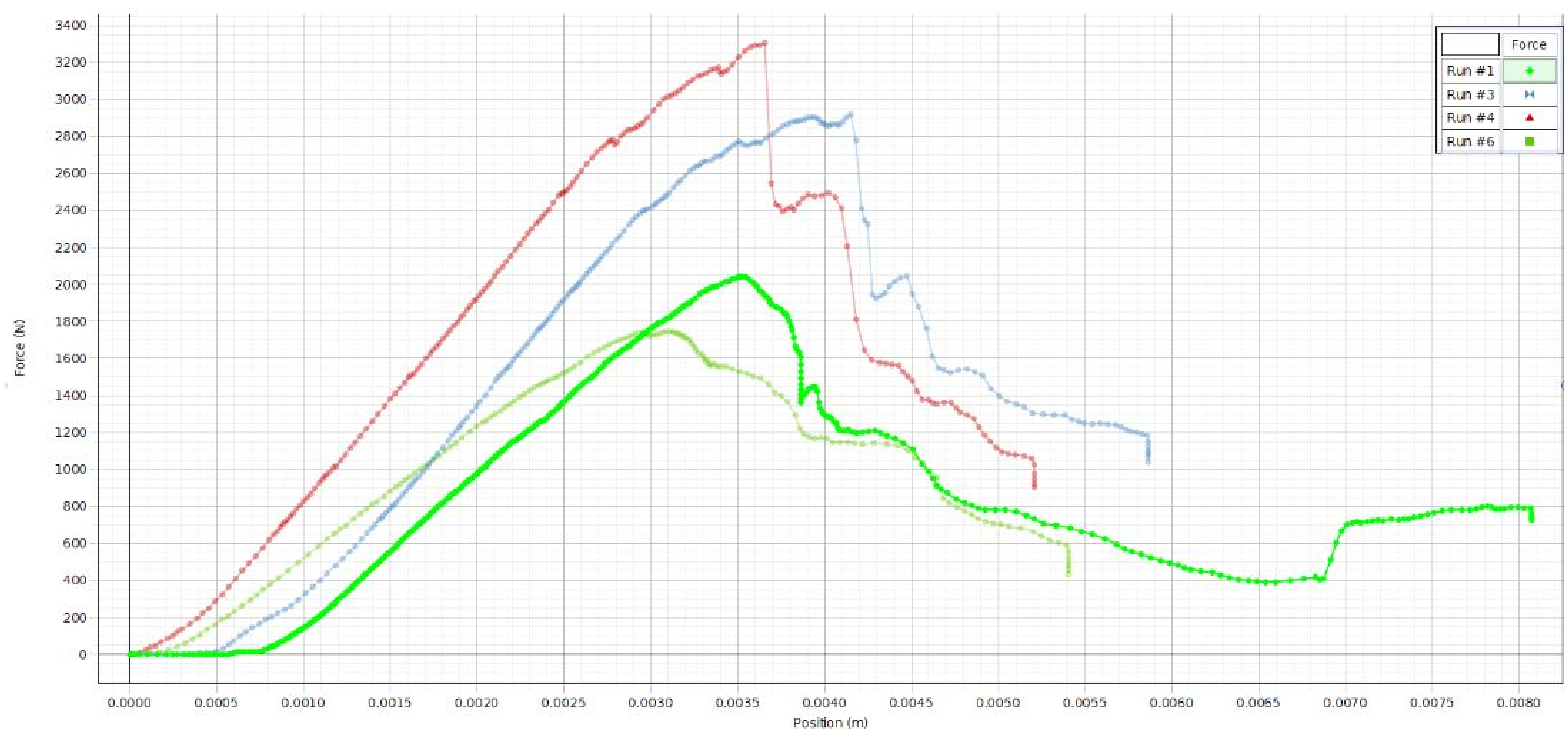


Figure I - (4) Test Sample Overlay for Comparison

8.1.1.4 Calculate Area Moment of Inertia

For a rectangular cross-section, the second moment of area about the neutral axis is:

$$I = \frac{bh^3}{12}$$

where:

- b = width of the beam cross-section
- h = thickness (height) of the beam cross-section

8.1.1.5 Apply Four-Point Bending Deflection Formula

For a beam in four-point bending, the mid-span deflection is given by:

$$\delta = \left(\frac{l_1^3 - l_2^3}{48EI}\right)F$$

This can be rearranged to express the relationship as:

$$\frac{F}{\delta} = K = \frac{48EI}{l_1^3 - l_2^3}$$

8.1.1.6 Solve for Modulus of Elasticity

Rearranging the equation from Step 3 to solve for E:

$$E = \frac{K(l_1^3 - l_2^3)}{48I}$$

Substituting the expression for I from Step 2:

$$E = \frac{K(l_1^3 - l_2^3)}{48\left(\frac{bh^3}{12}\right)}$$

Simplifying:

$$E = \frac{K(l_1^3 - l_2^3)}{4bh^3}$$

8.1.1.7 Data

	Run	**Max F**	**Max Linear F**	**delta**	**K = delta(F) / delta(Deflect)**	**h**	**b**	**I**	**L1**	**L2**	**E**
Units:	#	*N*	*N*	*mm*	*N/(mm)*	*mm*	*mm*	*mm^4*	*mm*	*mm*	*MPa*
1a	1	2043	2043	3.45	592.173	2.36	58.3	63.859	44.92	30.91	11805.375
1b	6	1743	1743	2.90	601.034	2.87	54.6	107.561	44.92	30.91	7113.709
2a	4	3305	3200	3.40	941.176	2.91	50.7	104.113	44.92	30.91	11508.513
2b	3	2922	2750	3.50	785.714	3.43	53.5	179.909	44.92	30.91	5559.856

Table 1 - Flexural Sample Calculation

8.1.1.8 Example Calculation

Given the following test data sourced from Test Sample 1a - Run #1:

- $K = 592.17$ N/mm (from linear regression)
- $l_1 = 44.92$ mm
- $l_2 = 30.91$ mm
- $b = 58.3$ mm
- $h = 2.36$ mm

Calculate E:

$$E = \frac{592.17 \times (44.92^3 - 30.91^3)}{4 \times 58.3 \times 2.36^3}$$

$$E = \frac{592.17 \times (90738.1 - 29537.5)}{4 \times 58.3 \times 13.14}$$

$$E = \frac{592.17 \times 61200.6}{3063.8} = 11,805\ MPa = 11.8\ GPa$$

8.1.1.9 Expected Values for Fiberglass/Epoxy Composites

Laminate Type	Flexural Modulus Range
Neat epoxy resin	2–4 GPa
Low fiber content (30–40% by volume)	10–15 GPa
Moderate fiber content (50–60% by volume)	20–30 GPa
High fiber content unidirectional (60–70% by volume)	40–50 GPa

Table 2 - Typical flexural modulus ranges for for glass-fiber reinforced epoxy laminates

Values depend strongly on:

- Fiber volume fraction
- Fiber orientation relative to loading direction
- Fabric type (woven, unidirectional, random mat)
- Resin type and cure quality
- Presence of voids or manufacturing defects

8.1.2 Tensile Testing - Fiberglass/Epoxy Shell & Epoxy Used to Join Propulsion Box

8.1.2.1 Equipment and Specimens

This experiment will utilize a PASCO Scientific Materials Testing System Model ME-8230 to conduct the tensile tests.

Figure J - The MTS Criterion Model 45 used in this experiment

Specimens - Fiberglass/Epoxy Shell:

Figure K - Fiberglass/Epoxy Shell Dogbone Samples

Specimens - Epoxy used to join propulsion box:

Figure L - Joining Epoxy Dogbone Samples

8.1.2.2 Test Setup and Variables

Tensile testing provides comprehensive material property data through the analysis of engineering stress-strain curves. The fundamental relationships governing this analysis include:

$$\text{Engineering Stress: } \sigma = \frac{F}{A_0}$$

where:

- σ = engineering stress
- F = applied force
- A_0 = pre-testing cross sectional area

$$\text{Engineering Strain: } \varepsilon = \frac{\Delta L}{L_0}$$

where:

- ε = engineering strain
- ΔL = change in length = final length minus initial length
- L_0 = pre-testing gage length

$$\text{Elastic Modulus (Young's Modulus): } E = \frac{\sigma}{\varepsilon}$$

The elastic modulus represents the slope of the linear portion of the stress-strain curve and quantifies material stiffness.

Material ductility can be quantified through two primary metrics:

$$\text{Percent Elongation: } EL\% = \frac{(G_f - G_0)}{G_0} \times 100\ \%$$

$$\text{Percent Reduction in Area: } RA\% = \frac{(A_0 - A_f)}{A_0} \times 100\ \%$$

Where subscripts f and 0 denote final (post-fracture) and original conditions, respectively.

8.1.2.3 Tensile Testing Data - Fiberglass/Epoxy Shell

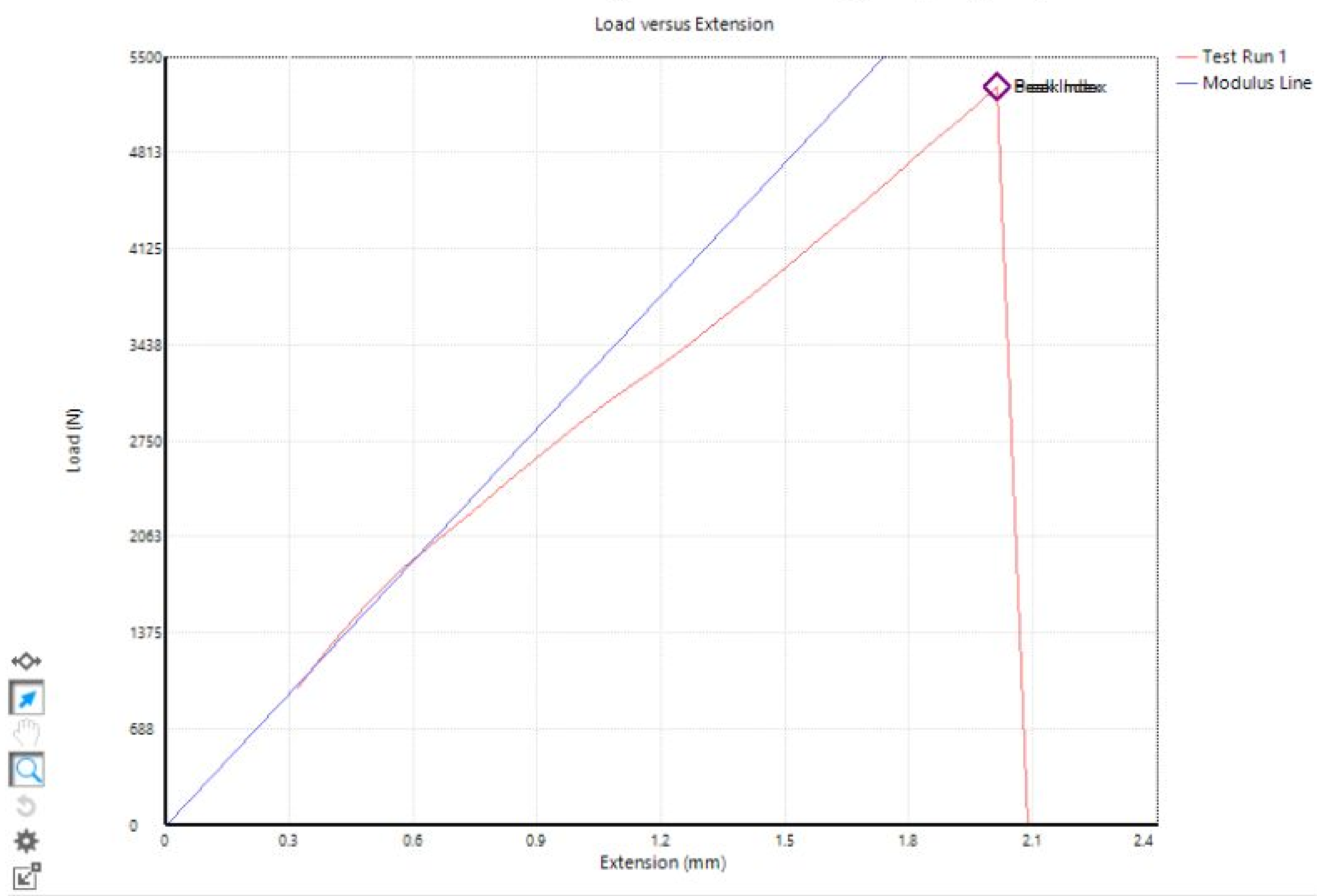


Figure M - Load versus Extension Plot of Control Sample for Fiberglass/Epoxy Shell Testing

Display Name	Value	Unit	Reset Value	Original Value	Description	Test-level
Test Run End Reason	Break Detected			Break Detected		No
Width	23.000	mm		23.000	The specimen wi	No
Thickness	2.500	mm		2.500	The specimen thi	No
Strain at Break	0.040	mm/mm		0.040		No
Peak Stress	91.9	MPa		91.9	The maximum str	No
Modulus	2791.230	MPa		2791.230	The Young's mod	No
Peak Load	5285.891	N		5285.891		No

Figure N - Testing Data of Control Sample for Fiberglass/Epoxy ShellTesting

8.1.2.4 Tensile Testing Data - Joining Epoxy

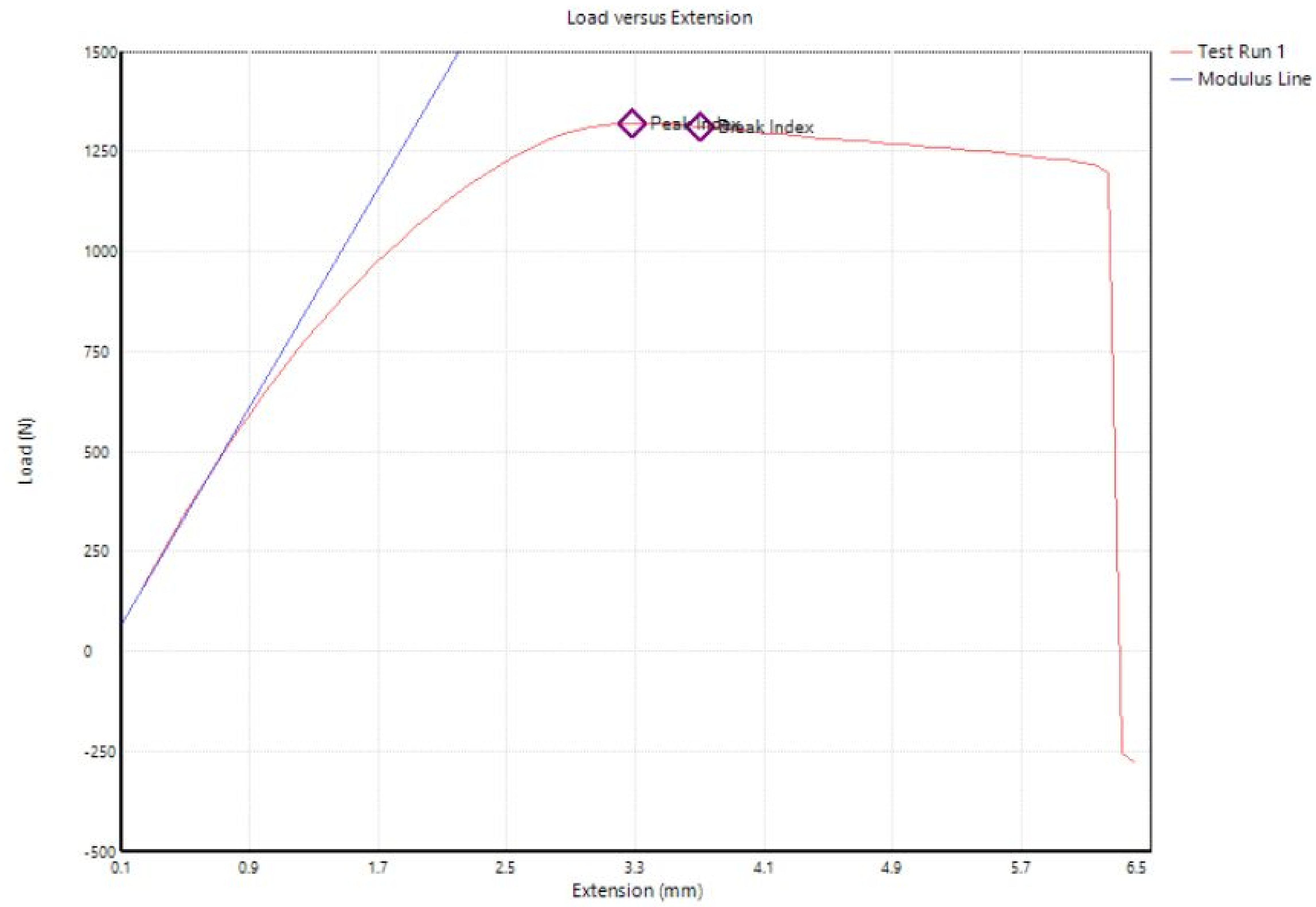


Figure O - Load versus Extension Plot of Control Sample for Joining Epoxy Testing

Drag a column header here to group by that column.

Display Name	Value	Unit	Reset Value	Original Value	Description	Test-level
Test Run End Reason	Break Detected			Break Detected		No
Peak Stress	4748.1	lbf/in²		4748.1	The maximum str	No
Peak Load	296.757	lbf		296.757		No
Strain at Break	0.073	in/in		0.073		No
Modulus	124568.257	lbf/in²		124568.257	The Young's mod	No
Width	0.500	in		0.500	The specimen wi	No
Thickness	0.125	in		0.125	The specimen thi	No

Figure P - Testing Data of Control Sample for Joining Epoxy Testing

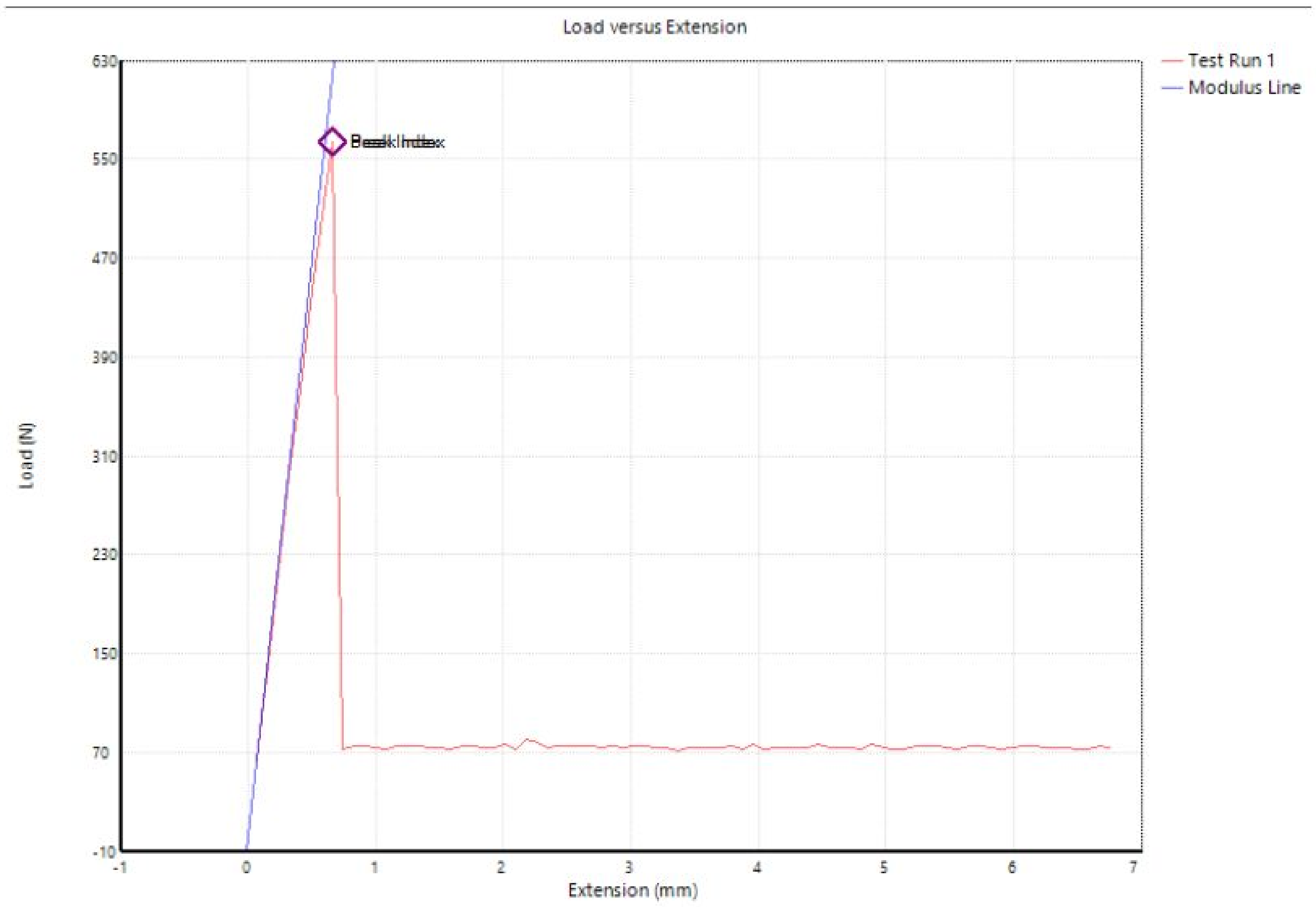


Figure Q - Load versus Extension Plot of Joining Epoxy Sample 1

Drag a column header here to group by that column.

Display Name	Value	Unit	Reset Value	Original Value	Description	Test-level
Test Run End Reason	Test Stopped			Test Stopped		No
Peak Stress	1569.2	lbf/in²		1569.2	The maximum str	No
Peak Load	126.719	lbf		126.719		No
Strain at Break	0.009	in/in		0.009		No
Modulus	181807.108	lbf/in²		181807.108	The Young's mod	No
Width	0.481	in		0.481	The specimen wi	No
Thickness	0.168	in		0.168	The specimen thi	No

Figure R - Testing Data of Joining Epoxy Sample 1

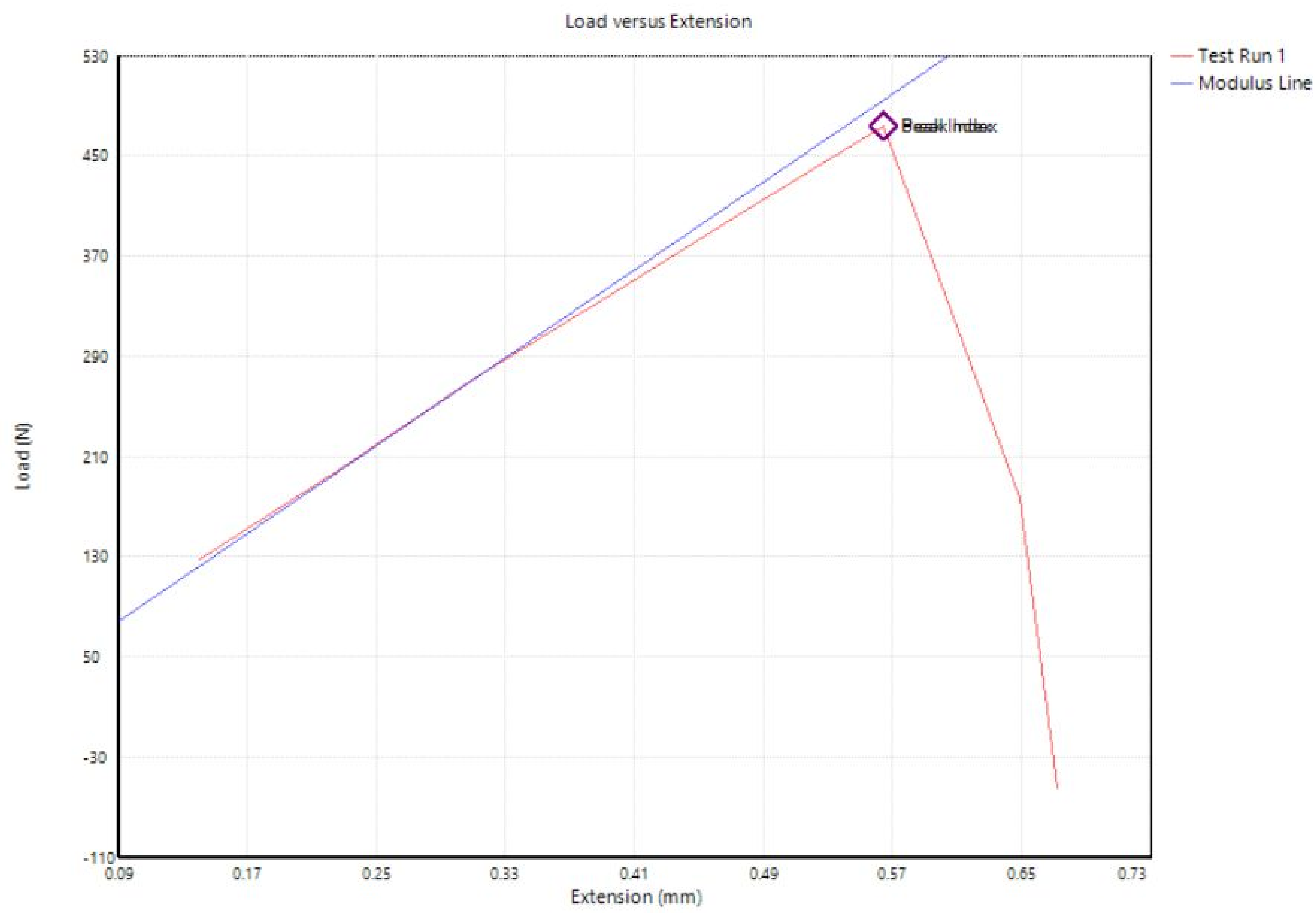


Figure S - Load versus Extension Plot of Joining Epoxy Sample 2

Drag a column header here to group by that column.

Display Name	Value	Unit	Reset Value	Original Value	Description	Test-level
Test Run End Reason	Break Detected			Break Detected		No
Peak Stress	1318.1	lbf/in²		1318.1	The maximum str	No
Peak Load	106.441	lbf		106.441		No
Strain at Break	0.008	in/in		0.008		No
Modulus	170892.794	lbf/in²		170892.794	The Young's mod	No
Width	0.481	in		0.481	The specimen wi	No
Thickness	0.168	in		0.168	The specimen thi	No

Figure T - Testing Data of Joining Epoxy Sample 2

8.1.2.5 Example Calculation

Parameter	**Control**	**Joining Sample 1**	**Joining Sample 2**	**Fiberglass Sample 1**
A_0 (mm^2)	8.966	49.72	49.72	57.5
Peak Load (N)	1320	563.7	473.5	5285.9
Peak Stress (MPa)	147.2	11.34	9.52	91.9

Table 3 - Sample Calculation

8.1.2.6 Discussion of Results

Tensile testing was conducted on three specimens to evaluate the performance of the joining epoxy relative to the control. The control sample, a circular cross-section specimen of PETG with a diameter of 3.378 mm and cross-sectional area of 8.966 mm^2, achieved an ultimate tensile stress of 147.2 MPa at a peak load of 1,320 N. This behavior is consistent with a relatively ductile fracture, confirmed by the gradual plateau and extended extension visible in the load-extension curve before final fracture.

Joining Epoxy Sample 1, with a rectangular cross-section of 49.72 mm^2, failed at a peak load of 563.7 N, yielding an ultimate tensile stress of 11.34 MPa — representing only 7.7% of the control's strength. Joining Epoxy Sample 2 performed similarly, failing at 473.5 N with an ultimate tensile stress of 9.52 MPa, or 6.5% of the control strength, indicating a highly brittle fracture mode consistent with the load drop observed in its load-extension curve.

Collectively, these results indicate that the joining epoxy, in its tested configuration, provides substantially reduced load-bearing capacity compared to the Control PETG material, with Sample 1 and Sample 2 failing at approximately 7.7% and 6.5% of the control's ultimate tensile strength, respectively. This significant strength reduction must be considered in the structural design of the propulsion box assembly.

8.1.2.7 Application of Joining Epoxy:

This tested epoxy was used to join the five printed segments of the propulsion housing box together. These results validate the need for adding structural support members between each of the segments to assist in the binding process and provide additional structural support.

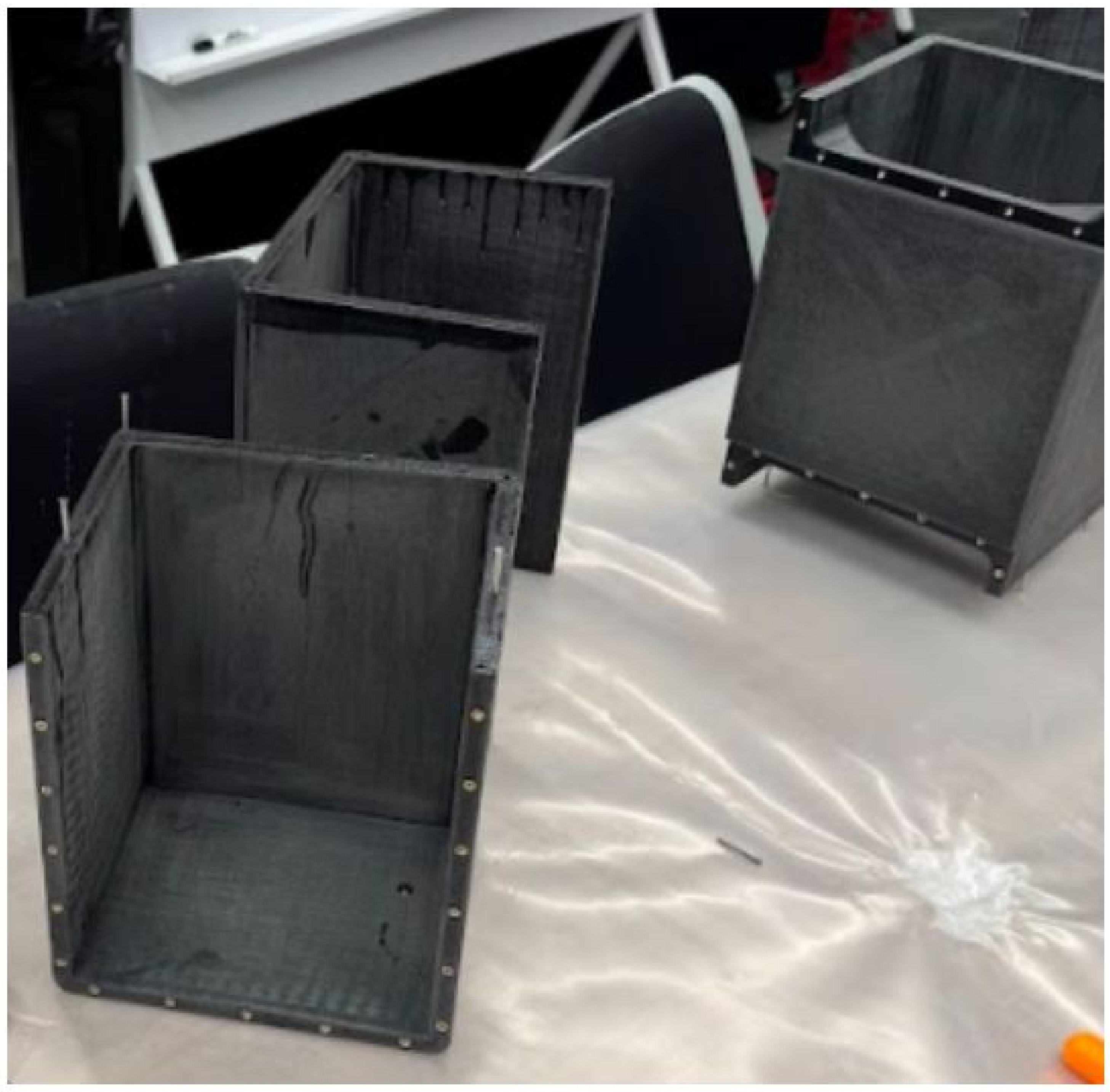


Figure U - Propulsion Box printed in segments, prior to being joined together

Figure V - Application of joining epoxy between each member

8.1.3 Factor of Safety - Fiberglass/Epoxy Shell

8.1.3.1 Assumptions

- The board is modeled as a simply supported beam of length $l = 2413\ mm$, with concentrated point loads applied at the rider position and propulsion unit location.
- The rider is centered on the board; their weight is divided equally between two support reactions.
- The propulsion unit is mounted at a fixed position along the board's length.
- The fiberglass/epoxy composite is treated conservatively as a homogeneous, isotropic material due to the limited available fatigue research on this material.

- Surface finish is assumed equivalent to a "Machined" finish for the purposes of the surface modification factor, based on published equation parameters.
- A 90% reliability target is selected, consistent with preliminary mechanical design practice for a non-safety-critical prototype.
- The motor operates at 90% of peak output to avoid driving electrical components to their absolute limits.
- Notch sensitivity $q = 1$ is applied conservatively, treating the composite as fully notch-sensitive.
- The endurance limit correction for the composite is taken as $S'_e = (0.3)\, S_{ut}$ rather than the standard $S'_e = (0.5)\, S_{ut}$ to account for material uncertainty.
- Loading is treated as fully reversed bending $\left(M_m = 0,\ T_a = 0\right)$; steady torque is applied from the driveshaft.

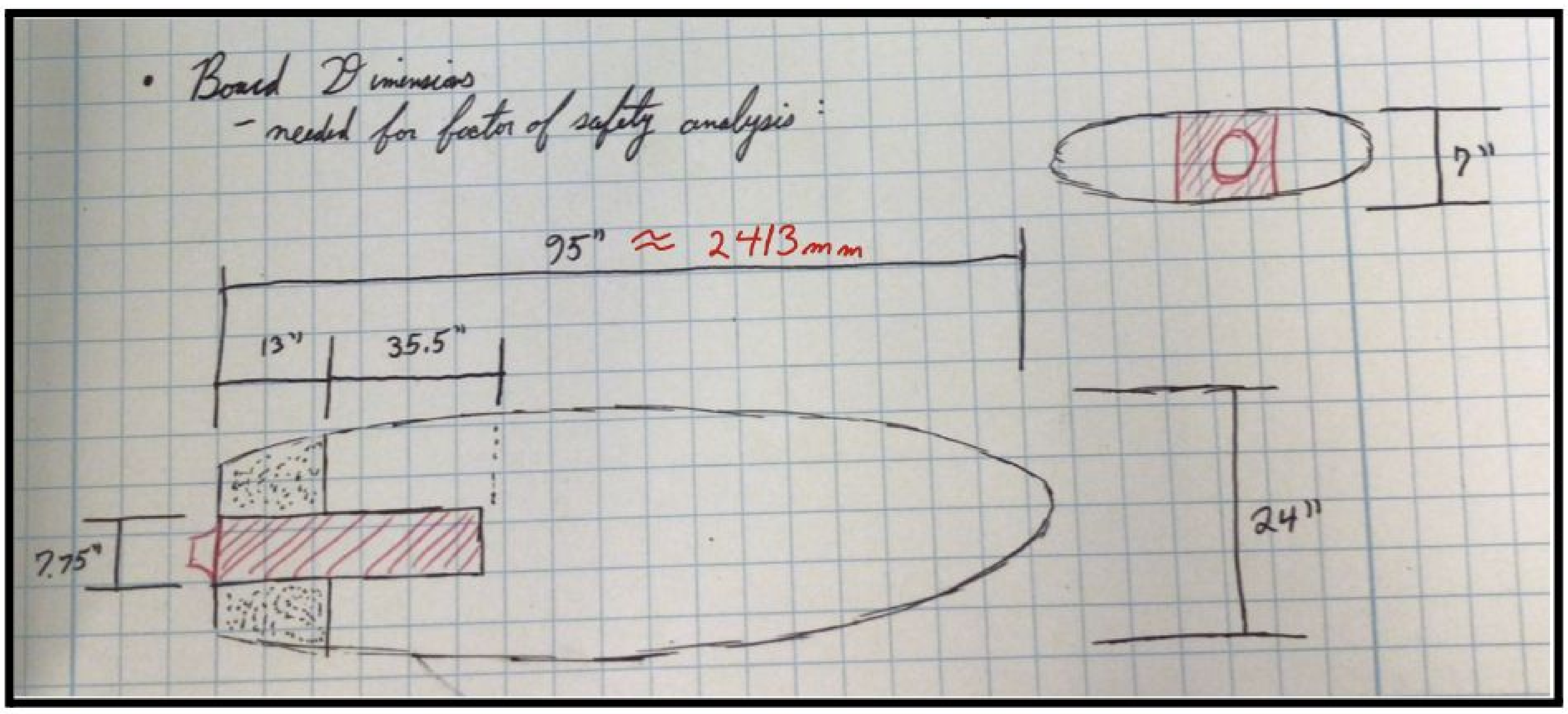


Figure W - Board Dimensions

8.1.3.2 Applied Loads

The two primary load sources are (1) the combined weight of the board and rider, and (2) the weight of the propulsion unit.

Board and Rider Weight:

$$W = \left(m_b + m_r\right)(g)$$

where:

- m_b = board mass
- m_r = rider mass
- $g = 9.81\ \frac{m}{s^2}$

$$W = \left(m_b + m_r\right)(g) = (86\ kg)\left(9.81\ \frac{m}{s^2}\right) = 843.66\ N \approx 844\ N$$

This load is divided equally between two reaction points:

$$F_1 = \frac{W}{2} = \frac{844}{2} = 422\ N$$

Propulsion Unit Weight:

$$F_p = m_p \cdot g$$

where:

- $m_p = 34\ kg$ is the mass of the propulsion unit.

$$F_p = m_p \cdot g = (34\ kg)\left(9.81\ \frac{m}{s^2}\right) = 333.54\ N \approx 334\ N$$

8.1.3.3 Section Properties and Bending Moment

Cross-Sectional Moment of Inertia

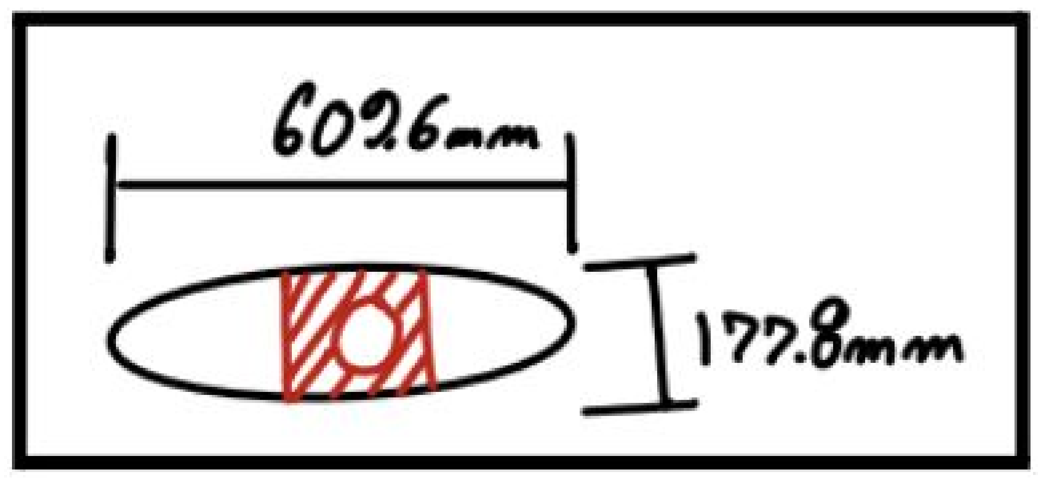


Figure X - Board Cross-Section Dimensioning

The critical cross-section is modeled as a solid rectangle with a width $b = 609.6\ mm$ and height $h = 177.8\ mm$. The area moment of inertia about the neutral axis is:

$$I = \frac{1}{12} bh^3$$

$$I = \frac{1}{12}(0.6096\ m)(0.1778\ m)^3 = 2.86 \times 10^{-4}\ m^4$$

Bending Moment Calculation (Superposition)

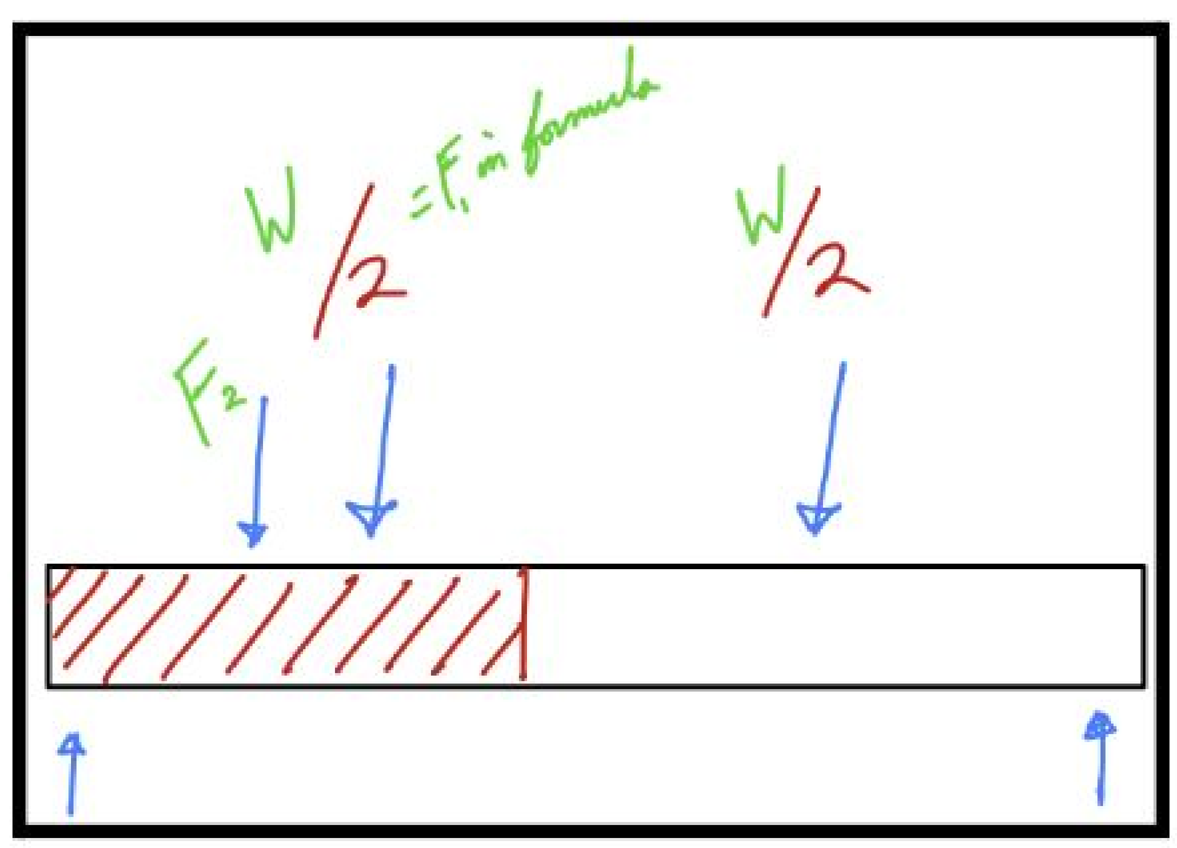


Figure Y - Simple Supports Diagram

Using beam superposition (Shigley's Table A-9), the bending moment at each load location is calculated separately and then combined.

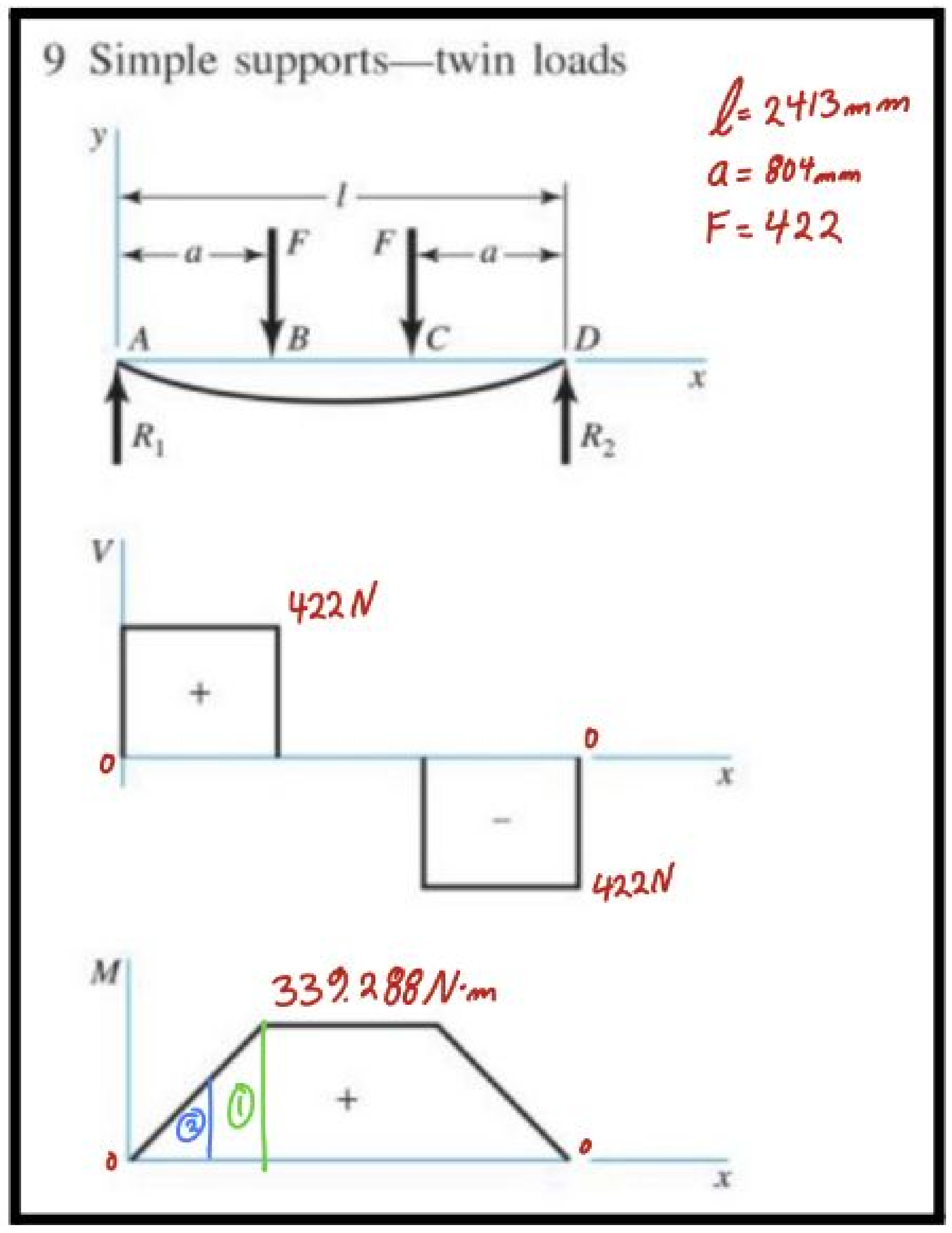


Figure Z - Load Case 1

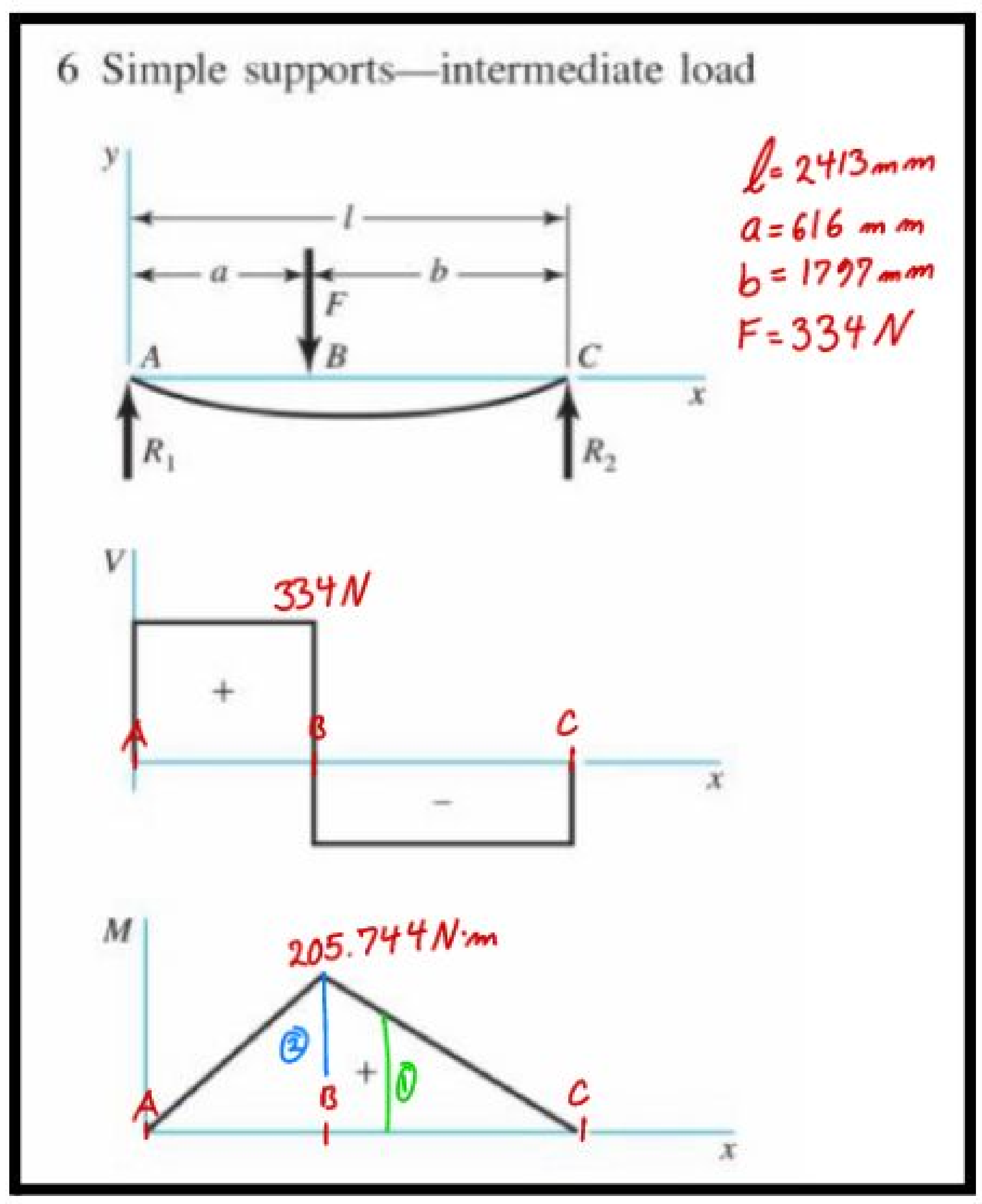


Figure AA - Load Case 2

where:

- $a_1 = 804\ mm$
- $a_2 = 616\ mm$
- $F_1 = 422\ N$
- $F_2 = 334\ N$
- $l = 2413\ mm$
- $b_2 = 1797\ mm$

Load Case 1 - Rider / Board Weight

$$M_1 @ 804\,mm = F_1 a_1 + \frac{F_2(a_2)}{l}(l - a_1) = 476.48\,N \cdot m$$

Load Case 2 - Propulsion Unit

$$M_2 @ 616\,mm = F_1 a_2 + \frac{F_2 \cdot a_2 \cdot b_2}{l} = 413.17\,N \cdot m$$

Maximum Bending Moment (Superposition):

$$M_1 = M_{max} \approx 477\,N \cdot m = 477,000\,N \cdot mm$$

Maximum Deflection

$$y_{max} = 0.0737\,mm$$

8.1.3.4 Shaft Torque

The peak motor output is $P_{peak} = 25\,kW$. To avoid driving electrical components to their absolute limit, 90% of peak power is used as the operating condition.

Operating Power:

$$P_{90} = (0.90)\,P_{peak} = (0.90)\,(25\,kW) = 22.5\,kW$$

Operating Shaft Speed:

$$N_{90} = (0.90)\left(N_{peak}\right) = (0.90)\,(9000\,rpm) = 8100\,rpm$$

Torque from Power and Speed (Shigley's Eq. 3-43):

$$T = \frac{P_{90} \cdot 60}{2\pi \cdot N_{90}}$$

where:

- T = torque in $N \cdot m$
- P_{90} is in W
- N_{90} is in rpm

$$T = \frac{(22{,}500)\,(60)}{2\pi\,(8100)} = 26.53\,N \cdot m = 26,530\,N \cdot mm$$

8.1.3.5 Marin Endurance Modification Factors

The modified endurance limit S_e is computed by applying a series of Marin modification factors to the correct material endurance limit S'_e:

$$S_e = k_a k_b k_c k_d k_e S'_e$$

Surface Factor (k_a)

Surface Finish	Factor a S_{ut}, kpsi	Factor a S_{ut}, MPa	Exponent b
Ground	1.21	1.38	−0.067
Machined or cold-drawn	2.00	3.04	−0.217
Hot-rolled	11.0	38.6	−0.650
As-forged	12.7	54.9	−0.758

Figure AB - Table 6-2, Curve Fit Parameters for Surface Factor

The Marin surface factor accounts for the difference between the test specimen surface finish and the actual component surface. It is given by (Shigley's Table 6-2):

$$k_a = a \cdot S_{ut}^{b}$$

where a and b are empirical constants for the assumed surface finish, and S_{ut} is the ultimate tensile strength of the material.

- Surface finish assumed: Machined (based on Marin curve-fit parameters)
- $a = 3.04\ MPa$
- $b = -0.217$
- $S_{ut} = 91.93\ MPa$ (determined from material testing)

$$k_a = (3.04)\,(91.93)^{-0.217} = 1.13$$

Since k_a cannot physically exceed 1.0, and to remain conservative, the following value is adopted:

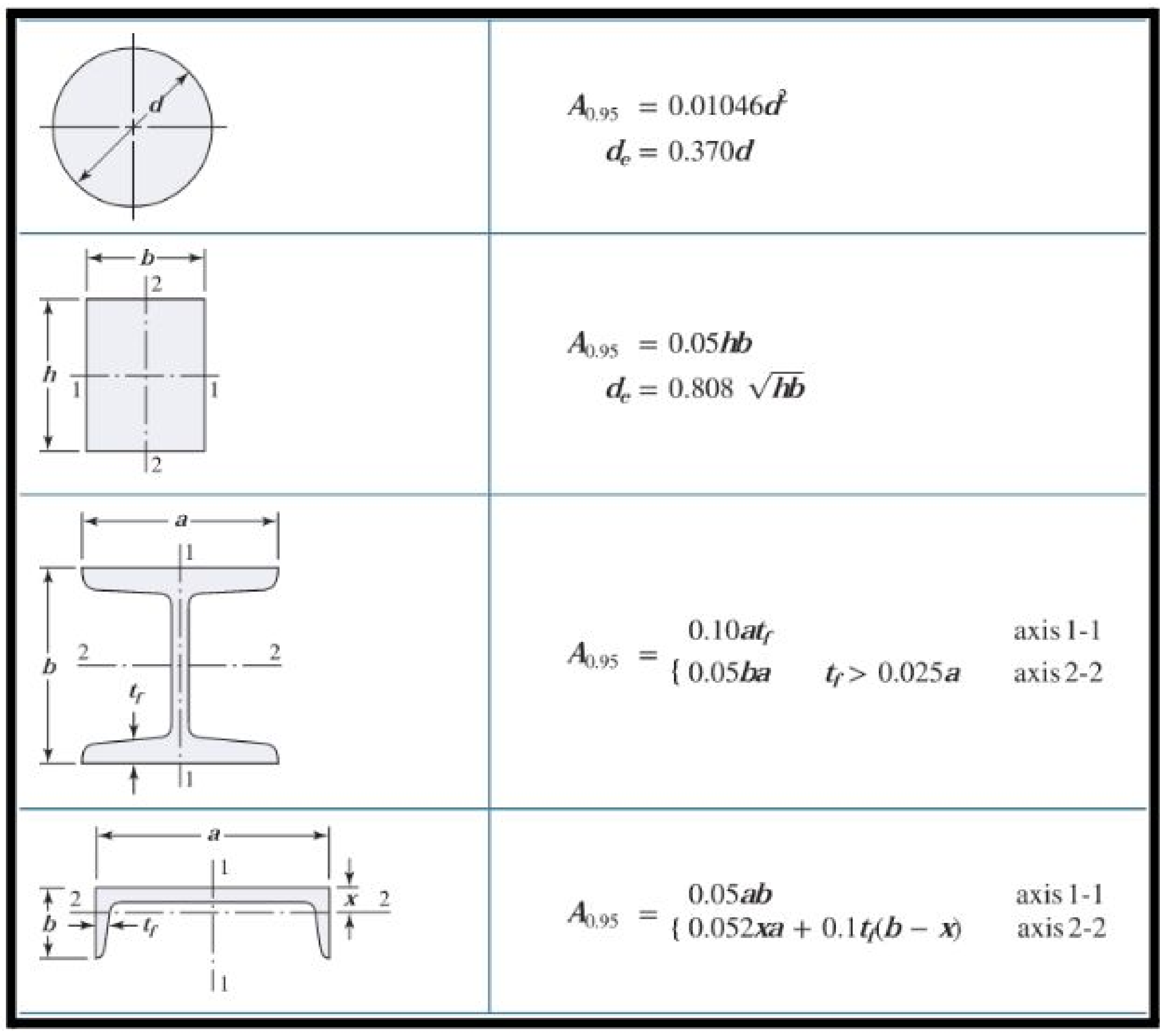
$$k_a = 0.9$$

Size Factor (k_b)

$$k_b = \begin{cases} (d/0.3)^{-0.107} = 0.879d^{-0.107} & 0.3 \le d \le 2 \text{ in} \\ 0.91d^{-0.157} & 2 < d \le 10 \text{ in} \\ (d/7.62)^{-0.107} = 1.24d^{-0.107} & 7.62 \le d \le 51 \text{ mm} \\ 1.51d^{-0.157} & 51 < d \le 254 \text{ mm} \end{cases}$$

Figure AC - k_b Surface Factor

For non-circular cross-sections loading in bending, an equivalent diameter d_e is used (Shigley's Table 6-3):



	$A_{0.95} = 0.01046d^2$ $d_e = 0.370d$
	$A_{0.95} = 0.05hb$ $d_e = 0.808\sqrt{hb}$
	$A_{0.95} = \begin{cases} 0.10at_f & & \text{axis 1-1} \\ 0.05ba & t_f > 0.025a & \text{axis 2-2} \end{cases}$
	$A_{0.95} = \begin{cases} 0.05ab & \text{axis 1-1} \\ 0.052xa + 0.1t_f(b-x) & \text{axis 2-2} \end{cases}$

Figure AD - Table 6-3, Areas of Common Nonrotating Structural Shapes Loaded in Bending

$$d_e = 0.808\sqrt{hb}$$

where:

- $b = 609.6\ mm$
- $h = 177.8\ mm$

$$d_e = 0.808\sqrt{(609.6)(177.8)} = 266\ mm$$

Since $d_e = 266\ mm$ falls outside the recommended range of the size factor equation (upper bound of 254 mm), k_b is evaluated at the upper bound of $d = 254\ mm$.

$$k_b = 1.51(d)^{-0.157}$$

$$k_b = 1.51(254)^{-0.157} = 0.633$$

Loading Factor (k_c)

$$k_c = \begin{cases} 1 & \text{bending} \\ 0.85 & \text{axial} \\ 0.59 & \text{torsion} \end{cases}$$

Figure AE - Loading Factor k_c

$$k_c = 1.0$$

Temperature Factor (k_d)

There is limited quantitative data on the fatigue behavior of fiberglass/epoxy composites under moderate temperature variation within the typical operational range of a watercraft. Therefore, no temperature correction is applied:

$$k_d = 1.0$$

Reliability Factor (k_e)

A 90% reliability level is selected. A 50% reliability target would carry a 50% probability of failure at the calculated stress level, which is unacceptable even for an initial prototype. 99% would be required if failure consequences were catastrophic; 90% is consistent with preliminary mechanical design practice for a non-safety-critical component (Shigley's Table 6-4):

Reliability, %	Transformation Variate z_a	Reliability Factor k_e
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702

Figure AF - Table 6-4, Reliability Factors k_e
Corresponding to 8 Percent Standard Deviation of the Endurance Limit

$$k_e = 0.897$$

Modified Endurance Limit (S_e)

$$S_e = \begin{cases} 0.5\,S_{ut} & S_{ut} \le 200 \text{ kpsi} (1400 \text{ MPa}) \\ 100 \text{ kpsi} & S_{ut} > 200 \text{ kpsi} \\ 700 \text{ MPa} & S_{ut} > 1400 \text{ MPa} \end{cases}$$

Figure AG - Estimating Endurance Limit

Because fiberglass/epoxy does not have abundant fatigue research, the corrected endurance limit S'_e is taken conservatively as:

$$S'_e = (0.3)\,S_{ut} = (0.3)\,(91.93\,MPa) = 27.579\,MPa$$

This is more conservative than the standard steel approximation of $(0.5)\,S_{ut}$, reflecting greater uncertainty in composite fatigue data.

Applying all Marin factors:

$$S_e = k_a\,k_b\,k_c\,k_d\,k_e\,S'_e$$

$$S_e = (0.9)\,(0.633)\,(1.0)\,(1.0)\,(0.897)\,(27.579)$$

$$S_e = 14.09\, MPa$$

Stress Concentration Factors (K_f) and (K_{fs})

Stress concentration factors are included to represent the propulsion unit exit cutout in the shell. Values for an mill-keyseat are used as a conservative geometric representation (Shigley's Table 7-1):

	Bending	Torsional	Axial
Shoulder fillet—sharp ($r/d = 0.02$)	2.7	2.2	3.0
Shoulder fillet—well rounded ($r/d = 0.1$)	1.7	1.5	1.9
End-mill keyseat ($r/d = 0.02$)	2.14	3.0	—
Sled runner keyseat	1.7	—	—
Retaining ring groove	5.0	3.0	5.0

Figure AH - Table 7-1, First Iteration Estimates for Stress Concentration Factors K_f and K_{fs}

<u>Loading Type</u>	<u>K_t (Theoretical)</u>
Bending	2.14
Torsional	3.0

Table 4 - First Iteration K_f and K_{fs} Utilized

The fatigue stress concentration factors K_f and K_{fs} are computed from the notch sensitivity q:

$$K_f = 1 + q\left(K_t - 1\right)$$

$$K_{fs} = 1 + q_{shear}\left(K_{ts} - 1\right)$$

Due to insufficient research on fiberglass and fiberglass/epoxy composites, notch sensitivity is set to $q = q_{shear} = 1$ (fully notch-sensitive) to treat the analysis conservatively:

$$K_f = 1 + (1)(2.14 - 1) = 2.14$$

$$K_{fs} = 1 + (1)(3.0 - 1) = 3.0$$

8.1.3.6 Factor of Safety: DE-Gerber

The Distortion Energy approach uses two amplitude groupings, with $T_a = 0$ and $M_m = 0$:

$$A = \sqrt{4\left(K_f M_a\right)^2 + 3\left(K_{fs} T_a\right)^2}$$

$$A = \sqrt{4\left((2.14)(477)\right)^2 + 0} = 2041.56\, N \cdot m$$

$$B = \sqrt{4\left(K_f M_m\right)^2 + 3\left(K_{fs} T_m\right)^2}$$

$$A = \sqrt{0 + 3\left((3.0)(26.53)\right)^2} = 137.854\, N \cdot m$$

$$\frac{1}{n} = \frac{8A}{\pi d^3 S_e} \left\{ 1 + \left[1 + \left(\frac{2BS_e}{AS_{ut}} \right)^2 \right]^{1/2} \right\}$$

$$n_{Gerber} = 22.21$$

8.1.3.7 Factor of Safety: DE-Goodman

The Distortion Energy approach uses two amplitude groupings, with $T_a = 0$ and $M_m = 0$:

$$\frac{1}{n} = \frac{16}{\pi d^3} \left\{ \frac{1}{S_e} \left[4\left(K_f M_a\right)^2 + 3\left(K_{fs} T_a\right)^2 \right]^{1/2} + \frac{1}{S_{ut}} \left[4\left(K_f M_m\right)^2 + 3\left(K_{fs} T_m\right)^2 \right]^{1/2} \right\}$$

$$n_{Goodman} = 21.82$$

8.1.3.8 Factor of Safety Conclusion

Both criteria yield factors of safety well above 1.0, confirming that the fiberglass shell structure has substantial margin against fatigue failure under the defined loading conditions. The conservative material assumptions (reduced S'_e, full notch sensitivity, machined surface penalty) indicate the true margin may be even larger once experimental fatigue data for this specific composite system becomes available.

8.1.4 Propulsion and Vehicle Performance Modeling

8.1.4.1 Assumptions

- Input power to shaft: P = 25 kW
- Overall hydraulic / jet efficiency: $\eta = 0.6-0.7$ (typical for small / hobby-level waterjet)

Hydraulic power to the jet:

- Case A: $P_{hA} = \eta P = (0.6)(25\ \text{kW}) = 15\ \text{kW}$
- Case B: $P_{hB} = \eta P = (0.7)(25\ \text{kW}) = 17.5\ \text{kW}$

8.1.4.2 Hydraulic Power Definition

Hydraulic power in the jet:

$$P_h = \frac{1}{2} \dot{m} u_j^2$$

where:

- $\dot{m}$ = mass flow rate
- u_j = jet exit velocity

8.1.4.3 Thrust From Jet

Thrust is calculated:

$$T = \dot{m} u_j$$

Rewriting the previous equations to eliminate ṁ:

$$P_h = \frac{1}{2}\dot{m}\,u_j^2 \Rightarrow \dot{m} = \frac{2\,P_h}{u_j^2}$$

$$T = \dot{m}\,u_j = \left(\frac{2\,P_h}{u_j^2}\right) u_j = \frac{2\,P_h}{u_j}$$

8.1.4.4 Mass Flow Rate and Jet Exit Area

Mass flow rate in terms of jet exit area:

$$\dot{m} = \rho\,A_j\,u_j$$

Jet exit area:

$$A_j = \pi\left(\frac{D_j}{2}\right)^2$$

where:

- ρ = water density (fresh water) = 1000 $\frac{kg}{m^3}$
- D_j = exit nozzle diameter = 70 mm = 0.07 m

then:

$$A_j = \pi\left(\frac{D_j}{2}\right)^2 = \pi\left(\frac{0.07}{2}\right)^2 = 3.85x10^{-3}\,m^2$$

8.1.4.5 Jet Velocity

Start from:

$$P_h = \frac{1}{2}\dot{m}\,u_j^2$$

Substitute $\dot{m} = \rho\,A_j\,u_j$:

$$P_h = \frac{1}{2}\left(\rho\,A_j\,u_j\right)u_j^2 = \frac{1}{2}\rho\,A_j\,u_j^3$$

Solve for u_j:

$$u_j^3 = \left(\frac{2\,P_h}{\rho\,A_j}\right) \Rightarrow u_j = \left(\frac{2\,P_h}{\rho\,A_j}\right)^{\frac{1}{3}}$$

Case A: P_{hA} = 15000 W

$$u_{j,A} = \left(\frac{2\,(15000)}{(1000)\,(3.85x10^{-3})}\right)^{\frac{1}{3}} \approx 19.8\,m/s \approx 44.3\,mph$$

Case B: P_{hB} = 17500 W

$$u_{j,A} = \left(\frac{2\,(17500)}{(1000)\,(3.85x10^{-3})}\right)^{\frac{1}{3}} \approx 20.9\,m/s \approx 46.8\,mph$$

8.1.4.6 Thrust

$$T = \frac{2P_h}{u_j}$$

Case A:

$$T_A = \frac{2\,(15000)}{u_{j,A}} \approx 1515\,N$$

Case B:

$$T_B = \frac{2\,(17500)}{u_{j,B}} \approx 1675\,N$$

8.1.4.7 Simple Drag Modeling for a Surfboard

Drag can be calculated as a function of speed:

$$D(V) = kV^2 = \frac{1}{2}\rho\, C_D\, A_f\, V^2$$

where:

- k = drag constant
- ρ = water density (fresh water)
- C_D = effective drag coefficient
- A_f = effective wetted frontal area
- V = board speed

Assumption: Use the widest cross-sectional area for the effective frontal area.

A_x = largest cross-sectional area of board from CAD model

$$A_x = 0.09455\,m^2$$

Effective wetted frontal area can be approximated as

$$A_f = \frac{V_{sub}}{V_{total}}\,(A_x)$$

where:

- V_{sub} = submerged volume at operating conditions
- V_{total} = total board volume

$$V_{total} = 0.1708\,m^3 + 0.02022\,m^3 = 0.19102\,m^3$$

Mass with an estimated rider weight of 190 lb

$$m_{total} = m_{board} + m_{box} + m_{rider}$$

- m_{total} = total riding mass
- m_{board} = board mass
- m_{rider} = rider mass (approximated at 170 lb)

$$m_{total} = 9\,kg + 34\,kg + 77\,kg = 120\,kg$$

Submerged volume approximated utilizing Archimedes Principles:

$$V_{sub} = \frac{m_{total}}{\rho} = \frac{120\ kg}{1000\ \frac{kg}{m^3}} = 0.12\ m^3$$

Archimedes Principle: a floating object displaces a weight of fluid equal to its own total weight.

Effective wetted frontal area:

$$A_f = \frac{V_{sub}}{V_{total}} (A_x) = \left(\frac{0.12\ m^3}{0.19102\ m^3}\right)\left(0.09455\ m^2\right) \approx 0.0594\ m^2$$

Effective drag coefficient was sourced based on the typical for a longboard at high speed:

$$C_D = 0.3$$

Drag coefficient k:

$$k = \frac{1}{2}\rho C_D A_f = \frac{1}{2}\left(1000\ \frac{kg}{m^3}\right)(0.3)\left(0.0594\ m^2\right) = 8.91\ \frac{kg}{m}$$

8.1.4.8 Speed from Thrust-Drag Balance

At steady max velocity:

$$\mathrm{T} = D(V) = kV^2$$

Solve for V:

$$V = \sqrt{\frac{T}{k}}$$

Case A:

$$V_A = \sqrt{\frac{T_A}{k}} = \sqrt{\frac{1515}{8.91}} \approx 13.04\ m/s \approx 29.2\ mph$$

Case B:

$$V_B = \sqrt{\frac{T_B}{k}} = \sqrt{\frac{1675}{8.91}} \approx 13.71\ m/s \approx 30.7\ mph$$

8.1.4.9 Operating "Top Speed" with Safety Factor

Assumption: operate at $V_{X,T} = 0.9\,V_{max}$ to avoid pushing the motor or battery to 100% of their respective limits.

Case A:

$$V_{A,T} = (0.9)\,(13.04\ m/s) = 11.736\ m/s \approx 26.3\ mph$$

Case B:

$$V_{B,T} = (0.9)\,(13.71\ m/s) = 12.339\ m/s \approx 27.6\ mph$$

8.1.4.10 Acceleration at Operating "Top Speed"

Starting from Newton's Second Law in the direction of motion:

$$m_{total}\,a_x = T_X - D_x = T_X - V_{X,T}^2$$

Solve for acceleration:

$$a_x = \frac{T_X - k\,V_{X,T}^2}{m_{total}}$$

where:

- $m_{total} = 120$ kg
- $k = 8.91\ \frac{kg}{m}$

Case A:

$$a_A = \frac{T_A - k\,V_{A,T}^2}{m_{total}} = \frac{1515 - 8.91\,(11.736)^2}{120} \approx 2.40\ m/s^2$$

Case B:

$$a_B = \frac{T_B - k\,V_{B,T}^2}{m_{total}} = \frac{1675 - 8.91\,(12.339)^2}{120} \approx 2.65\ m/s^2$$

8.1.4.11 Operating "Top Acceleration" with Safety Factor

Assumption: operate at $a_{X,T} = 0.9\ a_x$ to avoid pushing the motor or battery to 100% of their respective limits.

Case A:

$$a_{A,T} = (0.9)\left(2.40\ m/s^2\right) = 2.16\ m/s^2$$

Case B:

$$a_{B,T} = (0.9)\left(2.65\ m/s^2\right) = 2.39\ m/s^2$$

8.1.4.12 Time to "Top Speed" using "Top Acceleration"

Assuming a constant acceleration from rest $V_i = 0$ to the target operating speed of $V_{X,T}$:

$$t_{X,T} = \frac{V_{X,T} - V_i}{a_{X,T}} = \frac{V_{X,T}}{a_{X,T}}$$

Case A:

$$t_{A,T} = \frac{V_{A,T}}{a_{A,T}} = \frac{11.736\ m/s}{2.16\ m/s^2} \approx 5.43\ s$$

Case B:

$$t_{B,T} = \frac{V_{B,T}}{a_{B,T}} = \frac{12.339\ m/s}{2.39\ m/s^2} \approx 5.16\ s$$

See Appendix C - Propulsion and Vehicle Performance Modeling, for Detailed Hand Calculation Work

8.1.5 Battery Capacity and Runtime

To determine the battery capacity and runtime, we first calculated the battery capacity by multiplying the batteries volts by the amp hour rating. Then multiplying that by 0.9 to extend the battery life of our battery. We selected a lithium battery that runs at 72 volts and 300 amp- hours.

$$E = V * C * 0.9$$

Where V represents the voltage in (volts, V) and C represents the capacitance in (amp-hours, Ah. When multiplied it will give us E which represents energy in (watt-Hours, Wh).

$$E = 72V * 300Ah * 0.9 = 19440Wh$$

Next to calculate the runtime we used the energy value that we got previously and divided it by the motor maximum power so that we can determine the runtime in hours. The motor we selected was a 25kW motor. We will utilize 0.9 as a factor of limiting the peak motor output.

$$t = \frac{E}{P}$$

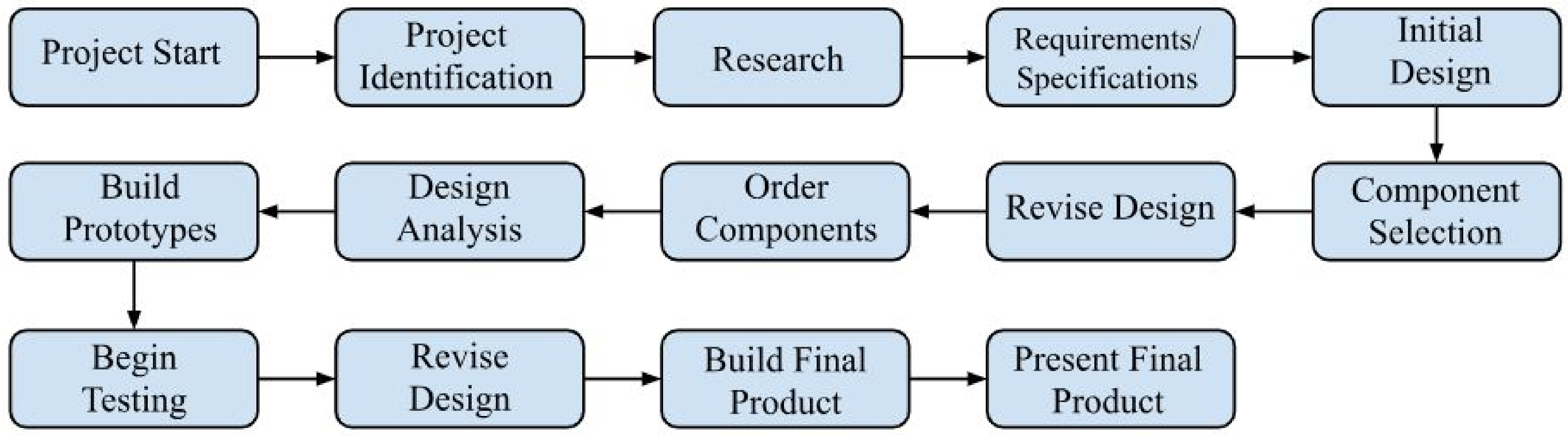
Where t represents the runtime of our battery (hours, h), E represents the energy in (watt-hours, Wh), and P represents the power in (watts, W).

$$t = \frac{19440Wh}{(25000W)\,(0.9)} = 0.864h$$

$$0.864h * \frac{60min}{1h} = 51.84min$$

So we should be able to have a runtime of about 51 minutes with the motor and battery combination that we have selected.

8.1.6 Flow Chart



8.1.7 Schematics

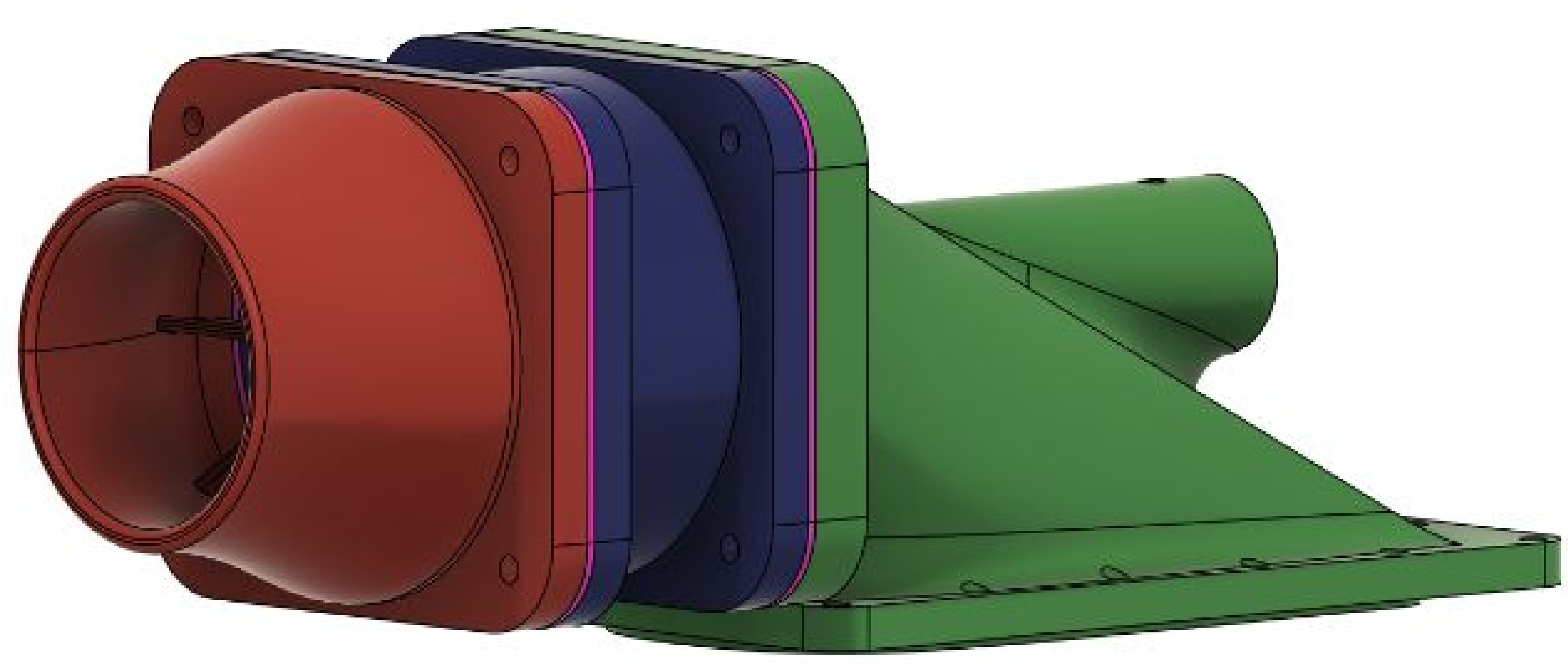


Figure A1 - Jet Pump Assembly

Figure 1 shows the assembled CAD model of the 100 mm jet pump used in the surfboard's propulsion system. Water enters through the large green intake housing, which directs the flow smoothly into the pump. The dark blue center section represents the main pump body where the impeller is housed and where the water is pressurized. After leaving the impeller, the flow enters the red nozzle section at the front. This nozzle contains the fixed stator vanes, which straighten the swirling flow coming off the impeller and help convert rotational energy into forward thrust. The nozzle then accelerates the water as it exits the pump. The mounting surfaces and flange geometry are designed to integrate tightly into the board's propulsion cavity while supporting the high loads produced by the 25 kW motor.

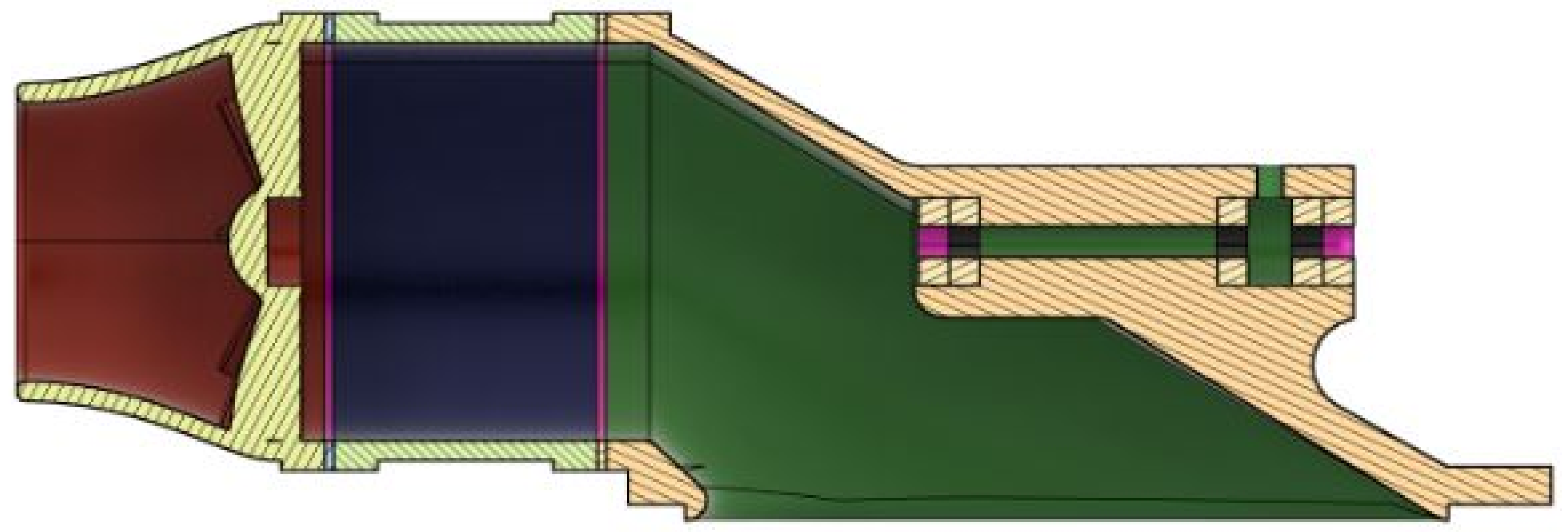


Figure AJ - Jet Pump Cross Section

Figure 2 shows a cross-section of the jet pump, highlighting the internal flow path and structural features. Water enters from the right through the green intake housing, where the gradually tapering shape guides the flow toward the impeller chamber. The impeller (housed inside the dark blue section) increases the water's energy as it passes through. Immediately downstream, within the red nozzle, the fixed *stator vanes* straighten the flow by removing rotational swirl left by the impeller. This straightened flow is then accelerated through the narrowing nozzle geometry to produce high-velocity thrust. The beige hatched areas show the structural walls, fastener locations, and reinforcement around the pump body. This cross-section clearly illustrates the correct sequence of intake, pressurization, flow straightening, and final jet acceleration.

Figure AK - Full board Prototype

Figure 3 shows the current CAD model of the full surfboard prototype, including the internal propulsion bay and the rear-mounted jet pump. The outer geometry of the board is shaped to provide high stability and smooth planing at speeds near 30 mph, with a tapered nose and wide mid-section to support rider balance. The large rectangular cavity in the center of the board houses the propulsion cartridge, which contains the battery pack, motor, ESC, and cooling system. This removable module design allows for easy maintenance and future upgrades without reconstructing the board structure. At the rear of the board, the jet pump assembly is fully integrated into the tail section, ensuring precise alignment between the pump outlet and the board's hydrodynamic flow path. The prototype captures the final proportions and layout that will be used for CNC machining the foam core and performing composite layups during fabrication.

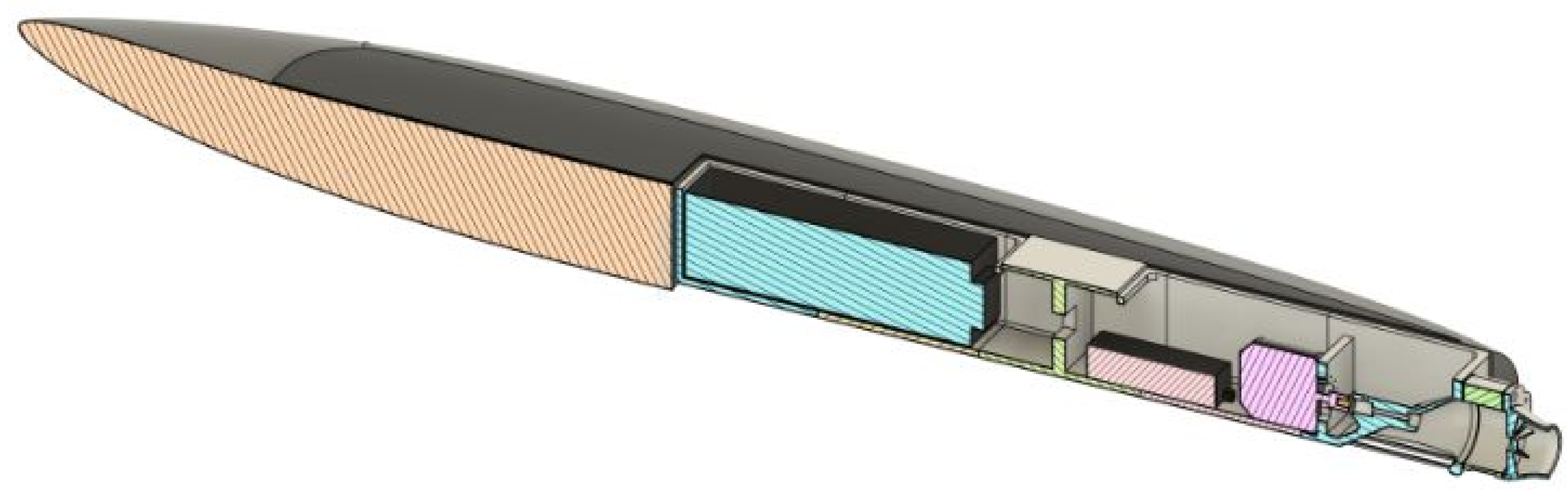


Figure AL - Full board cross-sectional view

Figure 4 shows a longitudinal cross-section of the full surfboard prototype, revealing the internal structure, buoyant core, and propulsion layout. The front portion of the board is composed entirely of solid XPS foam, providing buoyancy and stiffness while keeping weight low. Moving aft, the large rectangular cavity houses the removable propulsion cartridge, which contains the battery pack (blue), ESC compartment (pink), structural supports (green), and motor-to-pump alignment features. Each of these internal sections is shaped to maximize usable volume while preserving sufficient foam thickness around the perimeter for strength. At the tail of the board, the intake duct transitions smoothly into the integrated jet pump housing, ensuring clean flow from the intake channel to the impeller and out through the nozzle. This cutaway view illustrates how the electrical components, cooling pathways, and pump assembly fit within the hull, demonstrating the board's internal organization and the engineering decisions that balance buoyancy, structural integrity, and propulsion efficiency.

8.1.8 Simulations

Figure AM - Graph of Board speed vs Throttle input

Figure AM shows predicted board speed as a function of throttle setting based on a simplified drag–power model for a planing hull. The graph shows that speed increases nonlinearly with throttle, since the power required to overcome hydrodynamic drag scales approximately with the cube of velocity. As throttle increases, available motor power rises, allowing the board to reach higher steady-state speeds. The model predicts that cruising speeds in the mid-20s mph range occur at roughly 60–70% throttle, while maximum speeds approach the upper limit of the motor and jet pump capability. This plot provides a first-order estimate of performance prior to on-water testing.

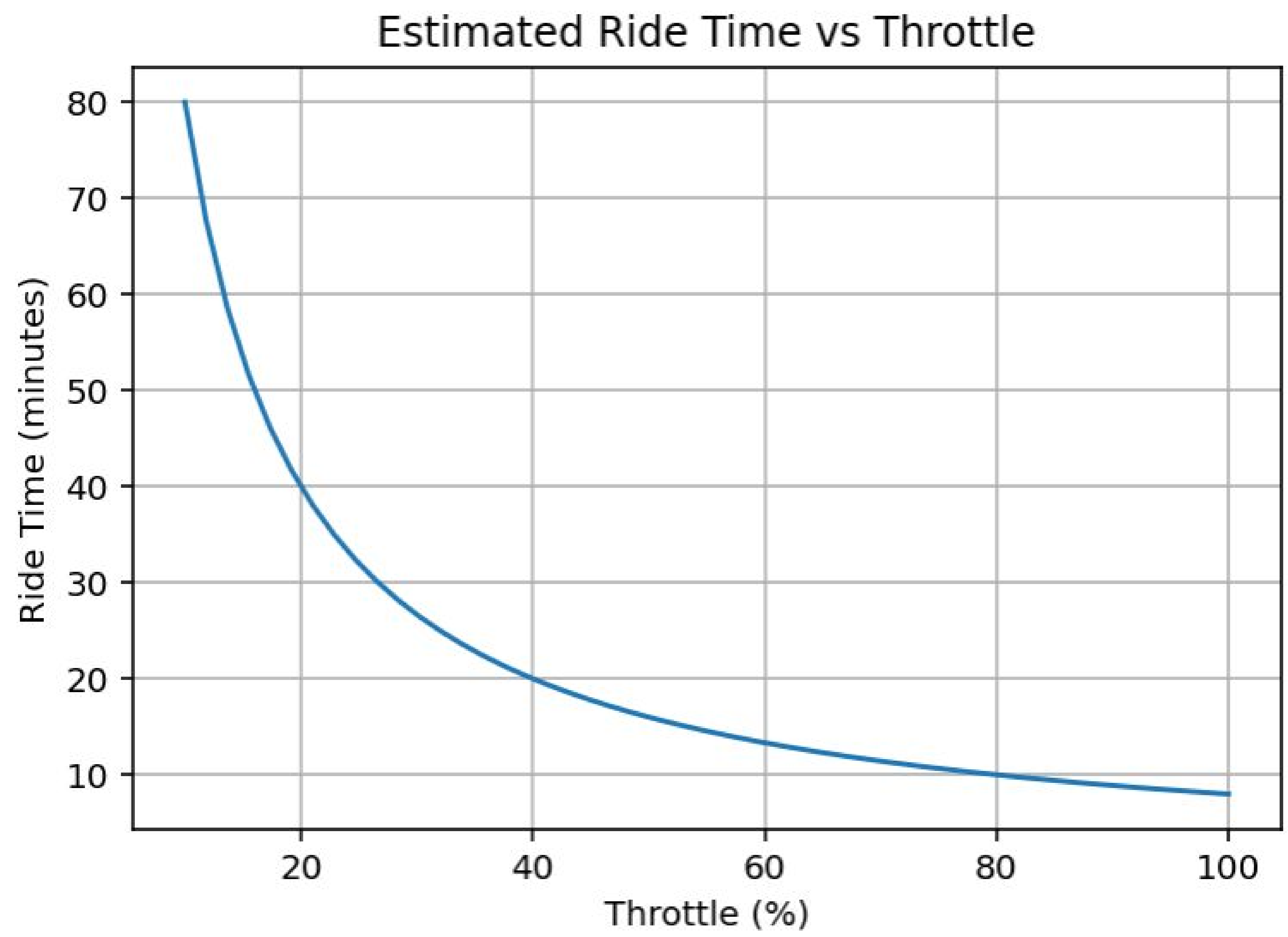


Figure AN - Graph of Ride time vs throttle input

Figure AN shows the estimated ride time of the surfboard as a function of throttle setting using total battery energy and electrical power draw. Because electrical power usage increases rapidly with higher throttle, ride time decreases sharply as throttle approaches maximum output. At moderate throttle settings around 50–60%, the model predicts ride times between 22 and 28 minutes, which aligns with expected performance for similar electric jet-propelled boards. This analysis helps establish realistic expectations for operating duration under different riding styles and power levels.

8.2 Comparison to Top Electric Surfboard

Metric	Unit	LIND Electric Surfboard	Designed Board	(incl. drag and 90% utility factor in water)
Board weight	lb	≈ 74	≈ 95	
Price	$	25000	≈ 4000	
Power	kW	20	25	
Power	hp	27	33	
Top speed	m/s	16.7	19.8–20.7	11.736–12.339 (2)
Top speed	mph	37	44.3–46.8	26.3–27.6 (2)
Max thrust	N	≈ 1200	1515–1675	
Acceleration	m/s^2	3.09 (1)	2.40–2.65	2.16–2.39 (2)
Zero-to-top-speed time	s	≈ 5.4 (1)		5.43–5.16 (2)

Table 5 - Comparison to Top Electric Surfboard on the Market

Footnotes:

(1) Assuming the LIND board produces half the drag constant of the designed board.
(2) Assuming zero planing, widest cross-section of the board for drag, and ~68% of the cross-sectional area submerged at full speed.

8.3 Alternate Designs Considered

8.3.1 Motor Mount

Throughout the project there were many design alternatives. With the motor mount initially the idea was to use a 3D printed plastic housing and create a thicker layer of 3D printed plastic inside the propulsion box as the mounting wall. However after inspecting the output torque of the motor there will now be an additional eighth inch aluminum sheet in front of the 3D printed mounting wall.

8.3.2 Jet Pump

The size and weight of the propulsion components lead to major swings in efficiency. We considered both 80 mm and 100 mm jet pumps. The 80 mm unit was lighter and less expensive, but thrust calculations showed it would struggle to push the board beyond 22 mph with the available motor power. The 100 mm pump provided better efficiency at the target speed range and could move more water, making it the better choice despite the added cost and weight.

8.3.3 Modular Propulsion Box

The propulsion components need to be accessible due to the battery needing to be charged after use and the ability to do general maintenance is necessary. For the connection between the electric propulsion box and the board there were three concepts. The first was to build an open spot in the top of the board similar to a hatch/compartment that would be closed with waterproof seals along the edge. However the biggest issue is ensuring that no water could enter the compartment because if it did it would ruin the battery. The second concept was to build the propulsion box separately and have a complete cutout in the board. Then once the board was assembled and the box was assembled there would be a channel to slide the box into. This idea was great for ease and access to the propulsion components but the biggest issue is the structural integrity of the connection between the propulsion box and the board. From the research it showed that this would weaken over time and fall apart due to the constant hitting on the waves and the thrust of the motor. The final solution is to create a sleeve the size of the propulsion box inside the board. This would allow the propulsion box to be slid inside the board and still have the structural integrity by having the whole board wrapped in fiberglass. As for the accessibility of the propulsion box it will be latched in the back so it is locked into the board. By having the sleeve built into the board it will distribute the force of the propulsion unit so it is not point loaded.

8.4 Future Iteration Considerations

8.4.1 Propulsion

Future work could produce a better product by increasing overall propulsive efficiency by optimizing the jet pump and hull hydrodynamics. One potential improvement is evaluating larger or higher-efficiency jet pumps that can generate the required thrust at lower shaft power, thus reducing current draw and easing thermal constraints on the motor and ESC.

The hull form could be refined using CFD-based analysis rather than relying on longboard drag concepts, with the goal of lowering the effective drag coefficient and wetted area at planing speeds.

Attention to intake design, including better alignment with oncoming flow, and a geometry that resists cavitation, would reduce inlet losses and improve flow uniformity into the impeller.

These changes could either raise the achievable top speed beyond the current 28–32 mph prediction or allow the same performance with improved runtime and reliability.

8.4.2 Structure and Materials

XPS foam was cut with a CNC machine, laminated with fiberglass-epoxy, to achieve a flexural modulus of roughly 11.8 GPa, which meets initial stiffness targets but leaves room for optimization in future iterations. Subsequent designs could adopt a more advanced sandwich structure with higher-stiffness skins, such as carbon fiber or hybrid glass/carbon layups, particularly in regions subjected to higher bending moments and local impact loads around the propulsion bay.

Core materials could also be reconsidered, exploring higher-density foams or locally reinforced inserts to improve fastener holding strength and resistance to fatigue at mounting points. The existing material characterization database from four-point bending and tensile testing provides a solid baseline for this optimization, enabling more accurate FEA-driven thickness and layup design.

Developing a method to cast or mold the propulsion box as a single continuous piece, eliminating the segmented assembly approach required by the build plate size limitations of available 3D printing equipment. The current multi-segment construction requires joining epoxy between sections, which tensile testing revealed to be a notable weak point failing at only 6–8% of the control material's ultimate tensile strength and exhibiting brittle fracture behavior. A monolithic propulsion box would remove these bond-line failure risks entirely and create a rigid, continuous structural backbone integrated into the board. This would not only improve the load-carrying capacity of the propulsion module but also enhance waterproofing reliability by reducing the number of joined interfaces that must be sealed.

Structural changes would reduce board deflection under rider load, improve durability in rough conditions, and potentially lower overall mass while preserving safety factors.

8.4.3 Modularity, Layout, and Waterproofing

The modular propulsion box is a key strength of the current design, but its connection with the hull and its sealing strategy could be refined substantially in future work. A more structured modular architecture might use integrated guide rails or indexed hardpoints so that the module can be inserted and locked with repeatable alignment and consistent compression of seals.

For waterproofing, replacing custom-printed features with proven marine-grade IP68 cable glands, double O-ring flanges, and standardized hatch interfaces would help move from "designed to IP-67" toward practically verifiable IP-68 performance in real operating conditions.

The geometry of the hull cutout and cartridge could be revised to minimize water pooling paths and improve drainage in the event of minor ingress.

These changes would make the system more robust to repeated assembly/disassembly cycles, simplify maintenance, and increase confidence in the protection of high-value electronics.

8.4.4 Material Testing

The material testing completed during this project included four-point bending and tensile testing. It established a functional baseline for the fiberglass-epoxy used in the shell, but the scope of testing was constrained by available time and budget. Future work would benefit significantly from a more comprehensive empirical dataset, particularly for fatigue behavior. Currently, the endurance limit used in the factor of safety analysis was estimated conservatively at 30% of the measured ultimate tensile strength $\left(S'_e = (0.3)\, S_{ut}\right)$ due to the lack of published S-N curve data specific to this composite. A fatigue testing dataset would allow that conservative assumption to be replaced with experimentally validated values, likely revealing a higher usable endurance limit and enabling more efficient material utilization in future designs.

Creating a matrix to include multiple layups, ply counts, fiber orientations, and fiber volume would provide a structured design optimization. Testing specimens fabricated under varying process conditions would quantify the sensitivity of mechanical properties and manufacturing variability.

Impact and shear strength testing would address two failure modes that current testing did not directly characterize but are relevant to a board that experiences hydrodynamic impacts, foot loading, and point loads around mounting hardware. Taken together, a more complete test program would reduce the conservative methods currently embedded in the structural analysis and provide the empirical foundation needed to confidently specify thinner, lighter, or higher-performance layups in subsequent design iterations.

8.5 Functional Block Diagram

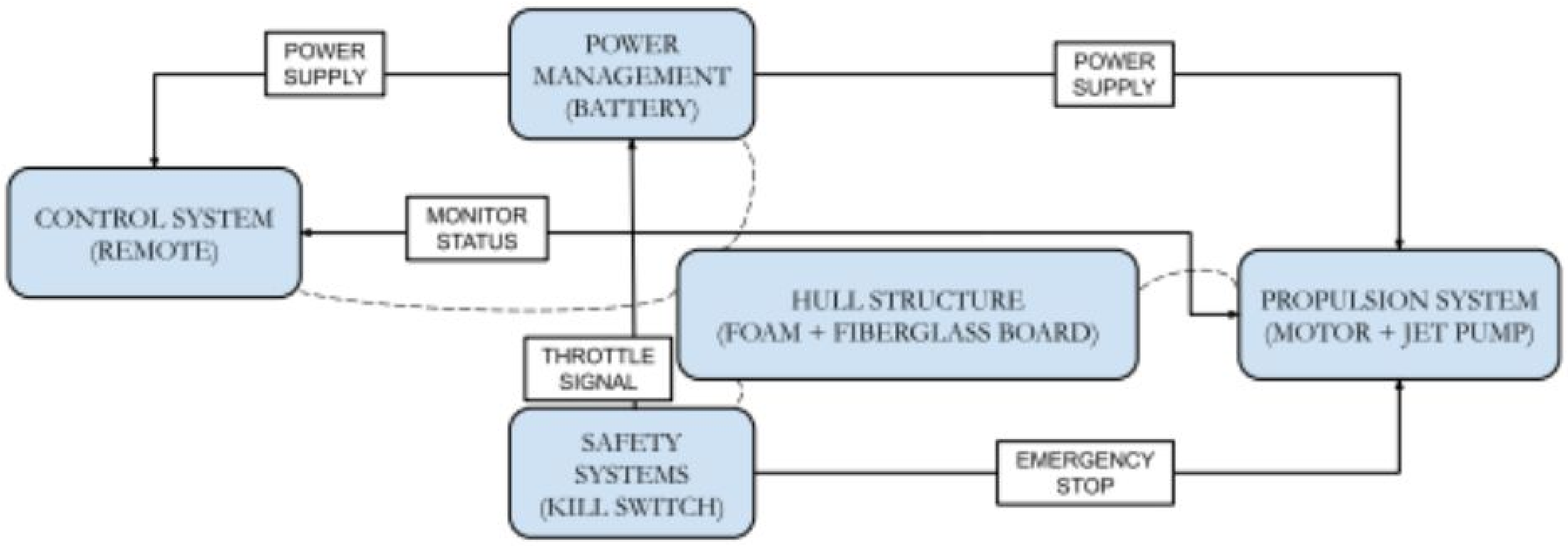


Figure AO - Jet surfboard block diagram

The electric jet-driven surfboard system is composed of five primary functional blocks that operate in an integrated manner to deliver controlled propulsion and safe operation. The Power Supply block forms the energy source for the entire system, delivering nominal voltage of 74 V at approximately 45 Ah capacity. This power is distributed through two primary pathways: to the Power Management (Battery) block, which monitors cell health, voltage balance, and thermal conditions through an integrated Battery Management System (BMS), and directly to the Control System (Remote) block, which receives wireless throttle commands and transmits control signals to the propulsion system. The Motor Status feedback signal creates a communication loop, allowing the ESC to report operational parameters back to the control system. Simultaneously, the Throttle Signal originating from the handheld wireless remote modulates the power delivered to the Propulsion System (Motor + Jet Pump) block, which converts electrical energy into mechanical thrust through a 25 kW brushless motor coupled to a 100 mm jet drive impeller.

The Safety Systems (Kill Switch) block operates as a critical protective layer, continuously monitoring system status and commanding an Emergency Stop signal that can rapidly shut down motor operation in the event of a fault, thermal event, or deliberate rider intervention. This emergency shutdown pathway provides a direct, independent control loop that bypasses normal throttle control to ensure fail-safe operation. The Hull Structure (Foam + Fiberglass Board) block serves as the central mechanical platform and integration point for all subsystems, housing the propulsion module, battery pack, and control electronics while maintaining structural integrity and buoyancy. Dashed lines connecting the hull block to peripheral systems indicate mechanical and thermal integration, including coolant pathways through the motor and ESC to manage heat dissipation during high-power operation. The architecture ensures that power flow, control signals, monitoring feedback, and safety commands all function through defined interfaces, allowing each subsystem to operate independently while maintaining system-level coordination and responsiveness to rider input.

9. Part List and Budget

Parts	Manufacturers	Quantity	Estimated Price
25kW Surfboard Power Kit	MA-3D	1	$1,414.17
Water Jet Pump Propulsion Unit 80mm (3d model)	Cults	1	$17.93
Polymaker ASA Filament 1.75mm Black	POLYMAKER	6	$29.99
uxcell Knurled Insert Nuts, 50pcs M4 x 7mm L x 7mm OD	uxcell	1	$7.98
608CE Full Ceramic Bearing 1PC 8x22x7mm ABEC-9 608	GUSIRUI	10	$10.99
600pcs M3 Screw Assortment Kit	VGBUY	2	$8.99
380 Pcs M4 Screw Nuts and Bolts Assortment Kit	MYWISH	2	$8.59
244Pcs M5 Screw Metric Screw Assortment Kit	MYWISH	1	$14.99
Shape 3D subscription	N/A	1	$95.47
Polycarbonate Clear Plastic Sheet 12" X 24" X 0.236" (1/4")	N/A	2	$23.99
Student Symposium Poster Board	N/A	1	$45
6oz Fiberglass Cloth	N/A	3	$89.23
ruthex M3 Threaded Inserts	N/A	1	$9.49
Foamular F250 4x8x2" foam board	N/A	5	$59.97
West Marine Epoxy 0.98 gallon	Westek	2	$114.03
West Marine Epoxy Hardener 0.86 Quart	Westek	2	$66.20
West Marine Epoxy Pumps	Westek	1	$27.60

Table 6 - Part List

10. Gantt Chart

Task_ID	Phase	Task_Name	Start_Date	End_Date	Dependencies
1	PROJECT INITIATION	Project Planning & Team Formation	11/01/2025	11/21/2025	
1.1	PROJECT INITIATION	Market Research	11/08/2025	11/28/2025	
1.2	PROJECT INITIATION	Determine Design Specifications	11/15/2025	11/28/2025	1.1
1.3	PROJECT INITIATION	Finalize Three Options for Materials	11/22/2025	12/05/2025	1.2
1.4	PROJECT INITIATION	OUR Funding Submittal #1	11/22/2025	11/28/2025	1.2
1.5	PROJECT INITIATION	CNC Research & Requirements Analysis	11/15/2025	11/28/2025	1.1
1.6	PROJECT INITIATION	Acquire Shape3D Software License	11/22/2025	11/28/2025	1.5
2	DESIGN PHASE	Multiple Perspective Design	11/29/2025	12/12/2025	1.3
2.0.1	DESIGN PHASE	Approximate Jet Drive Size	12/06/2025	12/15/2025	2
2.1	DESIGN PHASE	Hull Shape & Hydrodynamic Design	12/06/2025	12/26/2025	2,1.6
2.2	DESIGN PHASE	Initial Propulsion System Print	12/13/2025	12/19/2025	2.0.1
2.2.1	DESIGN PHASE	Initial Impeller Components Print	12/20/2025	12/26/2025	2.2
2.3	DESIGN PHASE	Rail System Design	12/20/2025	01/02/2026	2.1
2.3.1	DESIGN PHASE	Locking Mechanism Design	12/27/2025	01/09/2026	2.3
2.4	DESIGN PHASE	Battery Housing Design	12/20/2025	01/02/2026	2.2
2.5	DESIGN PHASE	Control System Design	12/27/2025	01/09/2026	2.2
2.6	DESIGN PHASE	CFD Analysis & Optimization	01/03/2026	01/16/2026	2.1,2.2
2.7	DESIGN PHASE	Safety System Design	01/10/2026	01/23/2026	2.4,2.5
2.8	DESIGN PHASE	Design Bench Testing Method	01/10/2026	01/19/2026	2.6
2.9	DESIGN PHASE	Semi-Final Propulsion System Print	01/17/2026	01/23/2026	2.6,2.2.1
2.1	DESIGN PHASE	Semi-Final Battery Housing Print	01/17/2026	01/23/2026	2.4,2.7
2.11	DESIGN PHASE	CNC Machining Program Development	01/17/2026	01/30/2026	2.3,1.5
3	MANUFACTURING	Material Procurement	01/24/2026	02/06/2026	1.3
3.1	MANUFACTURING	Foam Blank Shaping	01/31/2026	02/13/2026	3,2.1
3.2	MANUFACTURING	Fiberglass Layup Process	02/07/2026	02/20/2026	3.1
3.2.1	MANUFACTURING	Fiberglass Composition Testing	02/14/2026	02/20/2026	3.2
3.3	MANUFACTURING	Rail System Fabrication	01/31/2026	02/13/2026	3,2.3

Figure AP - Screenshot from Gantt Chart

11. Procedure Punchlist

11.1. Design

11.1.1. Determine void in market that can be filled

11.1.2. Create conceptual designs of boards and propulsion system

11.1.3. Run preliminary calculations on feasibility of product

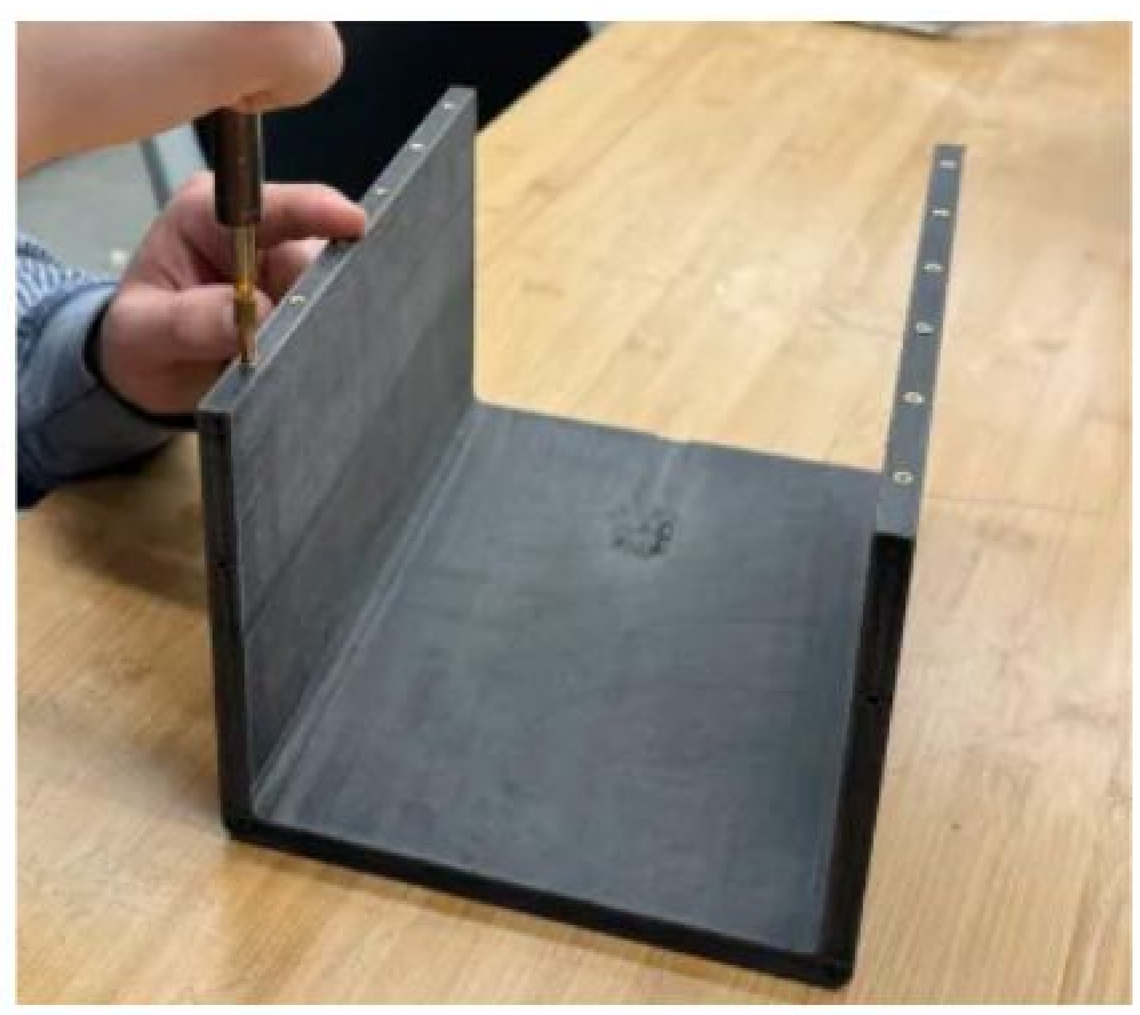
11.1.4. Iterate on conceptual design to create a compartmentalized construction set

11.2. Propulsion Box

11.2.1. Print propulsion box members

11.2.2. Sand propulsion box members


11.2.3. Install heat-sink threads into propulsion box members

11.2.4. Install support members and apply joining epoxy to join members together

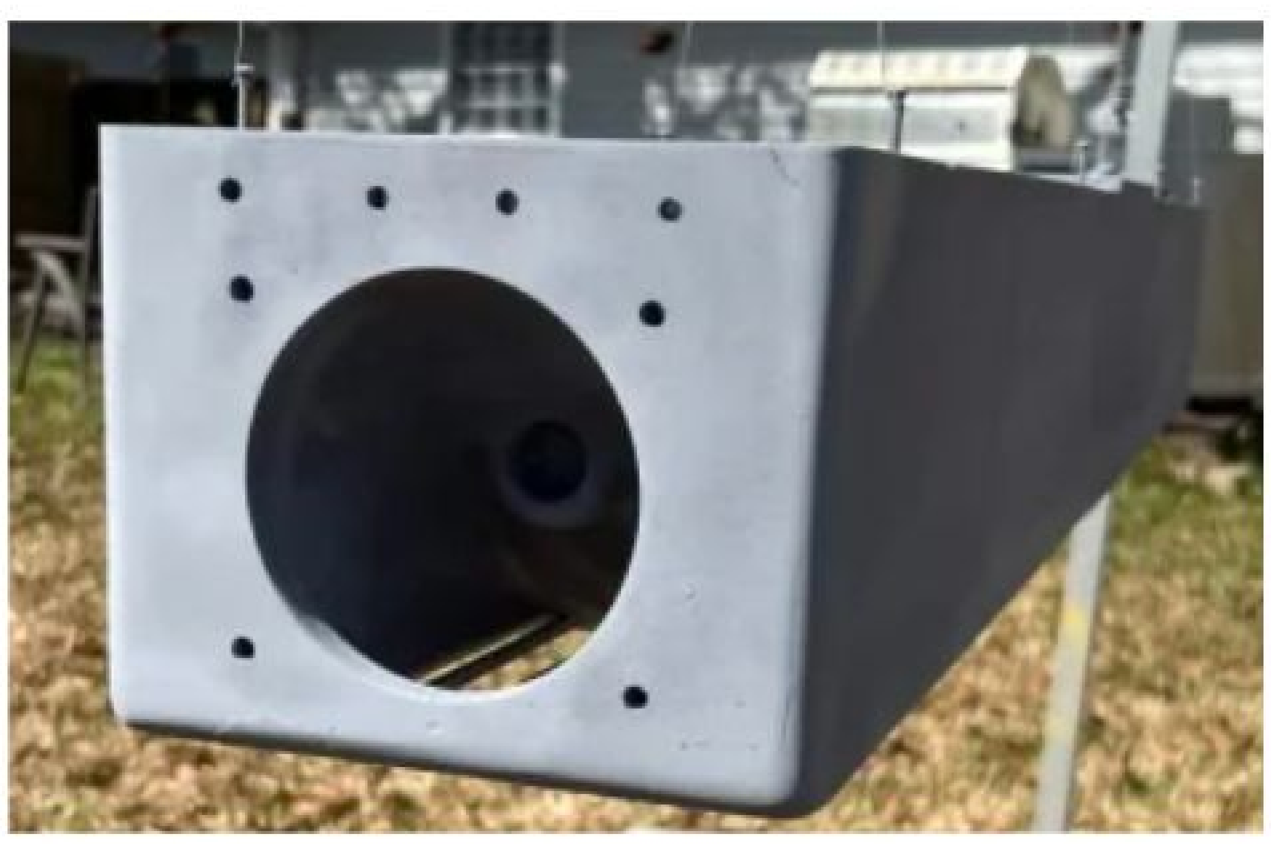
  

11.2.5. Sand excess joining epoxy

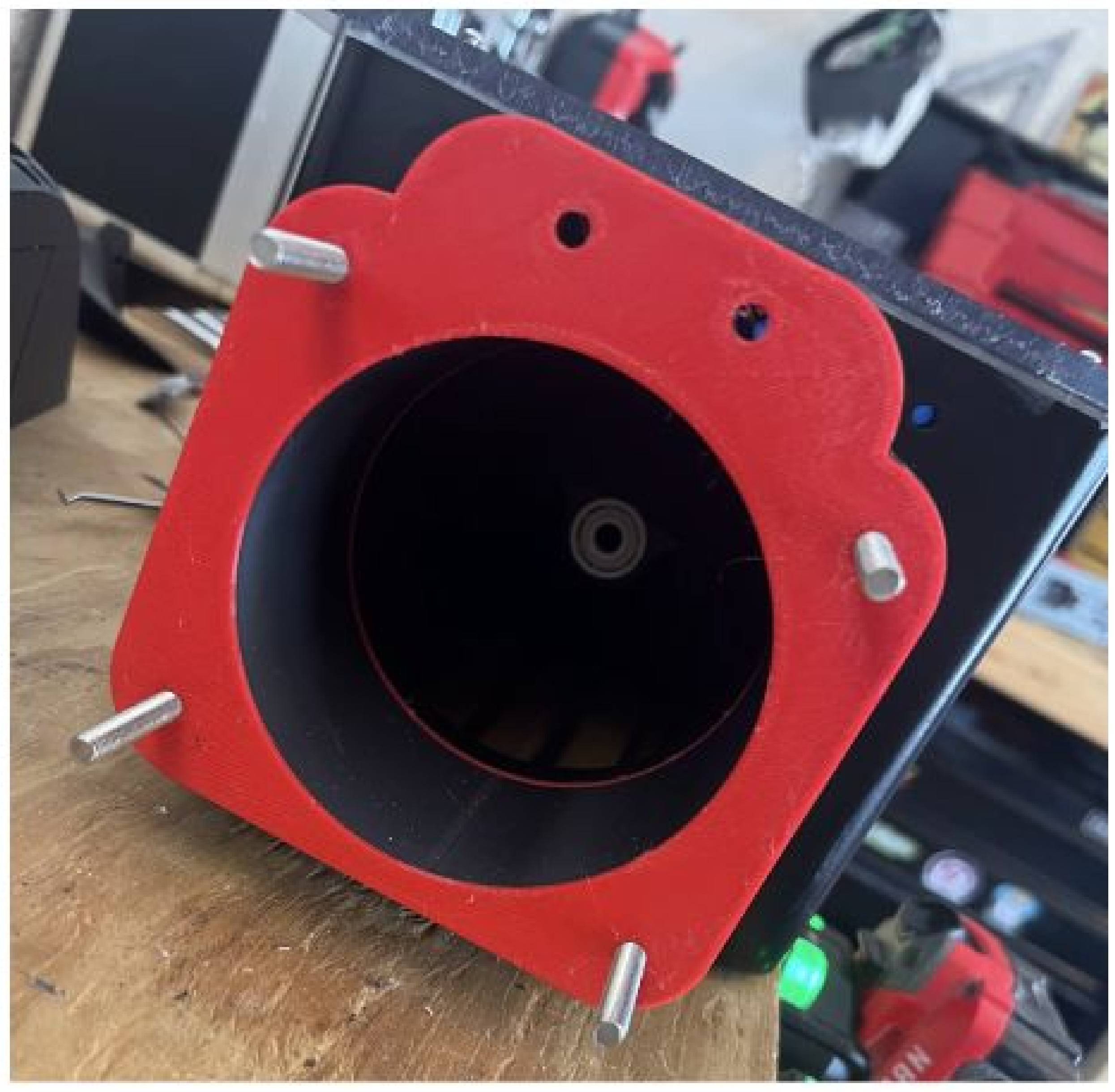
11.2.6. Apply three layers of fiberglass and epoxy, sanding and refinishing between each layer. Allow for time to set and harden respectively.

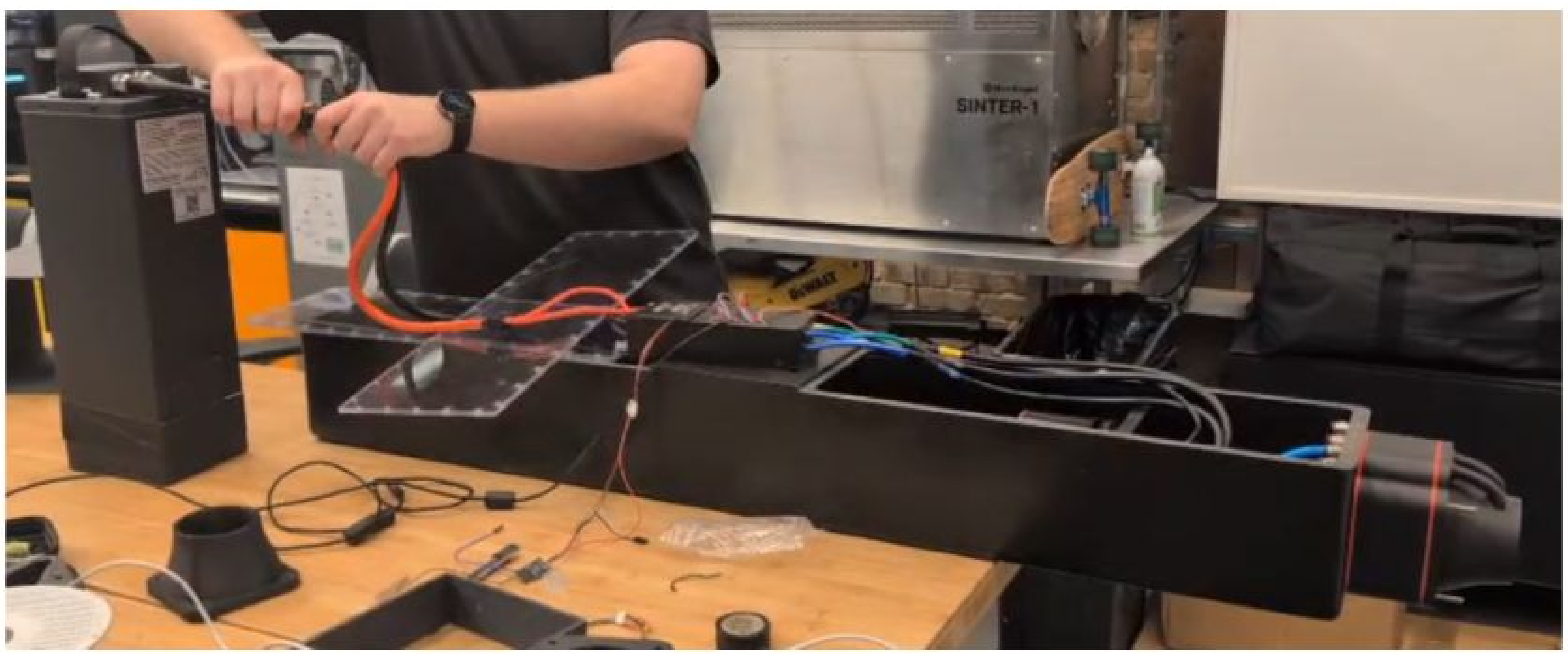
11.2.7. Prime, paint and clear coat on the propulsion box to provide a presentation quality surface



11.2.8. 3-D print seals for exhaust to create a seamless fit utilizing flexible material.



11.2.9. Dry connection testing, focusing on no heat loss or loose connections.



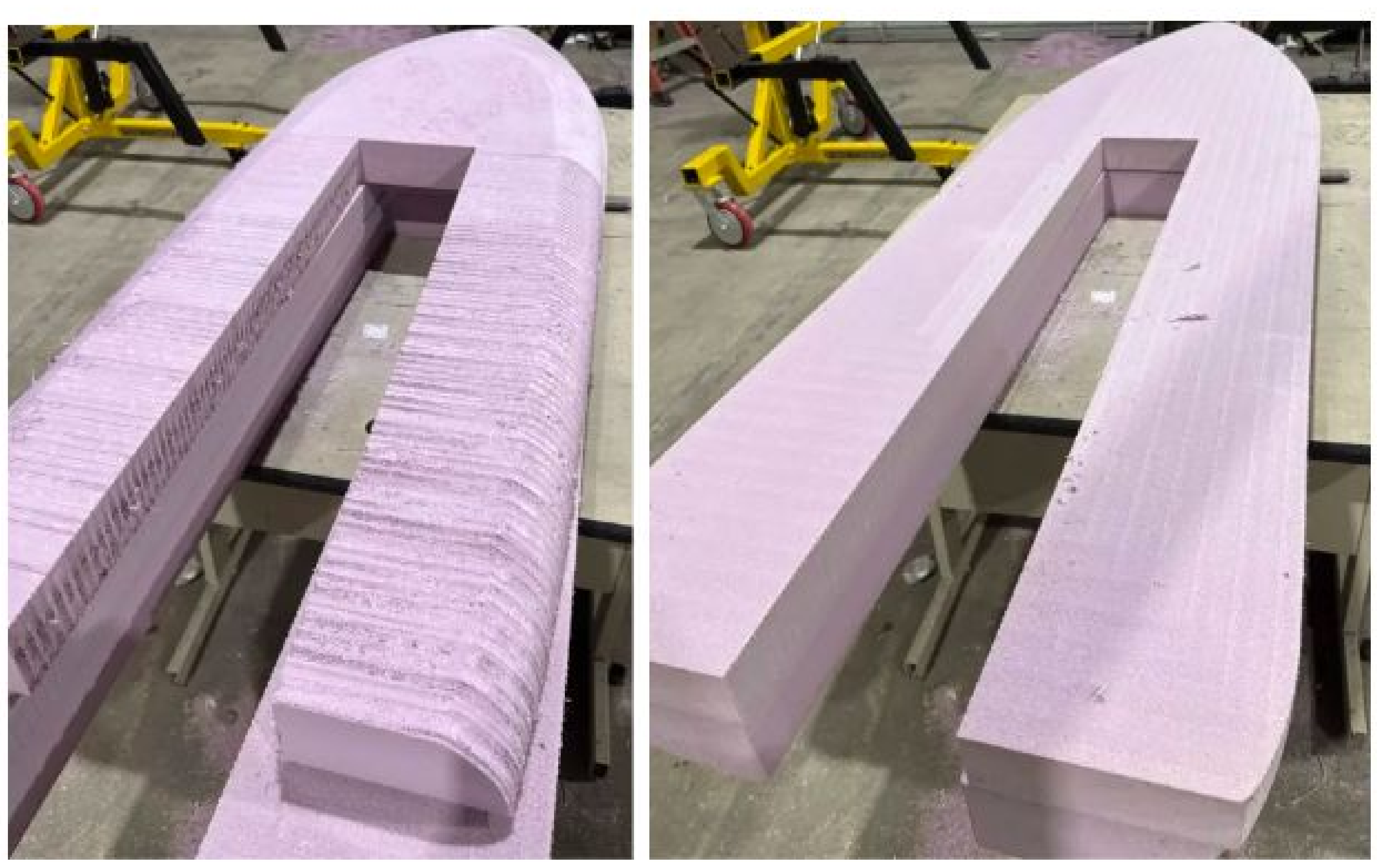
11.2.10. Installation of cooling lines and initial layout of wiring within the confined propulsion box area. Will take additional steps to focus on cable management and including failsafe features to ensure wires are unable to interfere with the motor.

11.2.11. Drill press polycarbonate lids to ensure all seals align.

11.2.12. Waterproof test propulsion unit, focusing on the lid and all sealing screw points.

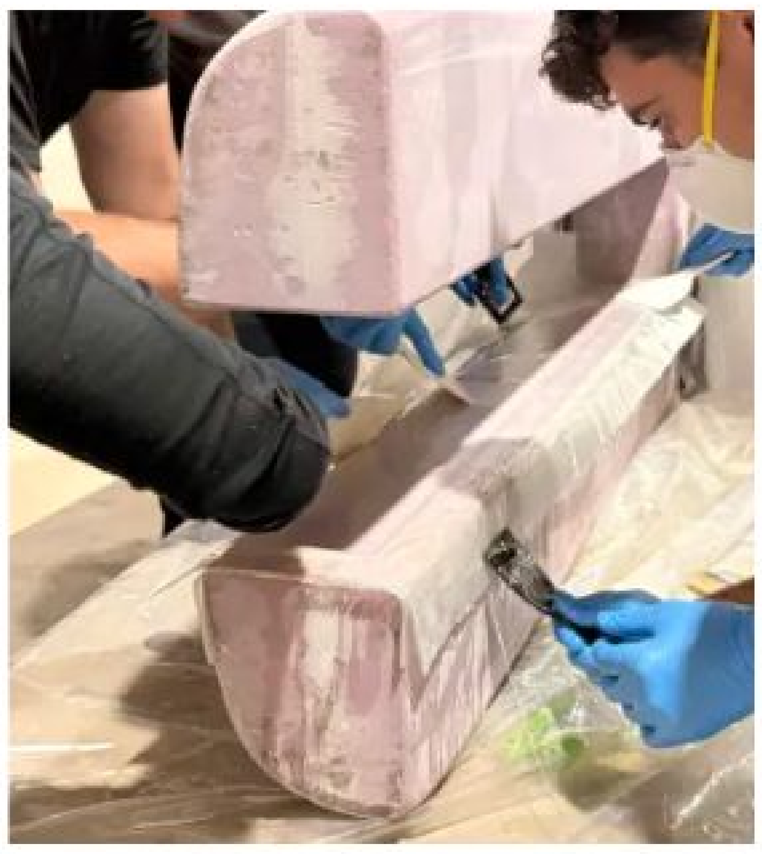
11.3. Board

11.3.1. CNC cut the board into two halves of closed cell foam.



11.3.2. Sand the surface of the foam to achieve the necessary surface finish required for epoxy and fiberglass to adhere.

11.3.3. Apply three layers of fiberglass and epoxy, sanding and refinishing between each layer. Allow for time to set and harden respectively. Starting with the interior cutout of the board where the propulsion box will sit.



11.3.4. Test fit propulsion box into board to ensure fiberglass has not caused a fitting or clearance conflict.

11.3.5. Print end-cap of propulsion box to ensure fit and feasibility of mounting points.



11.3.6. In-lay and adhere female propulsion unit to the board, ensure fit with male modular propulsion unit. Once hardened, sand to produce and flush finish between female box and exterior foam base.

11.3.7. Lay additional layers of fiberglass on the bottom of the board, beneath the propulsion female box for added stability.

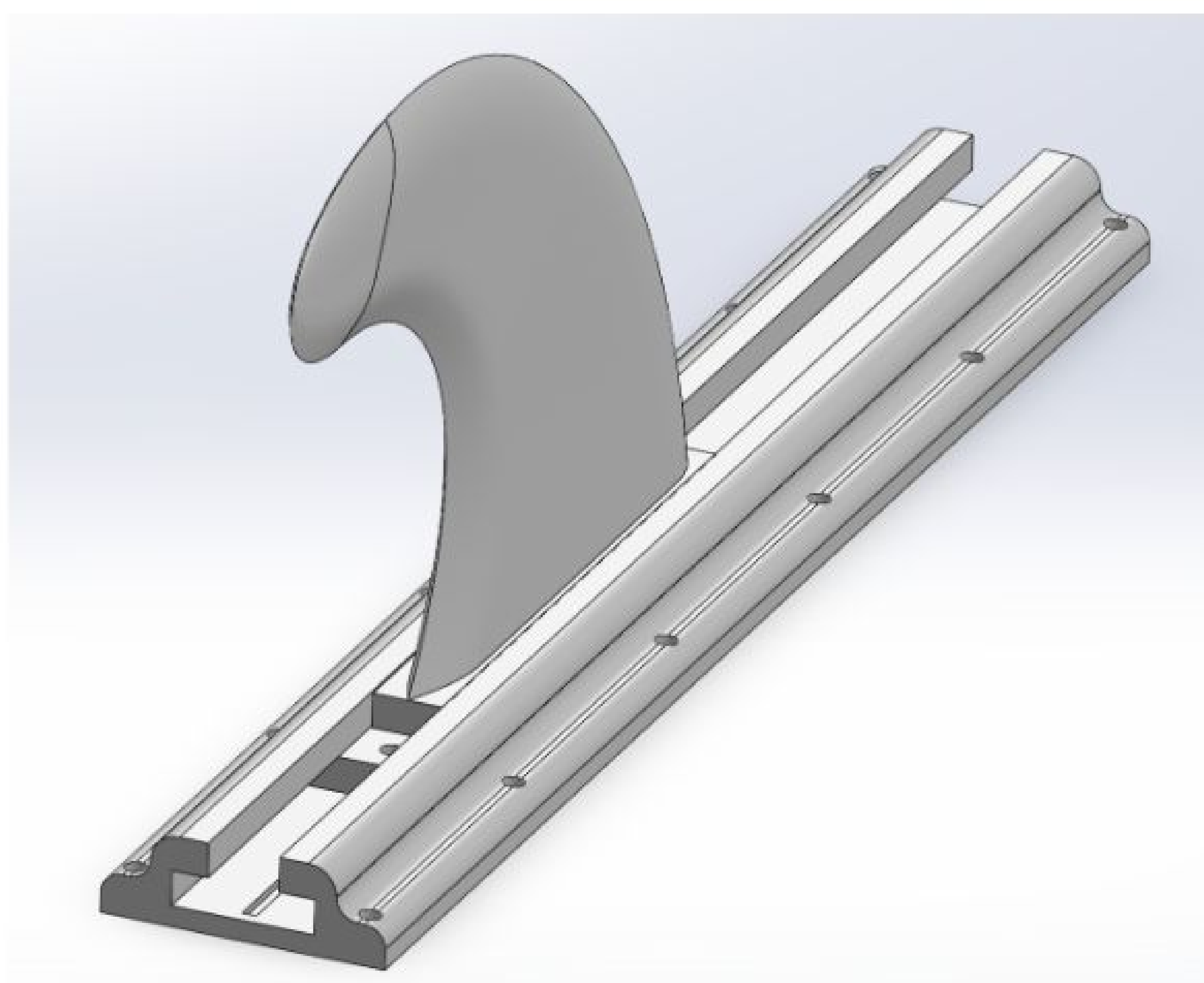
11.3.8. Apply three layers of fiberglass and epoxy, sanding and refinishing between each layer. Allow for time to set and harden respectively. Starting with the bottom of the board, rotating and trimming overhang/delamination between each layer.

11.3.9. Additional fiberglass putty added to top of board above the female propulsion box to create a flush transition and fill any voids.

11.3.10. Cut opening in fiberglass and female box to provide full flow into the intake of the propulsion unit.

11.3.11. Sanding final fiberglass layer on top/bottom of board.

11.3.12. Print out the fins and the rails they slide into

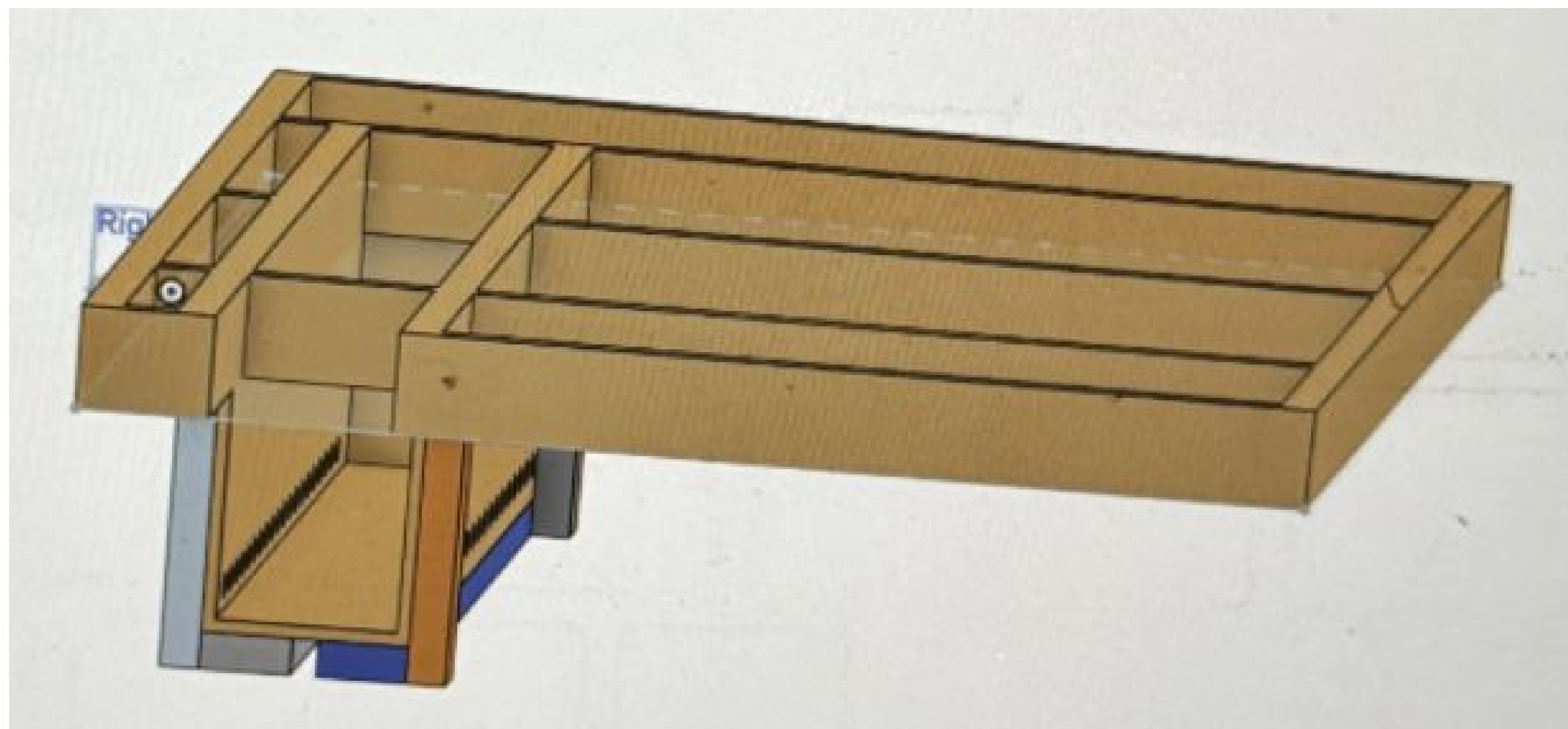


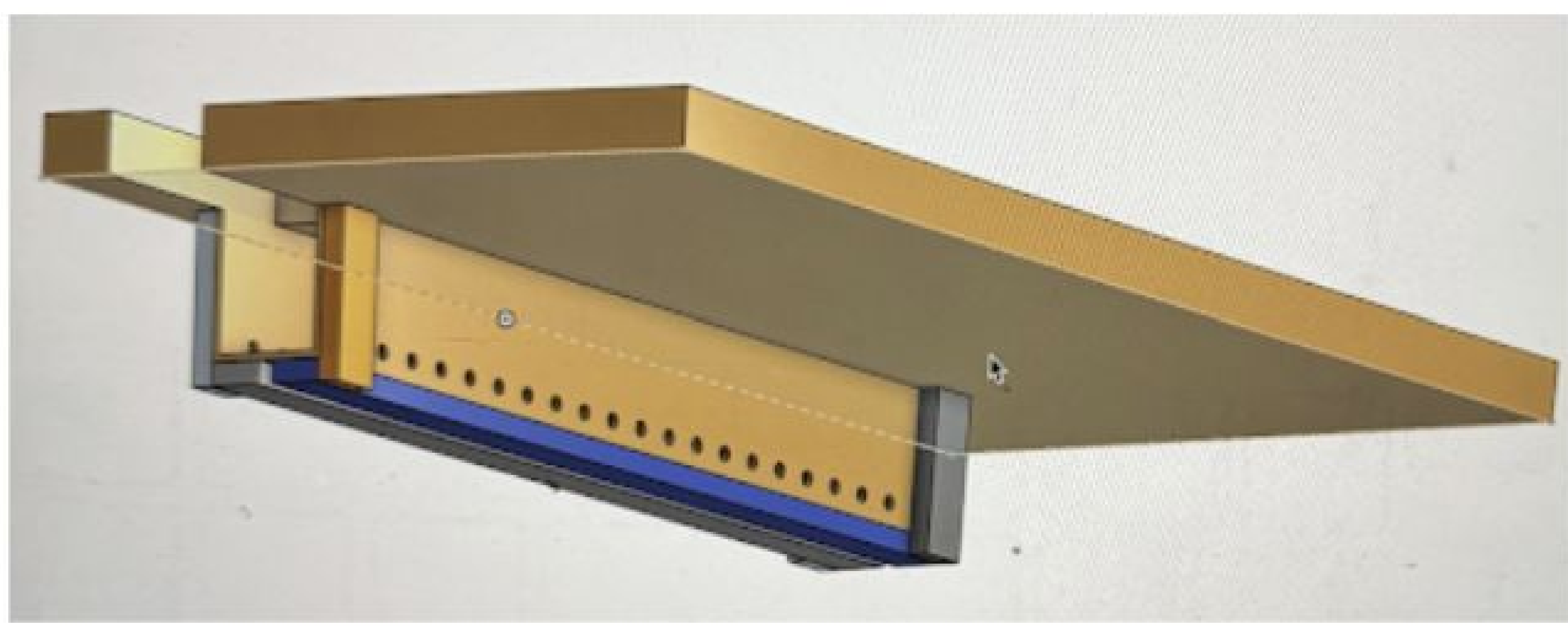
11.3.13. Prime, paint and clear coat on the board to provide a presentation quality surface.

11.3.14. Attach fin rail system, equip produced fin for testing.

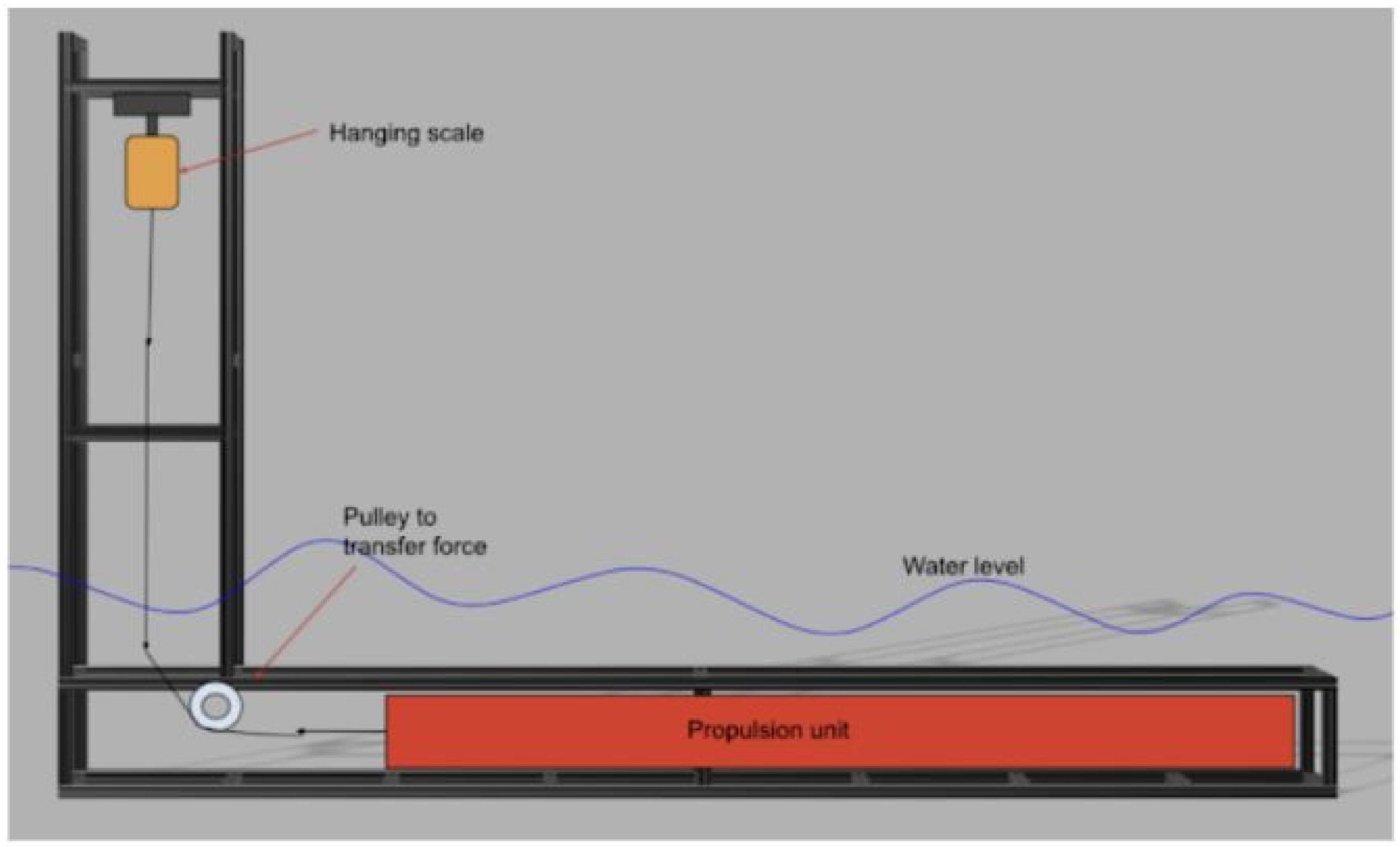
11.4. Newton Test Bench

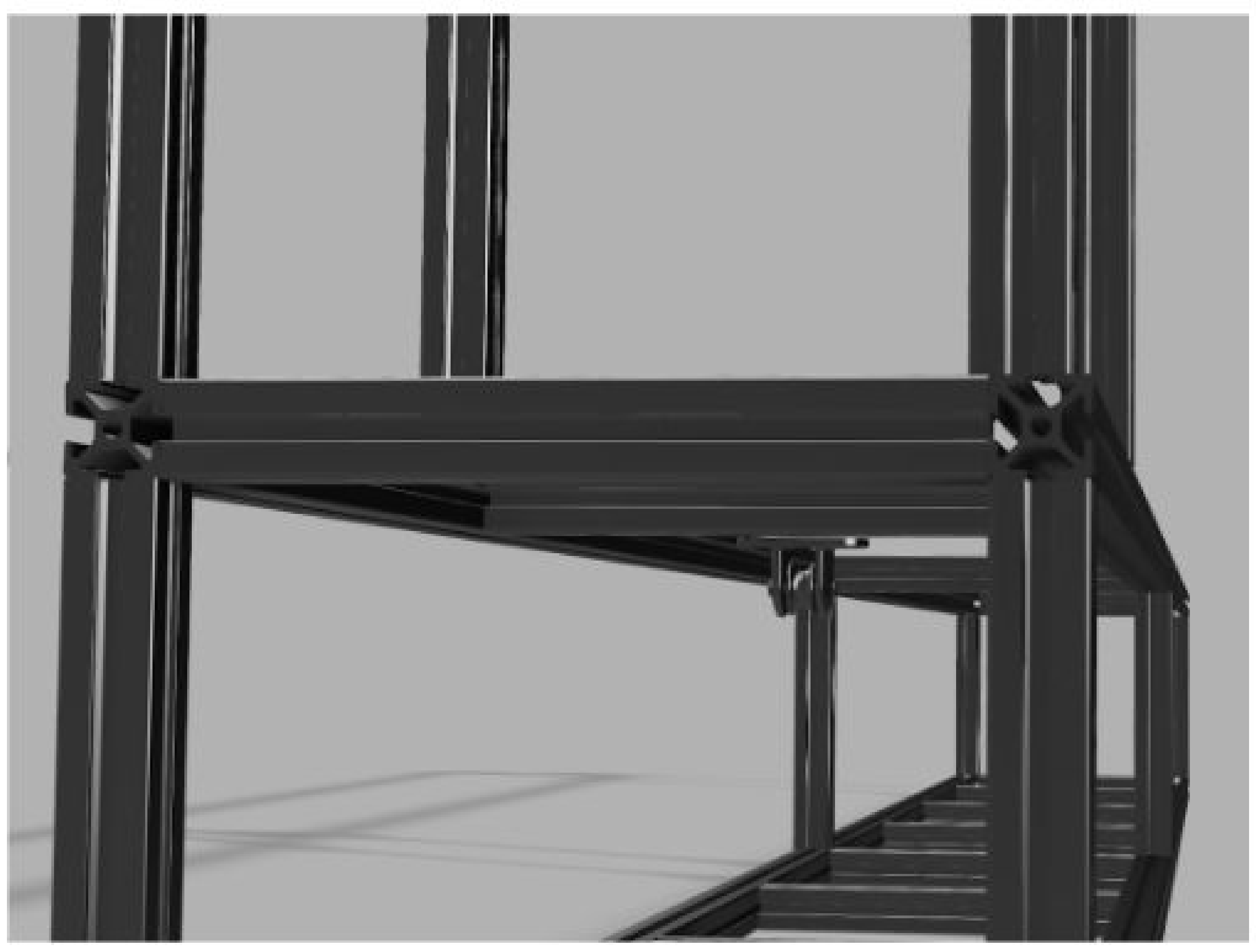
11.4.1. Initial design for the test bench, focusing on 1st-principles. The need is to have a "straw" that does not allow for the propulsion unit to yaw, rotate or translate in any direction other than a longitudinal linear motion. This "straw" needs to be held at a controlled depth, submerged in the water, as well as allow for the propulsion unit to be added and removed seamlessly. Utilizing any structural walls, edges of the test pool will help minimize control points.

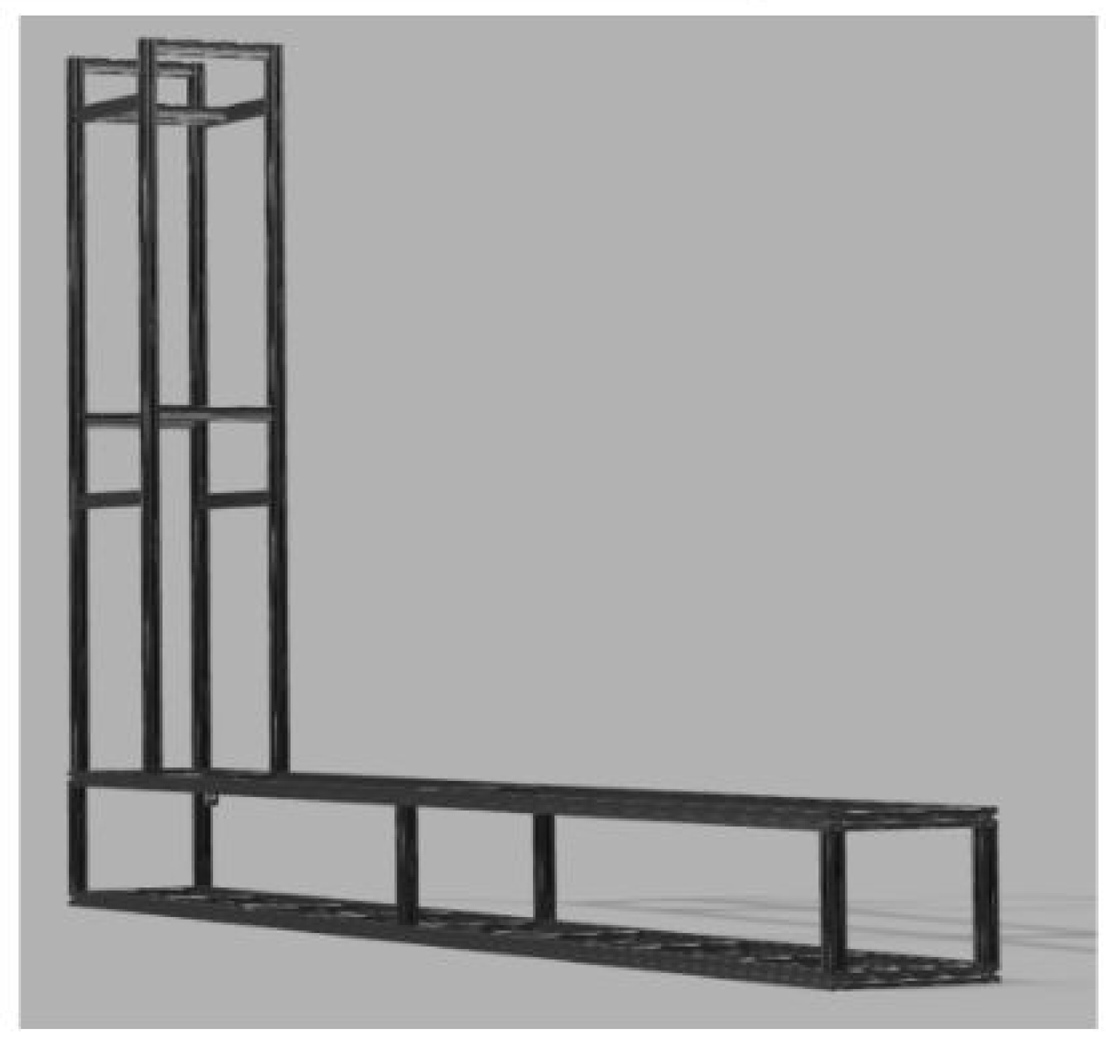




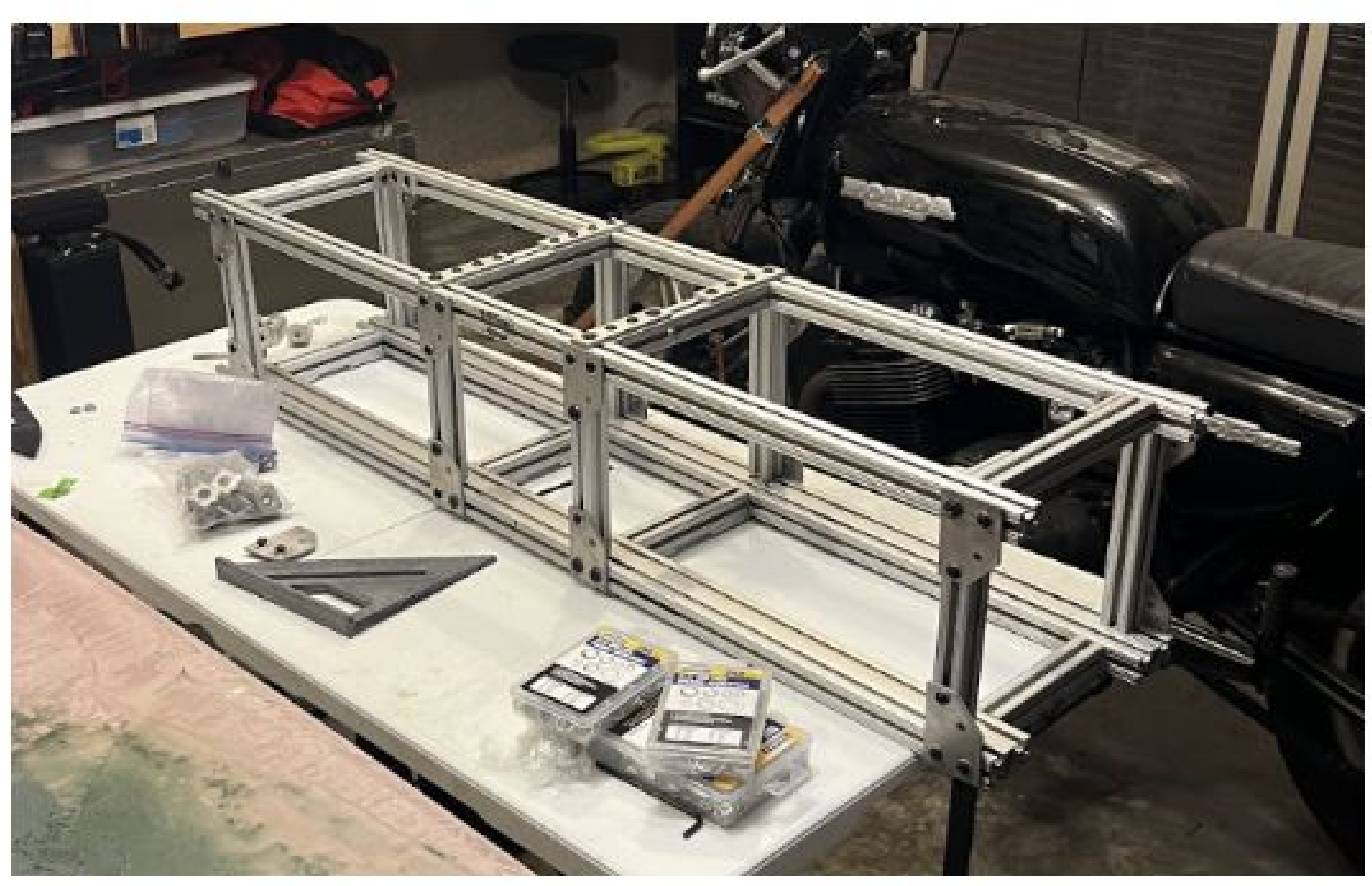
11.4.2. Further design consideration to account for reusability and strength led to design changes to utilize aluminum extrusion as the framework of the test bench.







11.4.3. Construction of the Aluminum Extrusion section.



11.4.4. Construction of the stabilization, anchoring plate.

11.4.5. Testing.

12. Testing Approach

The testing strategy for this project is divided into component-level testing in the fall semester and integrated system testing in the spring semester. Each test is designed to verify specific performance criteria and identify issues before water trials.

<u>Completed</u>

1. Battery Pack Assembly Testing
    a. After constructing the 20s10p pack, we measured individual cell voltages to verify balance. We performed a low-current discharge test (5 A) to confirm all parallel groups were functioning and that the BMS could monitor cell health. We also verified that fusing and wiring could handle expected currents by conducting a brief 50 A load test using a resistive load bank.
2. 3D Print Fit Testing
    a. We printed test sections of the jet pump housing and motor mount to verify dimensional accuracy and fit. This allowed us to identify tolerance issues and adjust CAD models before printing final parts.
3. CAD Model Validation
    a. We used the Shape3D software to model the hull shape and verify that the foam core could be CNC-machined within the capabilities of available equipment. We confirmed the propulsion cartridge would fit within the board interior without interfering with structural reinforcements.

<u>Planned</u>

1. Waterproof Enclosure Testing
    a. Anticipated: Before water trials, the completed propulsion module will be submerged in a test tank at 1m for 30 minutes. We will inspect for any water ingress and verify that all seals, gaskets, and cable pass-throughs remain watertight. This test must pass before any in-water operation. Upon passing the IP-67 rating test, we will test again for the IP-68 rating for being submerged at 3m for an extended period of time.
2. Static Thrust Testing
    a. Anticipated: The propulsion module will be mounted in a controlled test rig with the jet pump submerged in a water-filled container. We will measure thrust output at various throttle settings using a load cell. Success criteria: minimum 40 lbs of thrust at full throttle. This test verifies propulsion performance before mounting in the board.
3. Tethered Water Testing
    a. Anticipated: The assembled board will be tethered to a dock or boat and tested in shallow water (4–6 feet deep). A team member will stand on the board while it is held stationary, and throttle will be applied gradually to verify buoyancy, stability, and control responsiveness. This allows us to identify handling issues in a controlled environment.
4. Speed and Endurance Testing
    a. Anticipated: Once tethered tests are successful, the board will undergo open-water speed trials in a calm bay or lake. Using GPS tracking, we will measure maximum speed and record battery voltage, current draw, and motor temperature throughout the run. Success criteria: minimum 20 mph sustained speed and 15 minutes of runtime at cruising throttle.
    b. Actual:
5. Safety System Validation
    a. Anticipated: We will deliberately trigger emergency conditions (release the throttle suddenly, activate the kill switch) to verify that the motor stops within 5 seconds. We

will also monitor thermal sensors to confirm that the system shuts down if temperatures exceed safe limits.

6. Durability Testing
    a. Anticipated: After initial performance testing, the board will undergo repeated use over multiple sessions to identify wear patterns, seal degradation, or component failures. We will inspect the hull, propulsion module, and electrical connections after every five hours of operation.

13. Team Activity Report

This section should summarize the contributions of each member – who contributed to what sections of the project and the report.

	Team Members					
Sections	TL	ZT	BB	CC	NC	WE
Abstract		X	X		X	
1. Introduction				X		
2. Problem Definition	X				X	
2.1 Problem/Need	X				X	
2.2 Intended user(s) and use(s)	X		X			
2.3 Assumptions and limitations	X		X	X	X	
2.4 Design objectives	X					X
3. Design Specifications	X		X			
4. End-Product Description	X			X		X
5. Design Constraints		X			X	
5.1 Weight and Buoyancy		X	X			
5.2 Propulsion System		X		X		
5.3 Battery and Thermal		X				

5.4 Structural and Material		X		X		
5.5 Electrical and Safety		X		X		X
5.6 Manufacturing and Budget		X			X	X
5.7 Hydrodynamic and Stability		X		X		
5.8 Waterproofing and Durability		X				X
5.9 Economic		X	X		X	
6. Safety	X		X		X	
7. Engineering Standards			X			X
8. Technical Approach	X			X		
8.1 Technical Details of Implementation		X			X	
8.2 Comparison to Top Electric Surfboard		X				
8.3 Alternate Designs Considered		X				
8.4 Future Iteration Considerations		X				X
8.5 Functional Block Diagram		X			X	
9. Part List and Budget	X			X		X
10. Gantt Chart	X	X	X	X	X	X
11. Procedure Punchlist		X		X		X

11.1 Design		X				
11.2 Propulsion Box		X				X
11.3 Board		X	X		X	
11.4 Newton Test Bench	X	X	X	X	X	X
12. Testing Approach		X				
13. Team Activity Report		X		X		
Appendix A — Team Member Resumés	X	X	X	X	X	X
Appendix B — Graphing Script	X					X
Appendix C — Propulsion and Vehicle Performance Modeling		X				
Appendix D — Fiberglass Shell Factor of Safety		X	X			

Table 7 - Team Activity Report

Appendix A - Team Member Resumés

ZACHERY TAYLOR, EI

https://www.linkedin.com/in/zacheryalexandertaylor — Pensacola, FL
zach811taylor@gmail.com — (850) 261-1370

Professional Summary

Civil Engineering Designer with 5+ years of land development and infrastructure experience, FE Exam passed (Dec 2025). Expertise in subdivision design, site grading, stormwater management, and water/wastewater systems using AutoCAD Civil 3D and HydroCAD. Proven leadership through SAE Baja executive board role and concurrent management of 10+ municipal projects. Seeking to advance toward Professional Engineer (PE) licensure while driving impactful infrastructure solutions.

Education

Bachelor of Science in Mechanical Engineering — **May 2026**
Bachelor of Science in Mathematics, Minor in Statistics — **2020**
University of West Florida, Pensacola, FL
GPA (Program): 3.43 (3.90) | President's List (6 semesters), Dean's List (4 semesters)

Skills

Design & Drafting: AutoCAD Civil 3D, Inventor, AutoCAD, SOLIDWORKS (CSWA), Bluebeam
Analysis & Simulation: HydroCAD, AutoTurn, ANSYS Fluent, MATLAB, Multisim
Programming: Arduino (C++), Python, JavaScript, SAS, Microsoft Office Suite

Experience

Civil Engineering Designer — **07/2025 - Present**
Mullins, LLC – Civil Engineering Firm — **Pensacola, FL**

- Designed subdivision infrastructure; optimizing roadway layouts, site grading, stormwater management systems, and utility networks, ensuring compliance with local development standards and municipal codes.
- Coordinate with contractors during construction phases to address design modifications and field conditions, maintaining project momentum while upholding engineering standards.
- Managed multiple subdivision project timelines, prioritizing plan submittals, utility coordination, and permitting milestones to meet aggressive development schedules.

Designer — **05/2021 – 07/2025**
McKim & Creed – Civil Engineering Firm — **Pensacola, FL**

- Obtained direct experience under 8 Professional Engineers.
- Managed a concurrent workload of 10+ land development and water/wastewater design projects.
- Spearheaded utility conflict analysis for large-scale municipal projects, identifying interferences between proposed storm, sewer, and electrical networks prior to construction.

Computer Aided Design Drafter — **07/2020 – 05/2021**
FlynnBuilt – Custom Home Builder — **Pensacola, FL**

- Streamlined the design and development process, reducing design turnaround time by 50%.
- Creative freedom to develop and design over 20 new elevations and complete house floorplans.

Engineering Projects

SAE Baja – Executive Board / Treasurer — **09/2025 - Present**

- Secured $28,000 in funding (exceeding all historical team records) by authoring a successful SGA grant proposal and coordinating "spirit night" marketing initiatives.
- Established long-term stability by creating a UWF Foundation account and formalizing SGA club status, directly increasing non-engineering membership recruitment.

Electric Jet-Driven Surfboard — **2026**

- Designed and developed an electric jet-driven surfboard featuring a modular propulsion unit for simplified integration, maintenance, and future upgrades, combining electric powertrain design, water-jet propulsion, and board-level system packaging into a functional prototype.

Tanner S. Landry

Phone: (985)445-4478

Email: tannersabin@gmail.com

Professional Summary

Mechanical Engineering student with hands-on experience in additive manufacturing, CAD design, and prototyping. Skilled in developing and refining manufacturing workflows, performing quality control, and collaborating with multidisciplinary engineering teams. Eager to contribute to production optimization and continuous improvement initiatives.

Work History

10/2024 – Current **Additive Manufacturing Lab Technician**

Sea 3D - 3D printing Laboratory

- Operate and maintain industrial additive manufacturing systems (FDM, SLA, and composite reinforced printers)
- .Prepare build files, calibrate equipment, and troubleshoot print failures to ensure consistent production quality.
- Perform part post-processing operations including support removal, sanding, vapor smoothing, and finishing.
- Collaborate with clients, students, and research teams to translate conceptual designs into manufacturable parts.
- Evaluate materials and geometries to recommend optimal print settings.
- Manage print queues, production timelines, and material inventory to support lab scheduling and workflows.

04/2021 – 10/2024 **Additive Manufacturing Specialist**

Self Employed

- Analyze material properties and select appropriate materials for specific applications.

- Optimize printing parameters for quality, speed, and cost-effectiveness.
- Create detailed 3D models for additive manufacturing.
- Rapidly prototype concepts and iterate designs based on feedback.
- Perform mechanical and material testing (tensile, hardness, impact) on samples.
- Collaborate with cross-functional teams (engineers, designers, and researchers).

02/2020 – 04/2021 **Motorcycle Technician**

Rival Mx – Murrieta, Ca.

- Proficient in diagnosing complex mechanical and electrical issues in motorcycles.
- Utilize engineering principles to identify root causes and develop effective solutions.
- Employ specialized tools and testing equipment for accurate diagnostics.
- Disassemble, inspect, and rebuild motorcycle engines with meticulous attention to detail.
- Analyze motorcycle wiring diagrams and schematics.

Other Qualifications

- Extensive knowledge of mechatronics and robotics design
- Research experience in biomechanical design
- Leadership experience on multidisciplinary engineering research teams
- Experienced in various CAD programs (SolidWorks, AutoCAD, and Fusion 360)
- Skills in mesh modeling
- Proficient in C, C++, Java, and Python software tools

EDUCATION:

University of West Florida
11000 University Pkwy, Pensacola, FL 32514
B.S. Mechanical Engineering
Projected graduation: 05/2026

Northwest Florida State College
100 E College Blvd, Niceville, FL 32578

Bailey Burgis
1176 Don Bishop Road
Santa Rosa Beach, FL 32459
850-830-0814
reagan.burgis@gmail.com

Education

Bachelor of Science in Mechanical Engineering
Minor: construction Management
University of West Florida, Pensacola, FL
GPA: 3.67
Relevant Coursework: Mechatronics
Honors and Achievements:

- Dean's List: August 2023, January 2024, August 2025
- President's List: June 2025

Northwest Florida State College, Niceville, FL
Fall 2022 – Spring 2023
Associate of Arts completed

Relevant Engineering Projects **Mechatronics**

- Built a fully autonomous 4 way traffic light with sensors included for turn lanes. This one man project required construction capabilities, programming, and wiring knowledge
- Constructed a fully autonomous wall avoiding car. This project required ultrasound sensors to detect the walls and motors to steer away from the walls

Relevant Engineering Projects **Design**

- Constructed a fully functional offroad baja car. Through the means of welding, bending, cutting, and force analysis.
- Attended the competition in Arizona Spring of 2025 and placed in many of the trials

Relevant Engineering Projects **Capstone**

- Currently building an electric surfboard that will be fully 3D printed and fiberglassed
- This board will have a propulsion system similar to a jet ski and is expected to reach speeds of 20 mph

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Experience

Santa Rosa Boys – Lawn Mowing and Car Detailing
June 2018 – Present | Santa Rosa Beach, FL

- Owning a business taught many useful tools such as: negotiation, time management, communication, leadership, and financial management
- directing this business has given real-world workplace experience and dedication to hard work.

Emerald Coast Associates – Civil Engineering Internship
Summer 2025

- Utilized Civil 3D and hydrodynamics software allowed for a smooth transition into this internship
- performed the full process of project design in civil engineering and improving technical and communication skills.

Larry Batchelor Mechanical – Commercial HVAC Contractor
Summer 2024

- Oversaw project management, installation crews, and worked with technicians to understand every aspect of designing and installing commercial HVAC systems
- deepened an understanding of mechanical systems and teamwork in the field.

98 BBQ Restaurant – Line Cook
June 2021 – Present | Santa Rosa Beach, FL

- fast-paced kitchen taught me responsibility, communication, and time management to maintain order flow and quality

Back Beach BBQ – Cashier and Food Preparation
January 2020 – April 2021 | Panama City Beach, FL

- customer service, food preparation, and teamwork, especially in high-stress environments during busy hours

---

Leadership and Involvement

National Honor Society
Inducted sophomore year (2018–2021); active member throughout high school.

South Walton Link Crew
Selected by teachers to assist incoming freshmen by providing guidance and support during their transition into high school.

Certifications and Athletics

Certification in Word and Digital Imaging Technology (DIT)
Completed both certifications during freshman year of high school, and allowed for more knowledge in these fields.

South Walton Baseball Team
Member all four years of high school; developed strong time management balancing academics, clubs, and athletics. Played baseball from T-ball through middle school and high school.

University of West Florida Men's Volleyball Team

The UWF men's volleyball team increased skills in the sport as well as gained time management knowledge along the way.

Boaters Safety License

Experience and life-long passion for boating led me to obtain my Boater’s Safety License

CAMDEN CHAMBERS

400 Marsh Deer Run, Niceville FL 32578 | (850) 512-5207 |
camden_chambers_25@yahoo.com

PROFESSIONAL SUMMARY

Motivated Mechanical Engineering student at the University of West Florida with hands-on experience in CAD modeling, prototype development, and collaborative engineering projects. Strong foundation in mathematics and mechanical design principles, with an understanding of programming. Proven leadership experience and ability to manage multiple priorities while delivering high-quality technical work.

EDUCATION

University of West Florida (UWF), Florida

Bachelor of Science in Mechanical Engineering | Expected Graduation: May 2026

Relevant Coursework: Statics, Dynamics, Thermodynamics, Engineering Design, Fracture Mechanics, Fluid Dynamics, Programming for Engineers, Calculus I–III

Niceville High School, Florida

Graduated Magna Cum Laude | GPA: 4.16 | May 2022

ENGINEERING EXPERIENCE

Student Ambassador Intern | Hsu Educational Foundation, Fort Walton Beach, FL

May 2025 – August 2025

- Designed and developed engineering prototypes using CAD modeling software
- Assisted in drone design and gained working knowledge of FAA regulations and compliance
- Collaborated with engineers on technical design and hands-on engineering projects
- Gained exposure to project management and government contracting processes
- Mentored and supported STEM education initiatives for younger students

PROFESSIONAL EXPERIENCE

Bag Drop Attendant | Emerald Bay Golf Course, Destin, FL

May 2023 – August 2024

- Provided high-level customer service and coordinated with management and golf shop staff
- Managed equipment preparation and maintenance of golf carts and clubs

- Demonstrated strong communication, organization, and time management skills
- Maintained a professional environment under high customer volume conditions

TECHNICAL SKILLS

Engineering & Software:

- CAD Modeling (SolidWorks)
- C Programming
- Microsoft Excel (Data organization and analysis)
- Six Sigma Green Belt Certified

Engineering Strengths:

- Advanced Mathematics
- Prototype Development
- Mechanical Design Fundamentals
- Technical Collaboration
- Attention to Detail

LEADERSHIP & ACTIVITIES

- Founder, UWF Club Golf Program
- Founder, Niceville Men's Volleyball Program
- UWF Intramural Volleyball MVP (2022, 2023)
- Member, UWF Club Volleyball
- Member, Rocky Bayou Country Club

HONORS & AWARDS

- UWF Dean's Honor Roll (Fall 2022, Spring 2023, Spring 2025, Summer 2025)
- Florida Bright Futures Scholarship Recipient
- UWF Academic Scholarship
- Anne T. Mitchell Academic Honors (9th–12th Grade)
- Niceville High School Baseball – Most Improved Player (12th Grade)
- UWF Intramural Hall of Fame Inductee

VOLUNTEER EXPERIENCE

- Completed 120+ service hours through a four-year leadership program supporting school and community initiatives

Nacie M. Calarco

1919 Pineview Church Rd., Jay, FL 32565

224-523-6446 naciecalarco34@gmail.com

OBJECTIVE:

To earn a position at XYZ customs as an Engineer/Fabricator

Formal Education:

2023	CCAF AAS; Strategic Operations Management (app. graduation)
2022-2026	University of West Florida; Engineering Technologies (May 2026)
2020	Airman Leadership School

Relevant Courses: CAD, CSWP, Engineering Materials, Engineering Statistics, Physics ½, Capstone

Work History:

Department of Defense/ US Air Force

July 2016 – July 2022 — 19^{th} ASOS, Ft. Campbell, KY

Tactical Air Control Party (TACP)/Strike Team Supervisor/JTAC-I. Supervises the training, equipping and employment of 7 TACPs. Maintains Joint Terminal Attack Controller (JTAC) combat mission ready status; executes missions in support of ground commander's plan of maneuver. Advises and assists Army battalion ground commander and battle staff with close air support planning, integration, and execution. Expert in Air-Land Battle Doctrine; plans and conducts training to ensure TACP/JTAC personnel are combat mission ready.

July 2022 – 2024 — 238^{th} ASOS, Key Field, MS

Tactical Air Control Party (TACP)/Strike Team Supervisor. Air National Guard (Traditional)

October 2024 – Present — 284^{th} ASOS, Smokey Hill, KS

Tactical Air Control Party (TACP)/Strike Team Supervisor/JTAC-I. Air National Guard (Traditional)

Professional Training:

August 2025	Certified SolidWorks Professional
June-July 2021	SFAUC Course (Special Forces Advanced Urban Combat)
February 2021	JTAC-I Course (Joint Terminal Attack Controller Instructor)
May 2019	ISTC CQB Course (International Special Training Centre Close Quarters Combat)
July 2018	JTAC QC (JTAC Qualification Course)
March 2018	Army Air Assault
January 2017	Air Force SERE SV-80 (Survival Evasion Resistance and Escape)

Highlighted Experience:

Atlantic Resolve Deployment: Deployed to Poland, Germany, Estonia, Hungary, Czechia, and Romania assigned to the 2 EASOS from January 2019 to October 2019. Controlled fighter and helicopter attack sorties as well as artillery with partner nations from Hungary, Italy, Estonia, Belgium, United Kingdom, Poland, and United States. Assisted in the instruction and reconstruction of the Polish Joint Terminal Attack Controller Course. Participated in two Joint NATO exercises, integrating joint combat forces in a battle space. Attended advanced CQB course at the International Special Training Centre with special operation members from several NATO countries, Performing at the top of the class in all scored events.

5th Special Forces Group Liaison: Selected as sole Air Force liaison for 5th SFG. Worked in the ASC (advanced skills company) JTAC locker as an instructor and subject matter expert for ODAs and foreign partner forces. Assisted as an instructor for the group level JTAC/SOTAC program, certifying and maintaining currency for over 50 SOTAC (Special Operation Terminal Attack Controller) personnel. Graduated from SFAUC (Special Forces Advanced Urban Combat) course. Created an air to ground integration class for SFAUC. Instructed the class, as well as aided in the integration of live-fly aircraft for every SFAUC exercise. Created and instructed "near peer" classes to modernize the ground commander course, certifying 12 ODA commanders for deployment.

Air to Ground Integration West Africa Deployment: Deployed as a two man team as an Air advise to advise, assist, and enable the host nations military development, training, and operations. During this time, I reviewed applicable programs and restructured and advised on operations and professional development. I was a direct liaison between American agencies and General level host nation forces.

UWF Electric Jet Surfboard Capstone team: Member of the composite team for the Electric Jet Surfboard. During this time, I was charged with developing and designing the overall surfboard shape. Confirming fitment and support of the propulsion teams drive and battery box. As well as fabricating the board. This has given me experience in CNC, molding, fiberglass fabrication, and design for manufacture.

Additional relevant experience:

Push to Talk Mounting Bracket:

Designed, developed, and tested a PTT bracket that relocated the PTT to avoid interference and rigidly mount to body armor. This was a solution to a problem I had with gear. This product solved the problem and was used by multiple team members.

Vertical Grip:

Designed, developed, and tested a vertical grip for a rifle. This product is a custom design to be used as a barricade stop and vertical grip. I gained experience in additive manufacturing for both development and end use.

Cross Kart design:

Currently developing a 1000cc Cross Kart as a learning experience. Developing all aspects from scratch to include chassis, suspension, electrical, and engine fitment. The end goal is to fabricate and sell the design of this vehicle and use a motorcycle engine as the drivetrain donor. Status is in design with a large portion of the chassis and suspension already developed.

William Emiro | Student / Cadet / Pilot / Aspiring Engineer

850.610.4111 | woemiro@gmail.com

RECENT & ACTIVE SKILLS

- Mechanical and Electrical engineering skills from school and hands on experience
- Competent drone and RC aircraft operator, builder and maintainer
- In-training with FAA approved flight training program (FAR Part 61)

EDUCATION

- Currently attending the University of West Florida (UWF) Mechanical Engineering program with Bright Futures Florida Academic Scholarship, anticipated graduation May 2026
- Graduated with honors from Northwest Florida State College as a dual enrollment student, May 2023
- Graduated 3rd of over 120 students in academics at Baker High School, FL May 2023
- SAT score of 1340 on 1st attempt (94th percentile)
- ACT Composite score of 30 (93rd percentile)

PROFESSIONAL EXPERIENCE

AFMC 96th Test Wing PCIP Internship **05/2024 – 08/2025**

General Engineer/Summer Employee (GS-04)

- Gained experience integrating and programming Iridium GPS modems (2024)
- Gained experience working with Silvus radios via integration/replacement of the RQ-23a Tigershark long range radio system (2025)
- Familiar with flight operations for the RQ-23a Tigershark, including deployment, maintenance, module integration, and flight control software integration.
- Familiar with Air Force testing procedures and environments.

Rush Farms, Inc. **05/2018 - 2023**

Part Time Heavy Equipment Operator

- Operated heavy machinery including large tractors and combine harvesters
- Learned practical problem solving and fabrication skills including welding and operating machining tools

Arby's Inc. **08/2021 – 05/2022**

Team Member

- Food preparation and customer service
- Worked in multiple positions ranging from front register to backline operations

ACCOUNTS OF NOTE

- 2023/2024/2025 Premier College Internship Program (PCIP) finalist and participant
- SMART Internship Program 2023 semi-finalist
- Volunteer with non-profit organizations such as the Civil Air Patrol (CAP) & Swan Aviation Ministries
- Attended the US Navy National Flight Academy and learned how to plan and conduct missions in a realistic training environment
- Won the Choctawhatchee Electric Cooperative (CHELCO) Youth Tour to Tallahassee and Washington DC learning about the importance of co-ops in both the State and Federal Government's decisions regarding infrastructure

EXTRACURRICULARS

- **UWF Argonautics Aviation Team** **08/2023 – 05/2024**
  - Performed as team lead/captain
  - Designed and built a high-performance remote-control aircraft from the ground up to compete in the 2024/2025 American Institute of Aeronautics and Astronautics (AIAA) Design Build Fly (DBF) competition
  - Iterative analytical design process using a combination of mathematical and physical modelling and testing
  - Learned how to work with and make carbon fiber composite parts
  - Competed in the 2023 DBF competition
- **Junior Reserve Officers Training Corps (JROTC), LET-4** **08/2019 – 05/2023**
  - Oversaw the teaching of newer cadets, ensuring they learned principals of discipline, respect, and ceremonial activities like marching. Other duties included teaching more experienced cadets how to become successful leaders later in their JROTC careers
  - Marksmanship team captain (2 years); taught freshmen in the JROTC program principles of shooting and gun safety
  - Baker JROTC Drone team captain (2 years)
    Taught participating cadets, SUAS safety and piloting skills
    First place in competition (11/2019)
  - Baker JROTC Academic team captain (3 years)
  - Received basic practical first aid training through JROTC
- **Civil Air Patrol (CAP), SER-FL423 USAF Auxiliary** **09/2018 - Present**
  - Completed cadet program as Second Lieutenant (C/2dLt), Cadet commander (08/2022 – 06/2023)
    Currently transitioning to senior member program
  - As Cadet Commander at the Eglin Composite Squadron, I planned weekly meetings and special cadet activities like touring museums and participating in training operations. I also taught cadets how to march, conduct basic search and rescue, the principals and mechanics of flight and piloting an aircraft, and character development to help the younger cadets to develop their leadership skills and become excellent citizens
  - Ground Team Member (GTM) 3 certified; learned how to conduct search and rescue in the event of an emergency, such as a natural disaster
  - Mission Scanner certified, allowing participation in real-world airborne search and rescue missions
  - Received moderate field first aid training through Alabama Wing Emergency Services School (WESS)
- **Aviation / FAA**
  - Completed FAA Ground School and testing. Currently in training to obtaining a private pilot certificate
  - Completed over 50 hours of flight training so far, including long solo flights, and preparing for check ride examination for the completion of the private pilot certificate
  - Volunteer with Swan Aviation Ministries helping introduce youth to the basics of flight using a full-scale flight simulator
  - Also with Swan Aviation Ministries, I am learning how to operate and maintain aviation airframe, powerplant and avionics components.
- **Automotive and Mechanical**
  - 11 non-running cars completely restored to date. They were restored to roadworthy condition, including complete engine and transmission replacements, upgrades, rebuilds, safety systems and cosmetics

Appendix B - Graphing Script

Python script compiled in Spyder to generate graphs

```
import numpy as np

import matplotlib.pyplot as plt


# ---------------------------

# Parameters (tune these)

# ---------------------------

P_max = 25_000        # W, max electrical power from pack/ESC

rho = 1000            # kg/m^3, water density

S = 0.5               # m^2, effective wetted area in planing

C_D = 0.025           # dimensionless, effective drag coefficient


E_Wh = 3330           # Wh, battery energy (20s10p P45B)

E_Wh = float(E_Wh)


# Throttle range (10% to 100%)

throttle = np.linspace(0.1, 1.0, 50)


# ---------------------------

# Speed vs Throttle

# ---------------------------

P = P_max * throttle             # W


# Avoid zero power

V = (2 * P / (rho * S * C_D)) ** (1/3)   # m/s

V_mph = V * 2.23694                      # convert m/s → mph


# ---------------------------

# Ride Time vs Throttle

# ---------------------------

P_kW = P / 1000.0               # kW

time_hours = E_Wh / P_kW / 1000  # hr (E in kWh / kW)

time_minutes = time_hours * 60   # minutes


# ---------------------------

# Plot 1: Speed vs Throttle

# ---------------------------

plt.figure()

plt.plot(throttle * 100, V_mph)

plt.xlabel("Throttle (%)")

plt.ylabel("Speed (mph)")

plt.title("Predicted Board Speed vs Throttle")
```

```
plt.grid(True)

# ----------------------------
# Plot 2: Ride Time vs Throttle
# ----------------------------
plt.figure()
plt.plot(throttle * 100, time_minutes)
plt.xlabel("Throttle (%)")
plt.ylabel("Ride Time (minutes)")
plt.title("Estimated Ride Time vs Throttle")
plt.grid(True)
```

Detailed Hand Calculation Work

Propulsion & Vehicle Performance Modeling

Assumptions:

Input power (to shaft): $P = 25\,kW$

Overall hydraulic/jet efficiency $(\eta) = 0.6 - 0.7$

"typical for small/hobby level water jet"

① $P_h = \frac{1}{2} \dot{m} u_j^2$

u_j = water jet velocity; $\dot{m}$ = mass flow rate; P_h = Hydraulic power = $P\eta$

Case A: $(25\,kW)(0.6) = 15\,kW$

Case B: $(25\,kW)(0.7) = 17.5\,kW$

② $T = \dot{m} u_j$ $\quad$ T = thrust

$$\Rightarrow T = \frac{2P_h}{u_j}$$

③ $\dot{m} = \rho A_j u_j$

A_j = exit area $= \pi (D_j/2)^2$; ρ = water density $\approx 1000\,kg/m^3$ [fresh water]

D_j = exit nozzle diameter $= 70\,mm$

$A_j = 3.85 \times 10^{-3}\,m^2$

④ Converting ① to solve for u_j:

$$P_h = \frac{1}{2}[\rho A_j u_j] u_j^2 \Longrightarrow u_j = \left(\frac{2P_h}{\rho A_j}\right)^{1/3}$$

$$u_{j_A} = \left(\frac{2(15000)}{1000(3.85 \times 10^{-3})}\right)^{1/3} = 19.8\,m/s \approx 44.3\,mph$$

$$u_{j_B} = \left(\frac{2(17500)}{1000(3.85 \times 10^{-3})}\right)^{1/3} = 20.9\,m/s \approx 46.8\,mph$$

⑤ $T_A = \frac{2P_{h_A}}{u_{j_A}} = 1515\,N$

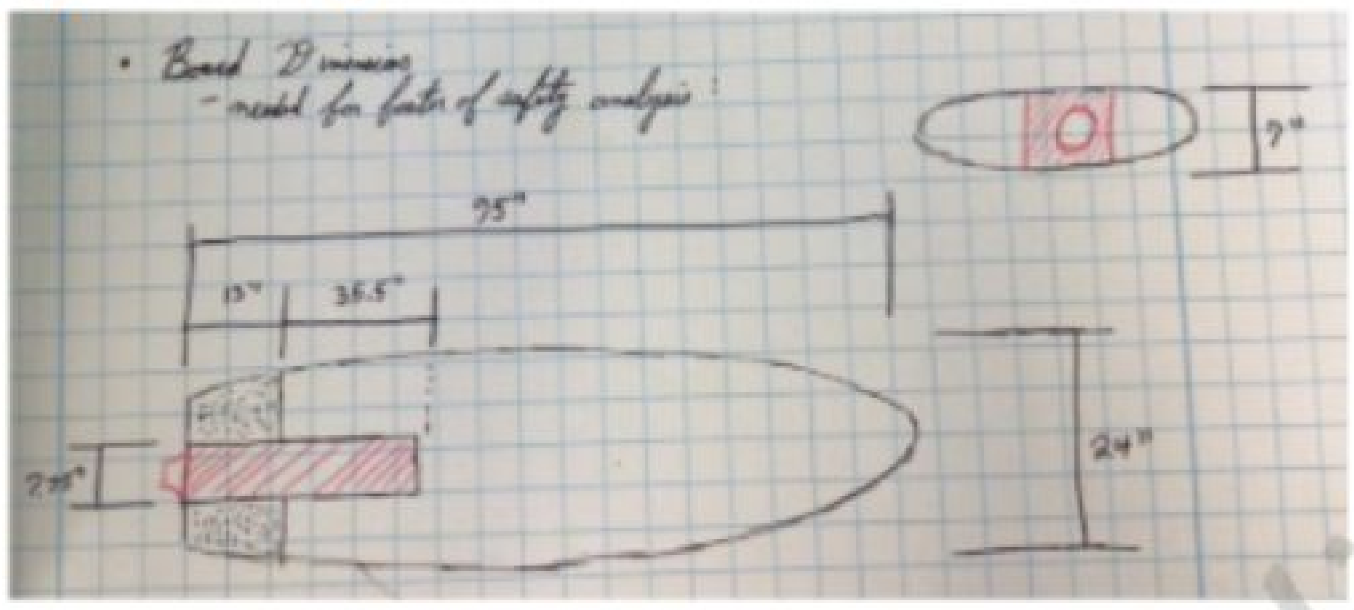
$T_B = \frac{2P_{h_B}}{u_{j_B}} = 1675\,N$

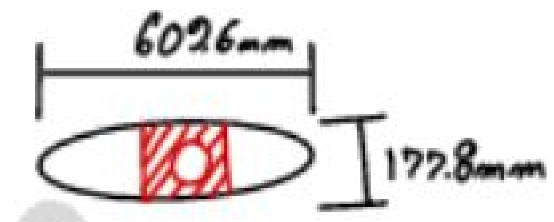
⑥ Simple Drag Modeling for a surfboard

"typical for low profile planing crafts"

$$D(V) = \tfrac{1}{2}\rho C_D A_f V^2$$

$\underline{A}$ = effective wetted frontal area





Assumptions: • will use widest cross section area

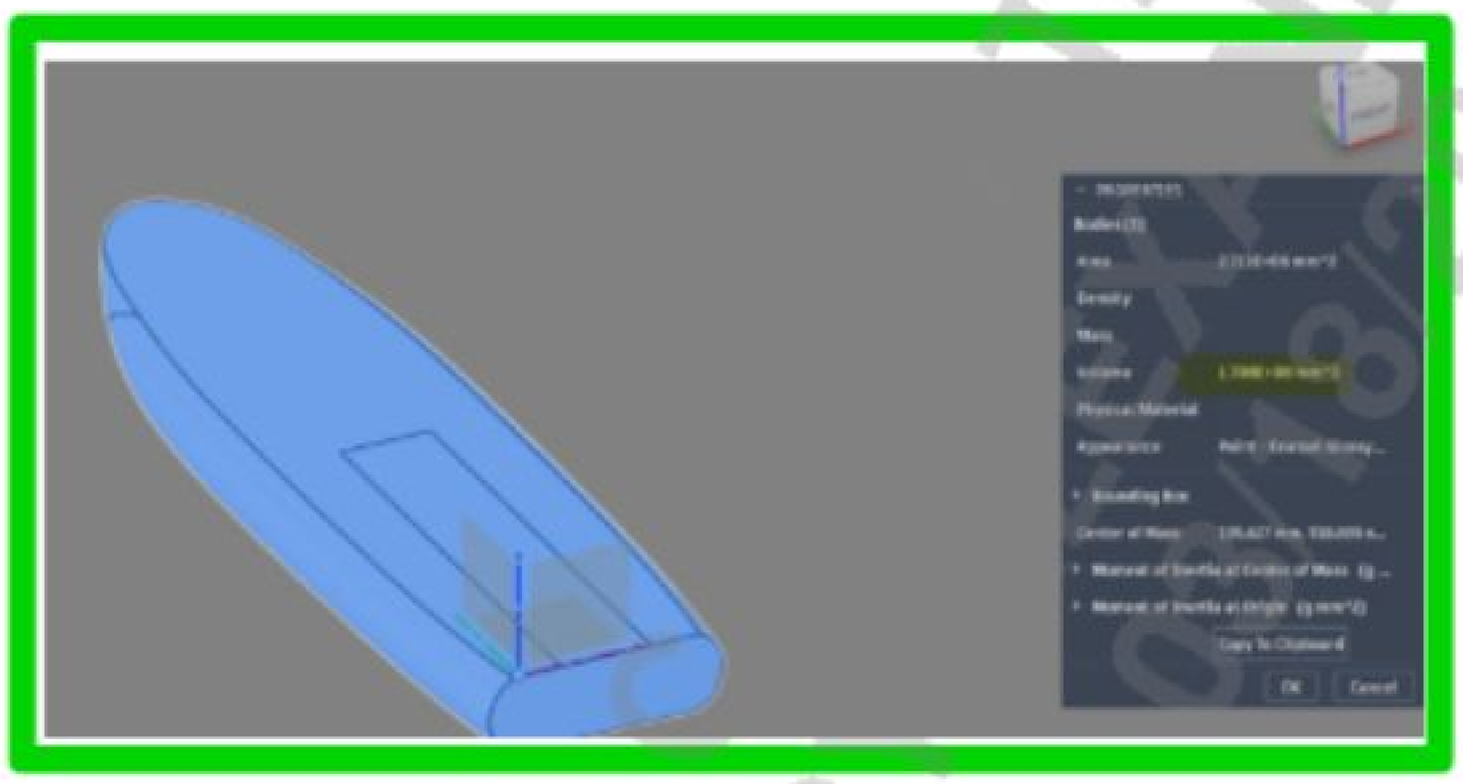
$\underline{A_x}$ = largest cross sectional area of board

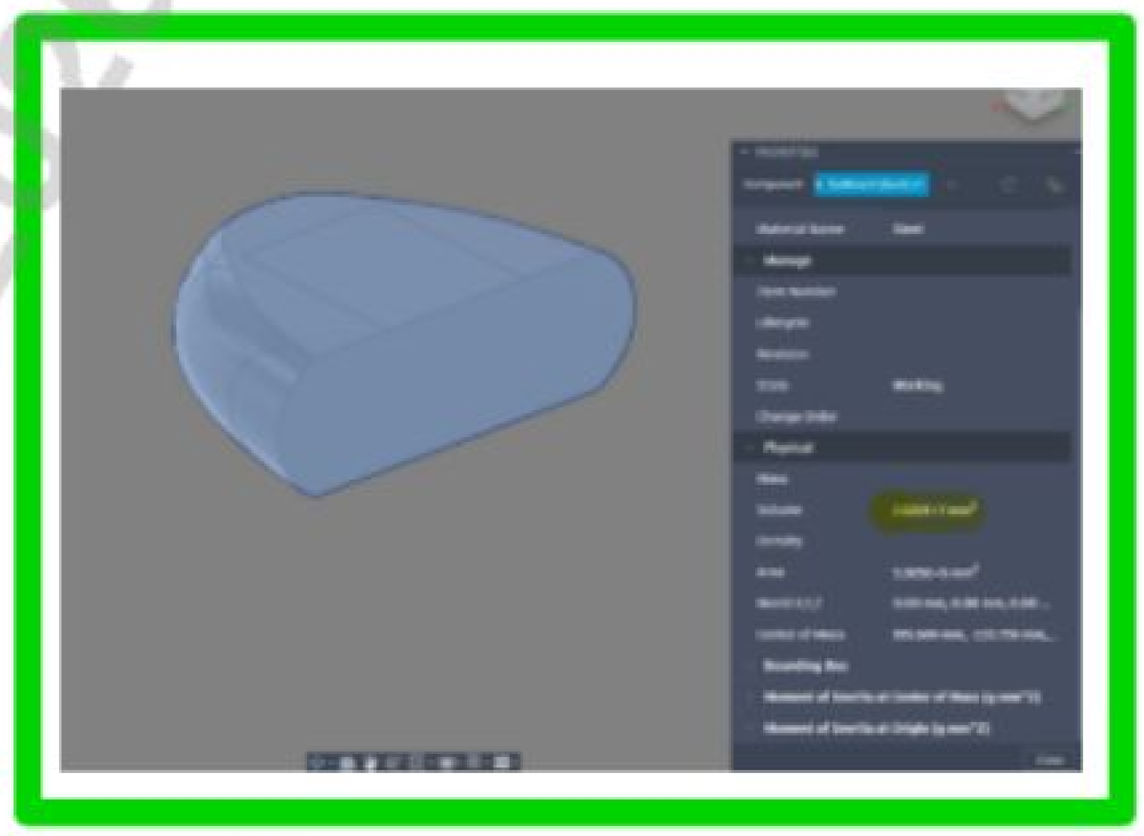
$$= (609.6\,mm)(177.8\,mm)$$

$$= 0.1084\,m^2$$

* from CAD Model, the exact area for the cross section $\longrightarrow 0.09455\,m^2$

$$A = \frac{V_{sub}}{V_{total}}(A_x)$$





$\underline{V_{total}} = 0.1708\,m^3 + 0.02022\,m^3 = 0.19102\,m^3$

$\underline{V_{sub}}$ = volume submerged $= \dfrac{m_{total}}{\rho}$ $\qquad V_{total}$ = total volume of board

$\underline{m_{total}}$ = Board + Box = 42.995 ≈ 43 kg
Rider = 190 lb ≈ 77 kg
Total = 120 kg

$$V_{sub} = \frac{m_{total}}{\rho} = \frac{120\,kg}{1000\,kg/m^3} = 0.12\,m^3$$

"Archimedes Principle: a floating object displaces a weight of fluid equal to its own total weight"

$$A = \frac{V_{sub}}{V_{total}}(A_x) = \left(\frac{0.12\,m^3}{0.19102\,m^3}\right)(0.09455\,m^2) = 0.0594\,m^2$$

C_D = effective drag coefficient $\longrightarrow$ typical for long board is 0.3 (high speed)

⑦ $D(V) = \mathcal{K} V^2$ "$\mathcal{K}$ is N of drag per V^2"

$$\Rightarrow \mathcal{K} = \tfrac{1}{2}\rho C_D A$$

$$\mathcal{K} = \tfrac{1}{2}(1000)(0.3)(0.0594\,m^2) = 8.91\ {kg}/{m}$$

$$\frac{kg \cdot m/s^2}{kg/m} \Rightarrow \frac{m^2}{s^2} = \sqrt{m^2/s^2} = m/s$$

⑧ $T = D(V) = \mathcal{K} V^2$

$$\Rightarrow V = \sqrt{T/\mathcal{K}}$$

$$V_A = \sqrt{T_A/\mathcal{K}} = \sqrt{1515/8.91} = 13.04\ m/s \approx 29.2\ mph$$

$$V_B = \sqrt{T_B/\mathcal{K}} = \sqrt{1675/8.91} = 13.71\ m/s \approx 30.7\ mph$$

⑨ "top speed" $\longrightarrow$ Assumptions: • will use $V_{xT} = (0.9)V_{max}$

"utilized as a safe guard to never push motor or battery to 100% of limit"

$$V_{AT} = (0.9)(13.04\ m/s) = 11.736\ m/s \approx 26.3\ mph$$

$$V_{BT} = (0.9)(13.71\ m/s) = 12.339\ m/s \approx 27.6\ mph$$

(10) $m a_x = T_x - D_x = T_x - K V_{xT}^2$

$$a_x = \frac{T_x - K V_{xT}^2}{m_{total}}$$

$$\left(\frac{m^2}{s^2}\right)\left(\frac{kg}{m}\right) \Rightarrow kg \cdot m/s^2$$

$$a_A = \frac{1515 - (8.91)(11.736\ m/s)^2}{120\ kg} = \frac{287.8\ N}{120\ kg} = 2.40\ m/s^2$$

$$a_B = \frac{1675 - (8.91)(12.339\ m/s)^2}{120\ kg} = \frac{318.4\ N}{120\ kg} = 2.65\ m/s^2$$

(11) "top acceleration" $\longrightarrow$ Assumptions: will use $a_{xT} = (0.9)\, a_x$

$$a_{AT} = (0.9)(2.40\ m/s^2) = 2.16\ m/s^2$$

$$a_{BT} = (0.9)(2.65\ m/s^2) = 2.39\ m/s^2$$

(12) time to "top speed" $\Rightarrow t_{xT} = \dfrac{V_{xT} - V_i}{a_{xT}}$

$$t_{AT} = \frac{11.736\ m/s}{2.16\ m/s^2} = 5.43\ sec$$

$$t_{BT} = \frac{12.339\ m/s}{2.39\ m/s^2} = 5.16\ sec$$

Comparison to LIND Electric Surfboard

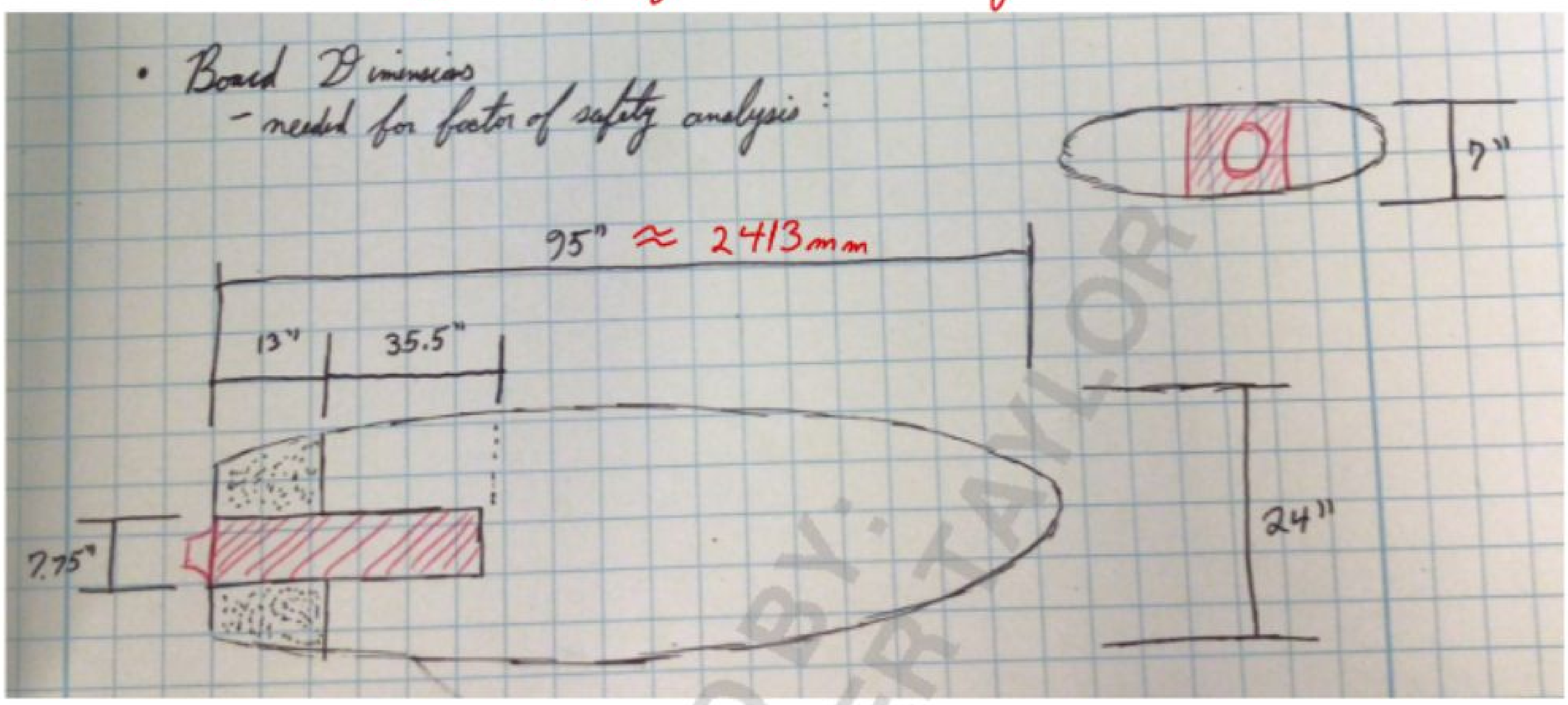
Known quantities from LIND		Our Board (including drag and 90% cubility factor in water)[2]
Board Weight (lb)	≈74	95
Price ($)	25000	≈4000
Power (kW)	20	25
Power (hp)	≈27	≈33
Top Speed (m/s)	16.7	19.8-20.7 (11.736-12.339)
Top Speed (mph)	37	44.3-46.8 (26.3-27.6)
Max Thrust (N)	≈1200	1515-1675
Acceleration (m/s^2)	3.09[1]	2.40-2.65 (2.16-2.39)
Zero to Top Speed Time (s)	≈5.4[1]	—— (5.43-5.16)

(1) Assuming LIND board can produce ½ the drag

(2) Assuming zero planing, widest section of board for drag, and ~68% of cross sectional area submerged at full speed

Detailed Hand Calculation Work

Factor of Safety - Fiberglass Shell



①

$\underline{W}$ = total weight of board + rider $= (m_b + m_r)(g)$

$= (9kg + 77kg)(9.81\ m/s^2)$

$= 843.66\ N \approx 844\ N$

$\underline{F_1} = W/2 = 422\ N$

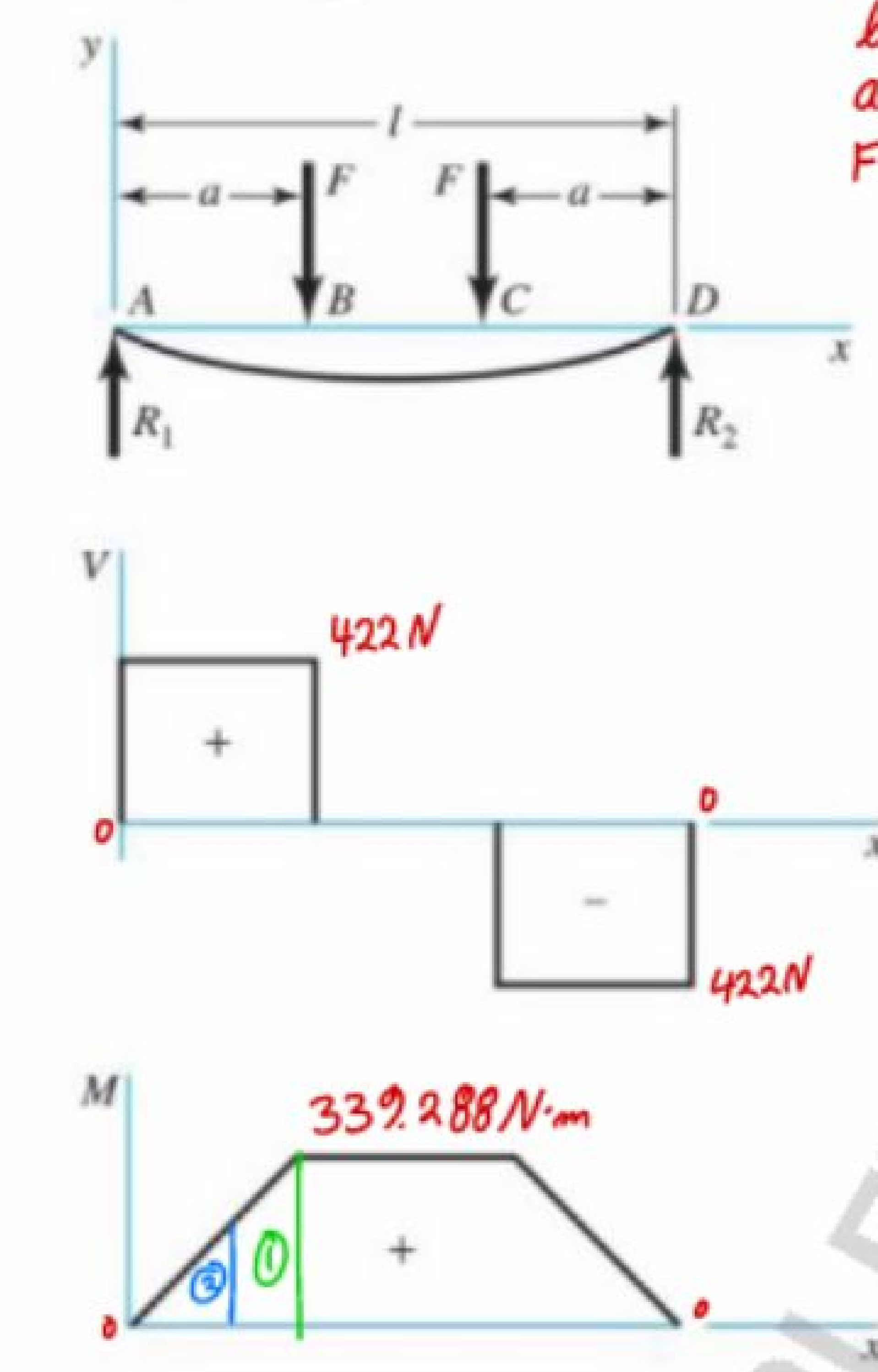
$\underline{F_2}$ = total weight of propulsion unit $= (m_p)(g)$

$= (34kg)(9.81\ m/s^2)$

$= 333.54\ N \approx 334\ N$

② Table A-9 #9 to get M_{max} @ middle and y_{max}

9 Simple supports—twin loads

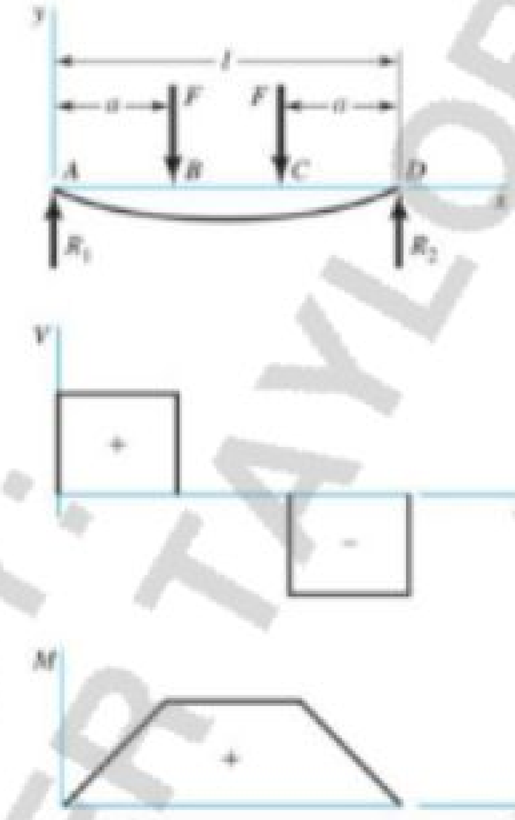


$l = 2413\,mm$
$a = 804\,mm$
$F = 422$

Table A–9 Shear, Moment, and Deflection of Beams *(Continued)*

(*Note:* Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 3–2.)

9 Simple supports—twin loads



$R_1 = R_2 = F \qquad V_{AB} = F \qquad V_{BC} = 0$

$V_{CD} = -F$

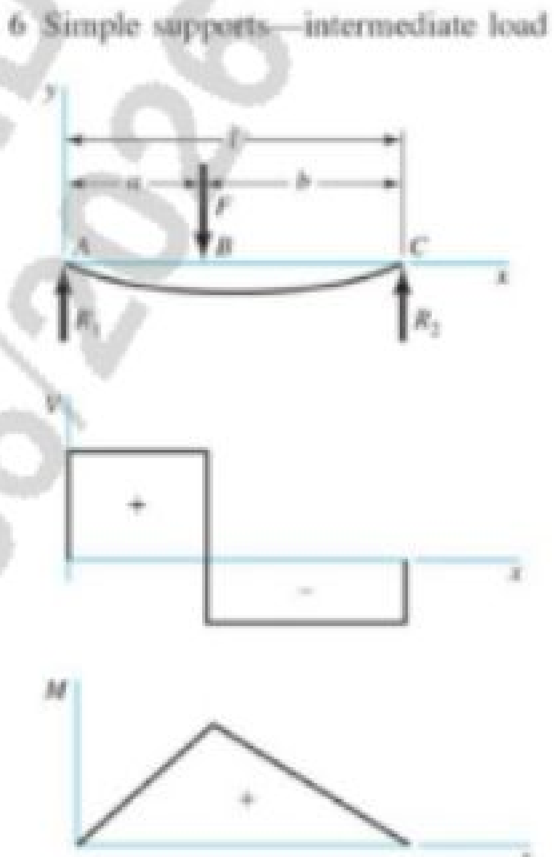
$M_{AB} = Fx \qquad M_{BC} = Fa \qquad M_{CD} = F(l - x)$

$$y_{AB} = \frac{Fx}{6EI}(x^2 + 3a^2 - 3la)$$

$$y_{BC} = \frac{Fa}{6EI}(3x^2 + a^2 - 3lx)$$

$$y_{max} = \frac{Fa}{24EI}(4a^2 - 3l^2)$$

6 Simple supports—intermediate load



$R_1 = \frac{Fb}{l} \qquad R_2 = \frac{Fa}{l}$

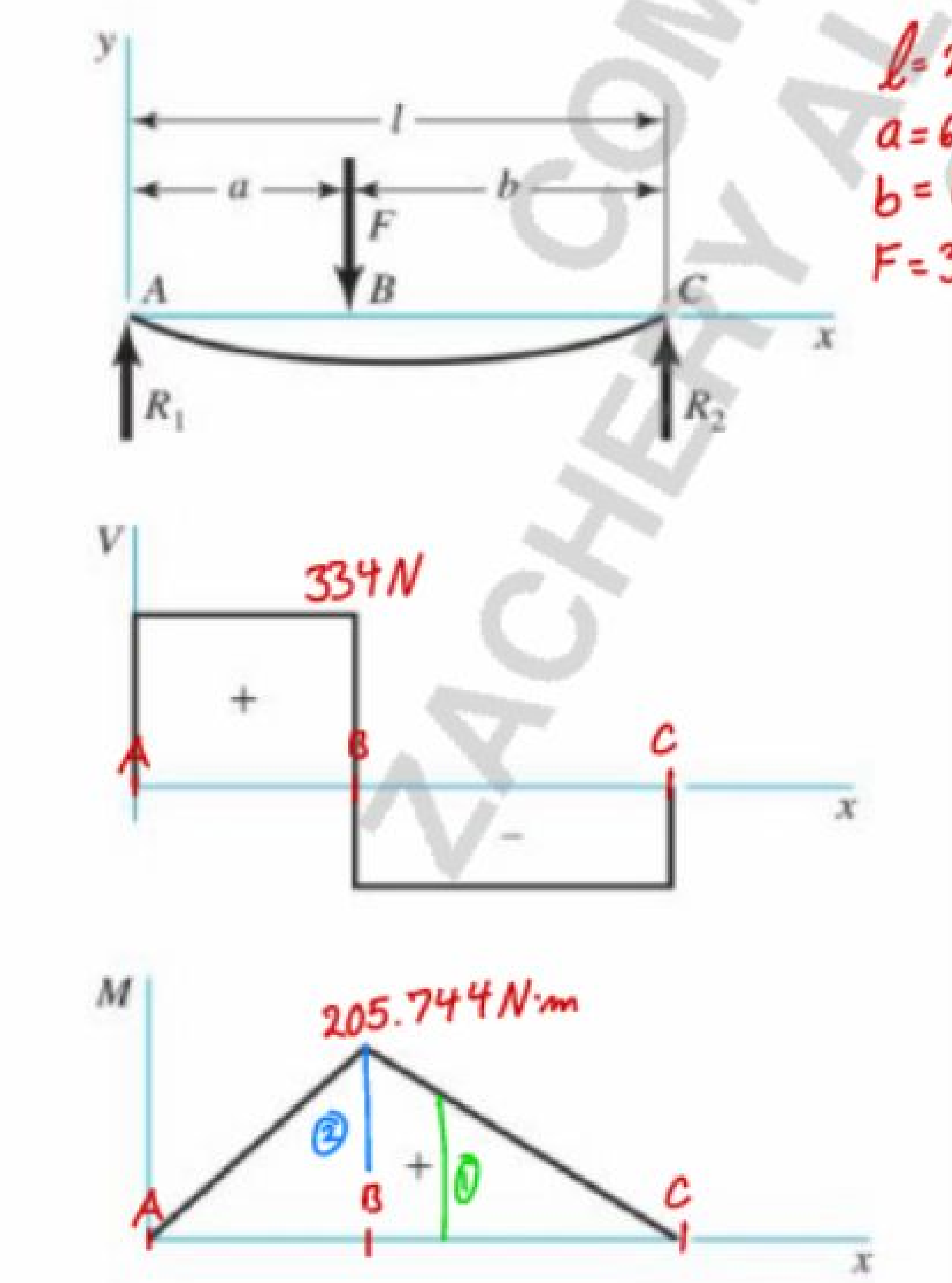
$V_{AB} = R_1 \qquad V_{BC} = -R_2$

$M_{AB} = \frac{Fbx}{l} \qquad M_{BC} = \frac{Fa}{l}(l - x)$

$$y_{AB} = \frac{Fbx}{6EIl}(x^2 + b^2 - l^2)$$

$$y_{BC} = \frac{Fa(l - x)}{6EIl}(x^2 + a^2 - 2lx)$$

6 Simple supports—intermediate load



$l = 2413\,mm$
$a = 616\,mm$
$b = 1797\,mm$
$F = 334\,N$

$a_1 = 804mm \quad F_1 = 422N$

$a_2 = 616mm \quad F_2 = 334N \quad b_2 = 1797mm$

$l = 2413mm \quad E = 11.8\ GPa$

$I = \frac{1}{12} bh^3 = 0.000286 m^4$

$$M_1 @ 804mm = F_1(a_1) + \frac{F_2(a_2)}{l}(l - a_1)$$

$$M_1 = 476.4791\ N \cdot m$$

$$M_2 @ 616mm = F_1(a_2) + \frac{F_2 b(a_2)}{l}$$

$$M_2 = 413.1729 N \cdot m$$

$$\boxed{M_{max} \approx 477 N \cdot m \approx 477000 N \cdot mm}$$

$$y_{max} = \frac{F_1(a_1)}{6EI}\left[(a_1)^2 + 3(a_1)^2 - 3l(a_1)\right] + \frac{F_2 a_2 (l-(a_1))}{6EIl}\left[(a_1)^2 + (a_2)^2 - 2l(a_1)\right]$$

$$\boxed{y_{max} = 0.0737 mm}$$

③ T

Peak motor output to driveshaft $= 25\,kW \approx 33\,hp$

* Will use 90% of peak as "top" to account for capping electrical components

Using formula (3-43)

$$H = \frac{T n}{63025}$$

H = power, hp $\quad$ n = shaft speed, rev/min

T = torque, lbf·in

* convert to metric *

$$P_{kW} = \frac{T_{N\cdot m}\, n}{9550} \Longrightarrow T_{N\cdot m} = \frac{(9550)(P_{kW})}{n}$$

$P_{kW,90}$ = power, kW = $P_{kW}(90\%)$ $\quad$ n_{90} = shaft speed, rev/min = $n(90\%)$

T_{90} = torque, N·m

$n_{90} = 9000\,rpm(90\%) = 8100\,rpm$ $\quad$ $P_{kW,90} = 25\,kW(90\%) = 22.5\,kW$

$$\boxed{T_{90} = \frac{(9550)(22.5)}{8100} = 26.53\,N\cdot m \approx 26530\,N\cdot mm}$$

④ k_a = Surface Factor

Using Table 6-2

$$k_a = a\, S_{UT}^{\ b}$$

S_{ut} = ultimate tensile strength of the material in MPa

* Surface Finish for this project will be assumed to be "Machined" based of the Curve Fit Parameters for Surface Factor

$a = 3.04\,MP_a$

$b = -0.217$ [exponent]

S_{ut} = determined based on testing data $= 91.93\,MP_a$

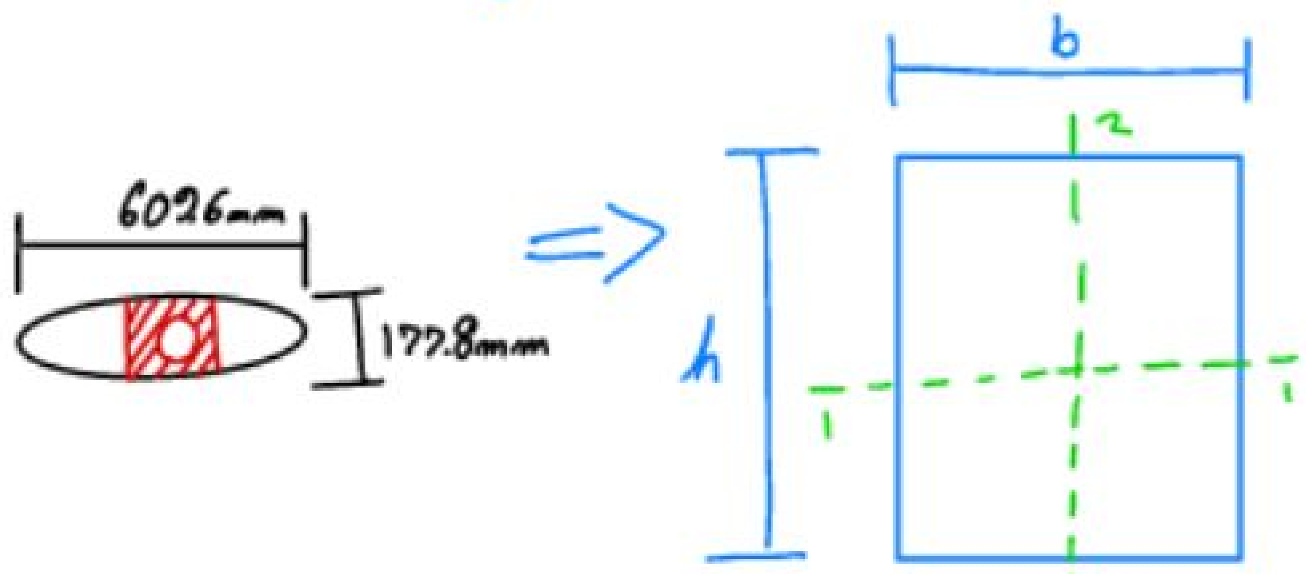
$$k_a = (3.04\,MP_a)(91.93\,MP_a)^{-0.217} \approx 1.13$$

k_a cannot be greater than 1.0 for research purposes we will use 0.9 to be conservative.

$$\boxed{k_a = 0.9}$$

⑤ k_b = Size Factor

Using Table 6-3



$d = d_e$ for Common Nonrotating Structural Shapes Loading in Bending

$$d_e = 0.808\sqrt{hb}$$

$$= 0.808\sqrt{(609.6\,mm)(177.8\,mm)}$$

$$d_e = 266\,mm$$

*Noting variable approximation

Since $d_e = 266\,mm$ is outside the recommended range, k_b is taken as the value at 254 mm and not reduced further.

Using (6-19)

$$k_b = \begin{cases} (d/7.62)^{-0.107} = 1.24\,d^{-0.107} & 7.62 \le d \le 51\,mm \\ 1.51\,d^{-0.157} & 51 \le d \le 254\,mm \end{cases}$$

$$k_b = 1.51(254)^{-0.157}$$

$$\boxed{k_b = 0.633}$$

⑥ k_c = Loading Factor

(6-25)

$$k_c = \begin{cases} 1 & \text{bending} \\ 0.85 & \text{axial} \\ 0.59 & \text{torsion} \end{cases}$$

$$\boxed{k_c = 1}$$

⑦ k_d = Temperature Factor

* Note

There is little quantitative data for how fiberglass/epoxy reacts to moderate temperature fluctuation within the typical working range.
Therefore 1 will be utilized for k_d.

$$\boxed{k_d = 1}$$

⑧ k_e = Reliability Factor

Reliability, %	Transformation Variate z_a	Reliability Factor k_e
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702
99.999	4.265	0.659
99.9999	4.753	0.620

Using 90% Reliability:

50% is too optimistic, with a 50% chance of failure at that stress level — this is not acceptable even for an initial prototype

>90% would be sufficient where failure has catastrophic consequences.

90% is consistent with preliminary mechanical design practice for a non-safety critical component.

⑨ $S_e = k_a k_b k_c k_d k_e (S_e') = k_a k_b k_c k_d k_e \left(\frac{S_{ut}}{2}\right)$

$k_a = 0.9$

$k_b = 0.633$

$k_c = 1$

$k_d = 1$

$k_e = 0.897$

$$S_e' = \begin{cases} 0.5 S_{ut}, & S_{ut} \le 1400 MPa \\ 700 MPa, & S_{ut} > 1400 MPa \end{cases}$$

* Note
To be extensively conservative with a material that does not have abundant research completed on it we will use $S_e' = 0.3 S_{ut} = 27.579 MPa$

$S_e = (0.9)(0.633)(1)(1)(0.897)(27.579)$

$\boxed{S_e = 14.09}$

(10) K_f & K_{fs} => First Iteration Estimate for Stress Concentration Factors K_t & K_{ts}

* Utilized to represent rear propulsion exit cut

Table 7-1	Bending	Torsional
Using End-mill keyseat	2.14	3.0

$$K_f = 1 + q(K_t - 1) \qquad K_{fs} = 1 + q_{shear}(K_{ts} - 1)$$

* $q = q_{shear} = 1$;

There is not sufficient research on fiberglass/epoxy. For analysis applying $q=1$ will remain conservative in the analysis; treating the composite material as fully notch-sensitive (no stress relief below K_t)

$$K_f = 1 + q(K_t - 1) \qquad K_{fs} = 1 + q_{shear}(K_{ts} - 1)$$

$$K_f = 1 + 1(2.14 - 1) \qquad K_{fs} = 1 + 1(3.0 - 1)$$

$$\boxed{K_f = 2.14} \qquad \boxed{K_{fs} = 3.0}$$

DE + Gerber

* Use Gerber since first design

Gerber $\frac{n\sigma_a}{S_e} + \left(\frac{n\sigma_m}{S_{ut}}\right)^2 = 1$

- **DE + Gerber**

$$\frac{1}{n} = \frac{8A}{\pi d^3 S_e}\left\{1 + \left[1 + \left(\frac{2BS_e}{AS_{ut}}\right)^2\right]^{1/2}\right\}$$

$$d = \left(\frac{8nA}{\pi S_e}\left\{1 + \left[1 + \left(\frac{2BS_e}{AS_{ut}}\right)^2\right]^{1/2}\right\}\right)^{1/3}$$

$$A = \sqrt{4(K_f M_a)^2 + 3(K_{fs} T_a)^2}$$

$$B = \sqrt{4(K_f M_m)^2 + 3(K_{fs} T_m)^2}$$

$A = \sqrt{4[(2.14)(477 N\cdot m)]^2}$ $*T_a = 0$

$A = 2041.56\ N\cdot m \approx 2041560\ N\cdot mm$

$B = \sqrt{3[(3.0)(26.53 N\cdot m)]^2}$ $*M_m = 0$

$B = 137.854\ N\cdot m \approx 137854\ N\cdot mm$

$\frac{8A}{\pi d^3 S_e} \approx 0.02251$ $\left\{1 + \left[1 + \left(\frac{2BS_e}{AS_{ut}}\right)^2\right]^{1/2}\right\} \approx 2.0$

$1/n \approx 0.045$

$\boxed{n = 22.21}$

DE + Goodman

- **DE + Modified Goodman**

$$\frac{1}{n} = \frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}}$$

$$\frac{1}{n} = \frac{16}{\pi d^3}\left\{\frac{1}{S_e}[4(K_f M_a)^2 + 3(K_{fs} T_a)^2]^{1/2} + \frac{1}{S_{ut}}[4(K_f M_m)^2 + 3(K_{fs} T_m)^2]^{1/2}\right\}$$

$$d = \left(\frac{16n}{\pi}\left\{\frac{1}{S_e}[4(K_f M_a)^2 + 3(K_{fs} T_a)^2]^{1/2} + \frac{1}{S_{ut}}[4(K_f M_m)^2 + 3(K_{fs} T_m)^2]^{1/2}\right\}\right)^{1/3}$$

$* T_a = 0 \; ; \; M_m = 0$

$$\frac{1}{n} = \frac{16}{\pi d^3}\left\{\frac{1}{S_e}\left[4(K_f M_a)^2\right]^{1/2} + \frac{1}{S_{ut}}\left[3(K_{fs} T_m)^2\right]^{1/2}\right\}$$

$$\frac{1}{n} = 0.0458$$

$$\boxed{n = 21.82}$$